EE 501 — Linear System Theory I

Lecture Notes

Based on the handwritten notes by Elif Topaloğlu
of Umut Orguner's lectures (Fall 2024)

LaTeX transcription, corrected and annotated; organized along the Fall 2026–2027 syllabus

How to read these notes. **Bold blue** marks a term at the place where it is defined; **bold** marks a key statement.

- Framed boxes: definitions, theorems, examples.
- Yellow "Explanation" boxes and ▷ *italic lines* are additions that are not in the original handwritten notes.
- The gray *p. N* in the margin gives the page of the original handwritten notes.
- Corrections to the original are listed in `CORRECTIONS.md`.

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Course Outline

Tentative schedule of EE501 (Fall 2026–2027) and the corresponding sections of these notes.

Week	Subject	Notes
1	Algebraic structures defined on sets, fields and vector spaces	§1.1–1.2
2	Subspaces, linear independence, basis	§1.3–1.7
3	Normed vector spaces, sequences, series and convergence in normed spaces, matrix norms	§2
4	Inner product spaces, orthogonality, Gram–Schmidt orthonormalization	§3
5	Linear transformations, null space, range space, matrix representations	§4
6	Direct sum decompositions, projection theorem, orthogonal projections	§5
7	Systems of linear equations of the form $Ax = b$	§6
8	Spectral analysis of linear transformations, eigenvalues and eigenvectors	§7.1–7.3
9	Invariance under operators, block diagonal forms	§7.4–7.5
10	Cayley–Hamilton theorem, characteristic and minimal polynomials	§8
11–12	Jordan canonical forms	§9
13	Hermitian matrices, positive definite matrices, positive semi-definite matrices	§10
14	Functions of a matrix	§11

References.

- S. Axler, *Linear Algebra Done Right*, Springer, 1997.
- C. A. Desoer, *Notes for a Second Course on Linear Systems*, Van Nostrand Reinhold, 1970.
- C. D. Meyer, *Matrix Analysis and Applied Linear Algebra*, SIAM, 2000.
- G. Strang, *Linear Algebra and its Applications*, 4th ed., Brooks/Cole, 2006.

1 Fields and Vector Spaces

Syllabus: Weeks 1–2 — algebraic structures, fields, vector spaces, subspaces, linear independence, basis

1.1 Fields

Definition 1.1. A **field** is a set $\mathcal{F}$ together with two mappings called **addition** and **multiplication** shown as:

$$\left.\begin{aligned}\oplus : \mathcal{F}\times\mathcal{F}\to\mathcal{F}\\ \odot : \mathcal{F}\times\mathcal{F}\to\mathcal{F}\end{aligned}\right\} \quad \mathcal{F} \text{ should be closed on both } \oplus \text{ and } \odot.$$

$\oplus$ and $\odot$ have the following properties (axioms):

Addition.

(A1) $a\oplus b = b\oplus a \quad \forall a,b\in\mathcal{F}$ (commutativity)

(A2) $a\oplus(b\oplus c) = (a\oplus b)\oplus c \quad \forall a,b,c\in\mathcal{F}$ (associativity)

(A3) There exists ($\exists$) an element in $\mathcal{F}$ denoted by $0_{\mathcal{F}}$ such that

$$a\oplus 0_{\mathcal{F}} = a \quad \forall a\in\mathcal{F} \qquad \text{(additive identity)}$$

(A4) For each $a\in\mathcal{F}$, $\exists$ an element in $\mathcal{F}$ denoted as $-a$ such that

$$a\oplus -a = 0_{\mathcal{F}} \qquad \text{(additive inverse)}$$

Multiplication.

(M1) $a\odot b = b\odot a \quad \forall a,b\in\mathcal{F}$

(M2) $(a\odot b)\odot c = a\odot(b\odot c) \quad \forall a,b,c\in\mathcal{F}$

(M3) $\exists\, 1_{\mathcal{F}}\in\mathcal{F}$ such that

$$a\odot 1_{\mathcal{F}} = a \quad \forall a\in\mathcal{F} \qquad \text{(multiplicative identity)}$$

(M4) For each $a\neq 0_{\mathcal{F}}$, $\exists$ an element in $\mathcal{F}$ shown as a^{-1} such that

$$a\odot a^{-1} = 1_{\mathcal{F}} \qquad \text{(multiplicative inverse)}$$

Distributivity (D). The operations $\oplus$ & $\odot$ satisfy

$$a\odot(b\oplus c) = (a\odot b)\oplus(a\odot c) \quad \forall a,b,c\in\mathcal{F} \qquad \text{(Distributive Law)}$$

Example 1.2. $\mathcal{F}=\{0,1\}$ with $\oplus = +$ and $\odot = \cdot$.

$$\oplus : \mathcal{F}\times\mathcal{F}\to\mathcal{F}, \qquad 1+1=2 \;\Rightarrow\; \text{not in Field}$$

Also no additive inverse holds: $1+a=0$ has no solution $a\in\{0,1\}$ (we would need $a=-1\notin\mathcal{F}$).

Remark

With arithmetic *modulo 2* ($1 \oplus 1 = 0$, $\odot$ the ordinary product) the set $\{0, 1\}$ *is* a field, usually denoted GF(2) or $\mathbb{Z}_2$: here $-1 = 1$ and $1^{-1} = 1$. So whether a set is a field depends on the operations, not only on the set.

Example 1.3. Set of real numbers $\mathbb{R}$ with the standard addition & multiplication.

$$\mathcal{F} = \mathbb{R}, \qquad \oplus = +, \qquad \odot = \cdot \qquad \Rightarrow \text{ A field}$$

Example 1.4. Let $\mathcal{F} = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ with p. 2

$$x = (x_1, x_2) \in \mathcal{F}, \qquad y = (y_1, y_2) \in \mathcal{F}$$

$$\begin{aligned} x \oplus y &= (x_1 + y_1,\ x_2 + y_2) \\ x \odot y &= (x_1 y_1 - x_2 y_2,\ x_2 y_1 + x_1 y_2) \end{aligned}$$

▷ *Go over all axioms and see if it is satisfied.*

$$\begin{aligned} 0_{\mathcal{F}} &= (0, 0) \\ -(x_1, x_2) &= (-x_1, -x_2) \end{aligned}$$

Assume as example (complex-number notation):

$$\begin{aligned} (x_1, x_2) &= x_1 + j x_2 \\ (y_1, y_2) &= y_1 + j y_2 \\ x \oplus y &= x_1 + y_1 + j(x_2 + y_2) \\ x \odot y &= x_1 y_1 - x_2 y_2 + j(x_2 y_1 + x_1 y_2) \end{aligned}$$

▷ *The multiplication rule is exactly complex multiplication:* $(x_1 + j x_2)(y_1 + j y_2) = x_1 y_1 - x_2 y_2 + j(x_2 y_1 + x_1 y_2)$ *since* $j^2 = -1$.

$$1_{\mathcal{F}} = (1, 0)$$

$$(x_1, x_2)^{-1} = \left(\frac{x_1}{x_1^2 + x_2^2},\ \frac{-x_2}{x_1^2 + x_2^2} \right), \qquad \frac{1}{x_1 + j x_2} = \frac{x_1 - j x_2}{x_1^2 + x_2^2}$$

∴ **This is a field.**

Explanation

Check: $(x_1, x_2) \odot (x_1, x_2)^{-1} = \left(\frac{x_1^2 + x_2^2}{x_1^2 + x_2^2},\ \frac{x_2 x_1 - x_1 x_2}{x_1^2 + x_2^2} \right) = (1, 0) = 1_{\mathcal{F}}$, defined for every $x \neq 0_{\mathcal{F}} = (0, 0)$. The remaining axioms (commutativity, associativity, distributivity) are inherited from those of complex numbers; this field is simply $\mathbb{C}$ written as pairs of reals.

Example 1.5. Let $\mathcal{F} = \mathbb{R}$ with $x \oplus y = xy$ and $x \odot y = x + y$. Is $\mathcal{F}$ a field?

$$x \oplus y = xy = yx = y \oplus x \quad \forall x, y \in \mathcal{F} \qquad \checkmark$$

A2 is satisfied $\checkmark$

Identities and inverses.

$$0_{\mathcal{F}} = 1: \qquad 0 \oplus a = 0_{\mathcal{F}} \;\Rightarrow\; \text{not satisfied } (0 \cdot a = 1\ \times)$$
$$1_{\mathcal{F}} = 0: \qquad a \in \mathcal{F} \Rightarrow a^{-1} = -a$$

▷ *The additive identity must satisfy* $a \oplus 0_{\mathcal{F}} = a \cdot 0_{\mathcal{F}} = a$ *for all* a*, so* $0_{\mathcal{F}} = 1$*. Then the element* 0 *has no additive inverse, since* $0 \oplus a = 0 \cdot a = 0 \neq 1$ *for every* a *(A4 fails). Likewise* $1_{\mathcal{F}} = 0$ *(as* $a + 0 = a$*), and* $a \odot (-a) = 0 = 1_{\mathcal{F}}$*, so (M4) holds.*

Distributivity.

$$x \odot (y \oplus z) \overset{?}{=} (x \odot y) \oplus (x \odot z)$$
$$x + yz \neq (x + y)(x + z) = x^2 + xz + xy + yz \quad \text{(in general; e.g. } x = y = z = 1\text{: } 2 \neq 4)$$

$\Rightarrow \therefore$ **Not a field, as A4 & D fail.**

Example 1.6. $\mathbb{R}^{2\times 2}$ with standard matrix addition & multiplication: for A, B there is no multiplicative commutativity.

$$\mathbb{R}^n: \begin{cases} n = 1, 2 & \Rightarrow \text{can be able to define fields} \\ n = 3, 4, \ldots & \Rightarrow \text{cannot be able to define any field.} \end{cases}$$

Precise meaning

Here "define a field on $\mathbb{R}^n$" means: componentwise addition and a multiplication that is compatible with real scaling (bilinear), so that $\mathbb{R} \subset \mathcal{F}$ sits inside as $(a, 0, \ldots, 0)$. For $n = 1$ we get $\mathbb{R}$, for $n = 2$ we get $\mathbb{C}$ (previous example). For $n \geq 3$ no such field exists (Frobenius/Hopf); e.g. for $n = 4$ the quaternions satisfy every axiom except commutativity (M1), exactly like $\mathbb{R}^{2\times 2}$ fails (M1) (and in addition (M4), as singular matrices have no inverse).

1.2 Vector Spaces

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Definition 1.7. A **linear space** (**vector space**) $\mathcal{V}$ is a set whose elements are called **vectors**, associated with a field $\mathcal{F}$ whose elements are called **scalars** and two operations $(\mathcal{V}, \mathcal{F})$: addition $\oplus_{\mathcal{V}}$ and scalar multiplication $\odot_{\mathcal{V}}$.

$$\left.\begin{aligned} \oplus &: \mathcal{V} \times \mathcal{V} \to \mathcal{V} \\ \odot &: \mathcal{F} \times \mathcal{V} \to \mathcal{V} \end{aligned}\right\} \quad \mathcal{V} \text{ should be closed on addition \& multiplication (closedness)}$$

The following axioms should hold for $\oplus$ and $\odot$:

Addition.

(A1) $x \oplus y = y \oplus x \quad \forall x, y \in \mathcal{V}$

(A2) $x \oplus (y \oplus z) = (x \oplus y) \oplus z \quad \forall x, y, z \in \mathcal{V}$

(A3) $\exists\, 0_{\mathcal{V}} \in \mathcal{V}$ s.t. $x \oplus 0_{\mathcal{V}} = x \quad \forall x \in \mathcal{V}$

(A4) For every element x, $\exists\, -x$ s.t. $x \oplus (-x) = 0_{\mathcal{V}}$

Scalar multiplication.

(SM1) $a \odot_{\mathcal{V}} (b \odot_{\mathcal{V}} x) = (a \odot_{\mathcal{F}} b) \odot_{\mathcal{V}} x \quad \forall \underbrace{a, b \in \mathcal{F}}_{\text{scalar}} \quad \forall \underbrace{x \in \mathcal{V}}_{\text{vector}}$

(SM2) $a \odot (x \oplus y) = (a \odot x) \oplus (a \odot y) \quad \forall a \in \mathcal{F} \quad \forall x, y \in \mathcal{V}$

(SM3) $(a \oplus b) \odot x = (a \odot x) \oplus (b \odot x)$

(SM4) With $1_{\mathcal{F}}$ being the multiplicative identity of the field $\mathcal{F}$, then we have

$$1_{\mathcal{F}} \odot x = x \quad \forall x \in \mathcal{V}$$

Example 1.8. Prove $0_{\mathcal{F}} \odot x = 0_{\mathcal{V}} \quad \forall x \in \mathcal{V}$.
Let $y \triangleq 0_{\mathcal{F}} \odot x$; then $y \in \mathcal{V}$ (as multiplication is closed).

$$\begin{aligned} y \oplus y &= (0_{\mathcal{F}} \odot x) \oplus (0_{\mathcal{F}} \odot x) \\ &\overset{\text{SM3}}{=} (0_{\mathcal{F}} \oplus 0_{\mathcal{F}}) \odot x \\ &\overset{\text{Additive identity}}{=} 0_{\mathcal{F}} \odot x = y \end{aligned}$$

▷ *SM3 read from right to left; then $0_{\mathcal{F}} \oplus 0_{\mathcal{F}} = 0_{\mathcal{F}}$ is (A3) of the field $\mathcal{F}$.*
$\Rightarrow \; y \oplus y = y$ is proven.

$$\begin{aligned} \text{A4} &\Rightarrow \exists -y \text{ s.t. } y \oplus -y = 0_{\mathcal{V}} \\ \text{add } -y \text{ to both sides of } y \oplus y = y &\Rightarrow (y \oplus y) \oplus -y = y \oplus -y \\ \text{A1, A2} &\Rightarrow (y \oplus -y) \oplus y = y \oplus -y \\ \text{A4} &\Rightarrow y \oplus 0_{\mathcal{V}} = 0_{\mathcal{V}} \\ \text{A3} &\Rightarrow \underline{y = 0_{\mathcal{V}}} \end{aligned}$$

Explanation

Since $y = 0_{\mathcal{F}} \odot x$, this proves $0_{\mathcal{F}} \odot x = 0_{\mathcal{V}}$. The same trick ($y \oplus y = y \Rightarrow y = 0_{\mathcal{V}}$) also gives $a \odot 0_{\mathcal{V}} = 0_{\mathcal{V}}$ for every $a \in \mathcal{F}$, and with SM3 one gets $(-1_{\mathcal{F}}) \odot x = -x$, because $x \oplus ((-1_{\mathcal{F}}) \odot x) = (1_{\mathcal{F}} \oplus -1_{\mathcal{F}}) \odot x = 0_{\mathcal{F}} \odot x = 0_{\mathcal{V}}$.

Example 1.9. Linear space $(\overset{\mathcal{V}}{\mathcal{F}^n}, \overset{\mathcal{F}}{\mathcal{F}})$; i.e., the vector space of n tuples in $\mathcal{F}$ over the field $\mathcal{F}$. p. 4

$$x = (x_1, x_2, \ldots, x_n), \qquad y = (y_1, y_2, \ldots, y_n), \qquad \text{where } x_i, y_i \in \mathcal{F} \; \forall i$$

Addition:

$$\begin{aligned} x \oplus_{\mathcal{V}} y &= (x_1 \oplus_{\mathcal{F}} y_1,\ x_2 \oplus_{\mathcal{F}} y_2,\ \ldots,\ x_n \oplus_{\mathcal{F}} y_n) \\ a \odot_{\mathcal{V}} y &= (a \odot_{\mathcal{F}} y_1,\ a \odot_{\mathcal{F}} y_2,\ \ldots,\ a \odot_{\mathcal{F}} y_n) \end{aligned}$$

$$(\mathbb{R}^n, \mathbb{R}) \checkmark \qquad (\mathbb{C}^n, \mathbb{R}) \checkmark \qquad (\mathbb{C}^n, \mathbb{C}) \checkmark \qquad (\mathbb{R}^n, \mathbb{C}) \times$$

▷ $(\mathbb{C}^n, \mathbb{R})$*: a real multiple of a complex vector is still complex, so it is a vector space (of dimension $2n$ over $\mathbb{R}$).*
▷ $(\mathbb{R}^n, \mathbb{C})$*: Scalar multiplication violates, take a real vector & multiply it with a complex scalar and we have a complex vector which is $\notin \mathcal{V}$.*

Example 1.10. Set of all real valued functions $t \mapsto f(t)$ defined over the real line.

$$\begin{aligned} \mathcal{F} &= \mathbb{R} \\ \mathcal{V} &= \text{real valued functions} \\ (f \oplus_{\mathcal{V}} g)(t) &= f(t) \oplus_{\mathcal{F}} g(t) \\ (a \odot_{\mathcal{V}} f)(t) &= a \odot_{\mathcal{F}} f(t) \end{aligned}$$

Example 1.11. Set of all polynomials with degree $n = 2$

$$\mathcal{V} = \{a_0 t^2 + a_1 t + a_2 \mid a_0, a_1, a_2 \in \mathbb{R} \ \& \ a_0 \neq 0\}$$

$\oplus$ and $\odot$ are standard polynomial addition and scalar multiplication.
$\therefore$ Not a vector space!

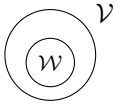
$$\begin{aligned} P_1(t) &= t^2 + 2t + 3 \in \mathcal{V} \\ P_2(t) &= -t^2 + 2t + 3 \in \mathcal{V} \\ (P_1 \oplus P_2)(t) &= 4t + 6 \notin \mathcal{V} \end{aligned}$$

Example 1.12. Set of all polynomials with degree $n \leq 2$ is a vector space.

$$\mathcal{V} = \{a_0 t^2 + a_1 t + a_2 \mid a_0, a_1, a_2 \in \mathbb{R}\}$$

1.3 Subspaces

Definition 1.13. Let $\mathcal{V}$ be a vector space over the field $\mathcal{F}$. A subset $\mathcal{W}$ of $\mathcal{V}$ is a **subspace** of $\mathcal{V}$ if and only if $(\mathcal{W}, \mathcal{F})$ is itself a vector space.



$$\left[\text{only show that} \quad \left.\begin{matrix} \oplus : \mathcal{V} \times \mathcal{V} \to \mathcal{V} \\ \odot : \mathcal{F} \times \mathcal{V} \to \mathcal{V} \end{matrix}\right\} \text{ holds}\right]$$

Remark. We can check whether a (nonempty) subset $\mathcal{W}$ is a subspace or not by checking **only closure under addition and scalar multiplication**:

(S1)	$w_1 \oplus w_2 \in \mathcal{W}$	$\forall w_1, w_2 \in \mathcal{W}$
(S2)	$a \odot w_1 \in \mathcal{W}$	$\forall w_1 \in \mathcal{W}, \ \forall a \in \mathcal{F}$

Why closure is enough

All axioms that are identities (A1, A2, SM1–SM4) hold in $\mathcal{W}$ automatically because they hold in the bigger set $\mathcal{V}$. The two existence axioms follow from (S2): taking $a = 0_{\mathcal{F}}$ gives $0_{\mathcal{F}} \odot w = 0_{\mathcal{V}} \in \mathcal{W}$ (A3), and taking $a = -1_{\mathcal{F}}$ gives $(-1_{\mathcal{F}}) \odot w = -w \in \mathcal{W}$ (A4). In particular, **every subspace contains** $0_{\mathcal{V}}$; a quick test is: if $0_{\mathcal{V}} \notin \mathcal{W}$, then $\mathcal{W}$ is not a subspace.

Example 1.14. $\mathcal{V} = \mathbb{R}^2, \mathcal{F} = \mathbb{R}, \quad \mathcal{W} = \left\{ \begin{bmatrix} a \\ 0 \end{bmatrix} \,\middle|\, a \in \mathbb{R} \right\}$ p. 5

$$w_1 \in \mathcal{W}: \quad w_1 = \begin{bmatrix} a \\ 0 \end{bmatrix} \qquad\qquad w_1 \oplus w_2 = \begin{bmatrix} a+b \\ 0 \end{bmatrix} \in \mathcal{W}$$

$$w_2 \in \mathcal{W}: \quad w_2 = \begin{bmatrix} b \\ 0 \end{bmatrix} \qquad\qquad c \odot w_1 = \begin{bmatrix} ca \\ 0 \end{bmatrix} \in \mathcal{W}$$

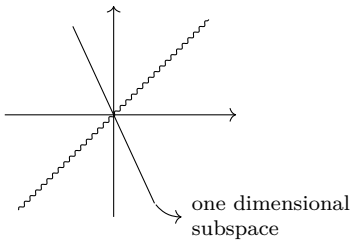
$\therefore \mathcal{W}$ is a subspace of $\mathcal{V}$.

Example 1.15. $\mathcal{V} = \mathbb{R}^2$, $\mathcal{F} = \mathbb{R}$, $\quad \mathcal{W} = \left\{ \begin{bmatrix} a \\ 1 \end{bmatrix} \,\middle|\, a \in \mathbb{R} \right\}$

$$w_1 = \begin{bmatrix} a \\ 1 \end{bmatrix}, \quad w_2 = \begin{bmatrix} b \\ 1 \end{bmatrix} \qquad \Rightarrow \qquad w_1 \oplus w_2 = \begin{bmatrix} a+b \\ 2 \end{bmatrix} \notin \mathcal{W}$$

Not closed under vector addition or scalar multiplication (e.g. $0 \odot w_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \notin \mathcal{W}$: $\mathcal{W}$ does not contain the zero vector).
$\therefore \mathcal{W}$ is <u>not</u> a subspace.

▶ $\{0_\mathcal{V}\}$: zero dimensional vector space $\Rightarrow$ subspace.



$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathcal{W}$$
$$-\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathcal{W}$$

$a \odot x = 0_\mathcal{V}$ with $a = 0_\mathcal{F}$

▷ *E.g. the line* $\mathcal{W} = \text{sp}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ *(the line* $x_2 = x_1$*): with* $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ *it also contains* $-\begin{bmatrix} 1 \\ 1 \end{bmatrix}$*, and* $0_\mathcal{F} \odot x = 0_\mathcal{V}$ *must lie in* $\mathcal{W}$.

$\Rightarrow$ **All subspaces should cross origin.**
$\Rightarrow$ For two dimensional subspace, include all $\mathbb{R}^2$.

Subspaces of $\mathbb{R}^2$

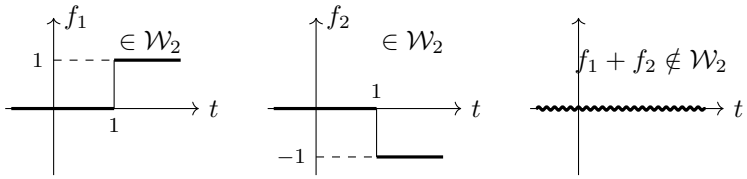
The subspaces of $\mathbb{R}^2$ are exactly: $\{0_\mathcal{V}\}$ (dimension 0), every line through the origin (dimension 1), and $\mathbb{R}^2$ itself (dimension 2). A line not passing through the origin (such as $\{[a, 1]^T\}$ above) is never a subspace.

Example 1.16. $\mathcal{V} = \mathbb{R}^{2\times 2}$, $\mathcal{F} = \mathbb{R}$, $\quad \mathcal{W} = \{\text{set of } 2 \times 2 \text{ real valued symmetric matrices}\}$

$$\begin{bmatrix} a_1 & b_1 \\ b_1 & c_1 \end{bmatrix} \oplus \begin{bmatrix} a_2 & b_2 \\ b_2 & c_2 \end{bmatrix} = \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ b_1+b_2 & c_1+c_2 \end{bmatrix} \in \mathcal{W}$$
$$\alpha \begin{bmatrix} a_1 & b_1 \\ b_1 & c_1 \end{bmatrix} = \begin{bmatrix} \alpha a_1 & \alpha b_1 \\ \alpha b_1 & \alpha c_1 \end{bmatrix} \in \mathcal{W}$$

$\therefore \mathcal{W}$ is a subspace of $\mathcal{V}$.

Example 1.17. $\mathcal{V}$: is a set of real valued functions defined over $\mathbb{R}$
$\mathcal{W}_1$: set of real valued continuous functions
$\mathcal{W}_2$: set of all real valued functions which are discontinuous at $t = 1$
$\therefore \mathcal{W}_1$ is a subspace. For $\mathcal{W}_2$:



▷ *f_1 and f_2 both jump at $t = 1$, but $f_1 + f_2$ is the zero function, which is continuous.*
$\therefore$ Not closed under vector addition & scalar multiplication (multiply with zero: $0 \cdot f_1$ is the zero function, which is continuous, so $\notin \mathcal{W}_2$). Hence $\mathcal{W}_2$ is not a subspace.

Explanation

$\mathcal{W}_1$ is a subspace because sums and scalar multiples of continuous functions are continuous (and the zero function is continuous). $\mathcal{W}_2$ fails already the quick test: the zero vector of $\mathcal{V}$ (the zero function) is not in $\mathcal{W}_2$.

1.4 Sums, Intersections and Products of Subspaces

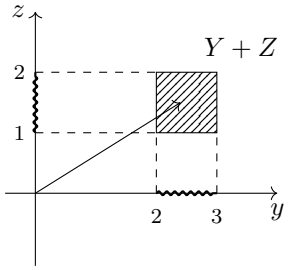
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Definition 1.18. The **sum** $Y + Z$ of two subsets Y & Z of a linear space is the set of all vectors $y + z$ where $y \in Y$ and $z \in Z$.

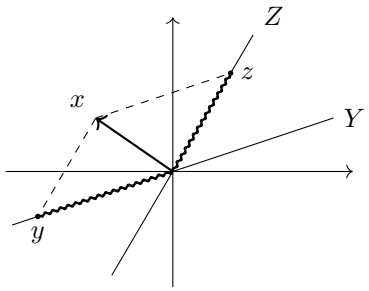
$$Y + Z \triangleq \{y + z \mid y \in Y,\ z \in Z\}$$

Example 1.19. $\mathcal{V} = \mathbb{R}^2$,

$$Y = \left\{ \begin{bmatrix} a \\ 0 \end{bmatrix} \,\middle|\, 2 \le a \le 3 \right\}, \qquad Z = \left\{ \begin{bmatrix} 0 \\ b \end{bmatrix} \,\middle|\, 1 \le b \le 2 \right\}$$



If instead Y and Z are two lines through the origin (subspaces), every vector x can be decomposed as $x = y + z$ with $y \in Y$, $z \in Z$:



All vectors I can do the same thing:

$$Y + Z = \mathbb{R}^2$$

$\Rightarrow$ sum of two subspaces end up with again a subspace.

Remark. If Y & Z are both subspaces, then $Y + Z$ **is a subspace**.

Proof. Let $x_1, x_2 \in Y + Z$:

$$x_1 = \underbrace{y_1}_{\in Y} + \underbrace{z_1}_{\in Z}, \qquad x_2 = \underbrace{y_2}_{\in Y} + \underbrace{z_2}_{\in Z}$$

$$x_1 + x_2 = y_1 + z_1 + y_2 + z_2 = \underbrace{y_1 + y_2}_{\in Y} + \underbrace{z_1 + z_2}_{\in Z}$$

▷ *as they are subspaces and closed under addition*

$$\underbrace{\alpha x_1}_{\in Y+Z} = \alpha(y_1 + z_1) = \underbrace{\alpha y_1}_{\in Y} + \underbrace{\alpha z_1}_{\in Z}$$

Closed under scalar multiplication.

$\therefore$ $Y + Z$ is a subspace. □

For the two lines Y, Z of the previous example:

$$Y \cap Z = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} \text{ is a subspace} \Rightarrow \text{not a coincidence!}$$

Remark. If Y & Z are subspaces, then $Y \cap Z$ **is a subspace**.

($Y \cup Z$ is closed under scalar multiplication but not closed under addition, which is the case in the previous example) $\Rightarrow$ $Y \cup Z$ **is not a subspace generally**.

Explanation

Proof for $Y \cap Z$: if $x_1, x_2 \in Y \cap Z$, then $x_1 + x_2 \in Y$ and $x_1 + x_2 \in Z$ (both are subspaces), so $x_1 + x_2 \in Y \cap Z$; similarly $\alpha x_1 \in Y \cap Z$.
Why $Y \cup Z$ fails: for the two lines, $y \in Y$, $z \in Z$ nonzero give $y + z$, which lies on neither line. In fact $Y \cup Z$ is a subspace only if $Y \subseteq Z$ or $Z \subseteq Y$. The sum $Y + Z$ is the **smallest subspace containing** $Y \cup Z$.

Definition 1.20 (Product space)**.** Let $(\mathcal{V}, \mathcal{F})$ & $(\mathcal{W}, \mathcal{F})$ be two linear spaces defined over the same field $\mathcal{F}$. The **product** of $(\mathcal{V}, \mathcal{F})$ & $(\mathcal{W}, \mathcal{F})$ is defined as:

$$\mathcal{V} \times \mathcal{W} = \{(v, w) \mid v \in \mathcal{V},\ w \in \mathcal{W}\}$$

$$(v_1, w_1) \underset{\mathcal{V}\times\mathcal{W}}{\oplus} (v_2, w_2) = (v_1 \oplus_{\mathcal{V}} v_2,\ w_1 \oplus_{\mathcal{W}} w_2)$$
$$\alpha \odot_{\mathcal{V}\times\mathcal{W}} (v_1, w_1) = (\alpha \odot_{\mathcal{V}} v_1,\ \alpha \odot_{\mathcal{W}} w_1)$$
$$\Rightarrow \quad (\underset{\mathcal{V}\times\mathcal{W}}{\mathbb{R}^2}, \underset{\mathcal{F}}{\mathbb{R}}) = (\underset{\mathcal{V}}{\mathbb{R}}, \underset{\mathcal{F}}{\mathbb{R}}) \times (\underset{\mathcal{W}}{\mathbb{R}}, \underset{\mathcal{F}}{\mathbb{R}})$$

▷ *In general* $\dim(\mathcal{V} \times \mathcal{W}) = \dim \mathcal{V} + \dim \mathcal{W}$*; e.g.* $\mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$.

1.5 Linear Combinations, Span and Linear Independence

Definition 1.21. A **linear combination** of n vectors $x_1, x_2, \ldots, x_n$ of a linear [space] $\mathcal{V}$ is a vector of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = \sum_{i=1}^{n} a_ix_i \qquad \text{where } a_i \in \mathcal{F} \text{ for } i = 1, \ldots, n$$

The set of all linear combinations of $x_1, \ldots, x_n$, i.e.

$$\left\{ \sum_{i=1}^{n} a_ix_i \;\middle|\; a_i \in \mathcal{F},\ i = 1, \ldots, n \right\}$$

is called the **span** of $\{x_1, \ldots, x_n\}$, denoted $\text{sp}\{x_1, \ldots, x_n\}$.

For $\mathcal{V} = \mathbb{R}^2$:

$$x_1 = \begin{bmatrix}1\\0\end{bmatrix},\ x_2 = \begin{bmatrix}0\\1\end{bmatrix}: \qquad a_1 \begin{bmatrix}1\\0\end{bmatrix} + a_2 \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}a_1\\a_2\end{bmatrix} \qquad \Rightarrow\ \text{sp}\{x_1, x_2\} = \mathbb{R}^2$$
$$x_1 = \begin{bmatrix}1\\0\end{bmatrix},\ x_2 = \begin{bmatrix}0\\1\end{bmatrix},\ x_3 = \begin{bmatrix}1\\1\end{bmatrix}: \qquad \Rightarrow\ \text{sp}\{x_1, x_2, x_3\} = \mathbb{R}^2$$

▷ x_3 *is redundant:* $x_3 = x_1 + x_2$ *is already a linear combination of* x_1, x_2*, so adding it does not enlarge the span.*

Definition 1.22. Vectors $x_1, \ldots, x_n \in \mathcal{V}$ are said to be **linearly independent** iff

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0_{\mathcal{V}} \quad \text{implies} \quad a_1 = a_2 = \cdots = a_n = 0_{\mathcal{F}}$$

If $\exists\, a_1, \ldots, a_n$ not all equal to zero s.t.

$$a_1x_1 + \cdots + a_nx_n = 0_{\mathcal{V}},$$

then $x_1, x_2, \ldots, x_n$ are **linearly dependent**.

Example 1.23. $x_1 = \begin{bmatrix}1\\0\end{bmatrix}, x_2 = \begin{bmatrix}0\\1\end{bmatrix}$:

$$a_1 \begin{bmatrix}1\\0\end{bmatrix} + a_2 \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}a_1\\a_2\end{bmatrix} \quad \Rightarrow \quad a_1 = a_2 = 0 \quad \Rightarrow \quad \text{linearly independent}$$

Example 1.24. $x_1 = \begin{bmatrix}1\\0\end{bmatrix}$, $x_2 = \begin{bmatrix}0\\1\end{bmatrix}$, $x_3 = \begin{bmatrix}1\\1\end{bmatrix}$:

$$a_1\begin{bmatrix}1\\0\end{bmatrix} + a_2\begin{bmatrix}0\\1\end{bmatrix} + a_3\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \;\Rightarrow\; \begin{matrix}a_1 + a_3 = 0\\ a_2 + a_3 = 0\end{matrix} \;\Rightarrow\; \text{e.g. } a_1 = a_2 = 1,\ a_3 = -1$$

$\therefore$ linearly dependent.

Example 1.25. Consider the linear space of polynomials degree $n \le 2$, and

$$P_1(t) = 1, \quad P_2(t) = 1 + t, \quad P_3(t) = 1 + t + t^2, \qquad P = \{P_1(t), P_2(t), P_3(t)\}.$$

Is P linearly independent?

$$\begin{aligned} &a_1 \cdot 1 + a_2(1 + t) + a_3(1 + t + t^2) = 0(t) && \text{(zero polynomial)}\\ \Rightarrow\ &(a_1 + a_2 + a_3) + (a_2 + a_3)t + a_3t^2 = 0(t) \;\Rightarrow\; a_1 = a_2 = a_3 = 0 \end{aligned}$$

▷ *A polynomial is the zero polynomial only if all its coefficients vanish: $a_3 = 0$ (from t^2), then $a_2 + a_3 = 0 \Rightarrow a_2 = 0$ (from t), then $a_1 + a_2 + a_3 = 0 \Rightarrow a_1 = 0$.*
$\therefore$ P is linearly independent.

Example 1.26. $\mathcal{V}$: set of all continuous functions defined over the real line. p. 8

$$S = \{\cos t,\ \sin t,\ \cos(t - \tfrac{\pi}{3})\}. \quad \text{Is } S \text{ linearly indep.?}$$

$$a\cos t + b\sin t + c \underbrace{\cos(t - \tfrac{\pi}{3})}_{\cos t \cdot \cos\frac{\pi}{3} + \sin t \cdot \sin\frac{\pi}{3}} = 0(t) \quad \forall t, \qquad \cos\tfrac{\pi}{3} = \tfrac{1}{2}, \quad \sin\tfrac{\pi}{3} = \tfrac{\sqrt{3}}{2}$$

Collecting terms: $(a + \frac{c}{2})\cos t + \left(b + \frac{\sqrt{3}}{2}c\right)\sin t = 0(t)$, which holds for $a = -\frac{c}{2}$, $b = -\frac{\sqrt{3}}{2}c$ with any c. Choosing $a = -\frac{1}{2}$, $b = -\frac{\sqrt{3}}{2}$, $c = 1$ gives the zero function.
$\Rightarrow \therefore$ S **is <u>not</u> linearly independent.**

$$A = [c_1 \mid c_2 \mid \cdots \mid c_n], \qquad Ax = \begin{bmatrix}0\\0\\\vdots\\0\end{bmatrix}$$

$$[c_1 \mid c_2 \mid \cdots \mid c_n]\begin{bmatrix}x_1\\x_2\\\vdots\\x_n\end{bmatrix} = \begin{bmatrix}0\\0\\\vdots\\0\end{bmatrix}$$

$$\underbrace{c_1x_1 + c_2x_2 + \cdots + c_nx_n}_{\text{zero}} = \begin{bmatrix}0\\0\\\vdots\\0\end{bmatrix}$$

▷ $c_1, c_2, \ldots, c_n \Rightarrow$ *linearly independent*

▷ *Ax is a linear combination of the columns of A with coefficients x_i. Hence:* ***the columns*** $c_1, \ldots, c_n$ ***are***

linearly independent $\iff$ *$Ax = 0$* ***has only the solution*** *$x = 0$; any nonzero solution x gives a dependence relation among the columns.*

1.6 Basis, Dimension and Coordinates

Definition 1.27. Let $\mathcal{V}$ be a linear space and set of vectors $B = \{x_1, \ldots, x_n\}$ be a subset of $\mathcal{V}$. B is said to be a **basis** for $\mathcal{V}$ iff

(i) $\text{sp}\{B\} = \mathcal{V}$

(ii) B is a linearly independent set.

For $\mathbb{R}^2$:

$$B_1 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} \qquad \times \text{ not a basis}$$

$$B_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \qquad \checkmark$$

$$B_3 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \qquad \times \text{ not linearly indep.}$$

$$B_4 = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \qquad \checkmark$$

$$B_5 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} \qquad \checkmark$$

▷ *The two basis vectors should not be on the same line through the origin (their spans intersect only at the origin); there are infinitely many bases and the number of vectors is always 2.*

▷ *E.g. for B_3: $\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. For B_1: $\text{sp}\, B_1$ is only the line $x_2 = x_1$, not $\mathbb{R}^2$.*

Remark. There are infinitely many basis sets for a linear space. **All of these basis sets have the same number of elements.** That number is called **dimension** of $\mathcal{V}$. If the dimension is finite, $\mathcal{V}$ is said to be **finite dimensional**. Otherwise, $\mathcal{V}$ is **infinite dimensional**.

Example 1.28. $\mathcal{V}$ is set of real valued symmetric matrices:

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

▷ *3 elements (dimension 3)*

Infinite dimensional linear space $\Rightarrow$ set of continuous functions.

▷ *For every n, the functions $1, t, t^2, \ldots, t^n$ are continuous and linearly independent, so no finite set can span this space.*

Definition 1.29. If $y \in \mathcal{V}$ and $B = \{x_1, x_2, \ldots, x_n\}$ is a basis set for $\mathcal{V}$, then there is a **unique** p. 9

n-tuple of scalars $(a_1, a_2, \ldots, a_n)$ such that

$$y = \sum_{i=1}^{n} a_i x_i$$

The scalars $(a_1, a_2, \ldots, a_n)$ are called **components** (**coordinates**), **representations** of y wrt basis B. We show the representation of y wrt B as $[y]_B$.

With B the basis of symmetric matrices above:

$$y = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \in \mathcal{V}, \qquad [y]_B = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{\text{unique}}, \qquad y \in \mathcal{V}, \quad [y]_B \in \mathbb{R}^3$$

► Why should representation be unique?

Proof. $[y]_B$ is unique. (Proof by contradiction)

Suppose $[y]_B$ is <u>not</u> unique:

$$[y]_B = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}, \qquad [y]_B = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}, \qquad \exists\, i \text{ s.t. } \alpha_i \neq \beta_i$$

By the representation definition,

$$y = \sum_{i=1}^{n} \alpha_i x_i, \qquad y = \sum_{i=1}^{n} \beta_i x_i \qquad \Rightarrow \qquad 0 = \sum_{i=1}^{n} (\alpha_i - \beta_i) x_i$$

▷ *linear combination of the basis vectors equal to zero with some of the coefficients nonzero, as* $\alpha_i - \beta_i \neq 0$

$\therefore$ x_i are linearly dependent $\Rightarrow$ contradiction. □

Example 1.30. $\mathcal{V} = \text{sp}\{\cos t, \sin t\}, \quad B = \{\cos t, \sin t\}$

$$y \triangleq \cos(t - \tfrac{\pi}{3}) = \tfrac{1}{2}\cos t + \tfrac{\sqrt{3}}{2}\sin t \qquad \Rightarrow \qquad [y]_B = \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}$$

1.7 Change of Basis

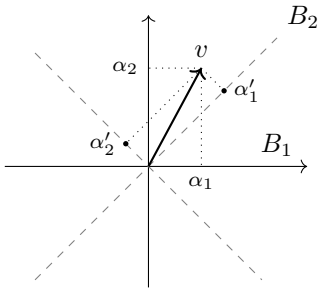
Theorem 1.31 (Change of basis)**.** Let $\mathcal{V}$ be an n-dimensional linear space over $\mathbb{R}$. Let B_1 and B_2 be two basis sets for $\mathcal{V}$. Then, there exists an $n \times n$ **real invertible matrix** P s.t.

$$[v]_{B_1} = P\,[v]_{B_2} \quad \forall v \in \mathcal{V}$$

$$B_1 = \left\{ \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix} \right\}, \qquad B_2 = \left\{ \begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix} \right\}$$

$$v = \alpha_1 \begin{bmatrix}1\\0\end{bmatrix} + \alpha_2 \begin{bmatrix}0\\1\end{bmatrix}$$

$$[v]_{B_1} = \begin{bmatrix}\alpha_1\\\alpha_2\end{bmatrix}, \qquad [v]_{B_2} = \begin{bmatrix}\alpha_1'\\\alpha_2'\end{bmatrix}$$



Finding P for this example

Write v in terms of B_2: $v = \alpha_1' \begin{bmatrix}1\\1\end{bmatrix} + \alpha_2' \begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix} \begin{bmatrix}\alpha_1'\\\alpha_2'\end{bmatrix}$. Since the B_1-coordinates are just the entries of v,

$$[v]_{B_1} = \underbrace{\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}}_{P} [v]_{B_2}, \qquad P^{-1} = \frac{1}{2}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}.$$

In general, **the j-th column of P is $[$$j$-th vector of $B_2\,]_{B_1}$**. P is invertible because its columns are coordinates of a basis, hence linearly independent. For the drawn vector $v = \begin{bmatrix}0.7\\1.3\end{bmatrix}$: $[v]_{B_2} = P^{-1}[v]_{B_1} = \begin{bmatrix}1\\-0.3\end{bmatrix}$.

Example 1.32. Consider $\mathcal{V}$: polynomials with degree $n \le 2$.

$$B_1 = \{\overset{v_1}{1},\ \overset{v_2}{1+t},\ \overset{v_3}{1+t+t^2}\}, \qquad B_2 = \{\underset{w_1}{1},\ \underset{w_2}{t},\ \underset{w_3}{t^2}\}$$

Find P such that $[v]_{B_1} = P\,[v]_{B_2} \quad \forall v \in \mathcal{V}$.

p. 10

$$v = w_1: \quad [w_1]_{B_2} = \begin{bmatrix}1\\0\\0\end{bmatrix} \Rightarrow P \begin{bmatrix}1\\0\\0\end{bmatrix} = \text{1st column of } P = [w_1]_{B_1} = \begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$v = w_2: \quad [w_2]_{B_2} = \begin{bmatrix}0\\1\\0\end{bmatrix} \Rightarrow P \begin{bmatrix}0\\1\\0\end{bmatrix} = \text{2nd column of } P = [w_2]_{B_1} = \begin{bmatrix}-1\\1\\0\end{bmatrix}$$

$$v = w_3: \quad [w_3]_{B_2} = \begin{bmatrix}0\\0\\1\end{bmatrix} \Rightarrow P \begin{bmatrix}0\\0\\1\end{bmatrix} = \text{3rd column of } P = [w_3]_{B_1} = \begin{bmatrix}0\\-1\\1\end{bmatrix}$$

▷ *The coordinates come from writing B_2 in terms of B_1: $w_1 = 1 = v_1$, $w_2 = t = (1+t) - 1 = v_2 - v_1$, $w_3 = t^2 = (1+t+t^2) - (1+t) = v_3 - v_2$.*

$$P = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

▷ *upper triangular matrix with diagonal entries 1. Diagonals are the eigenvalues* $\Rightarrow \det(P)$ *is multiplication of eigenvalues* $= 1 \neq 0$*, so* P *is invertible.*

$$[v]_{B_2} = Q\,[v]_{B_1} \;\Rightarrow\; Q = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad P = Q^{-1}$$

▷ *Columns of* Q *are* $[v_j]_{B_2}$*:* $v_1 = 1$*,* $v_2 = 1 + t$*,* $v_3 = 1 + t + t^2$ *give* $(1,0,0)$*,* $(1,1,0)$*,* $(1,1,1)$*.*

Check

$$PQ = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

E.g. $v = 3 + 2t + t^2$: $[v]_{B_2} = (3,2,1)$ and $P[v]_{B_2} = (1,1,1)$, i.e. $v = 1 \cdot 1 + 1 \cdot (1 + t) + 1 \cdot (1 + t + t^2) = 3 + 2t + t^2$ ✓.

2 Normed Vector Spaces

Syllabus: Week 3 — normed spaces, sequences and convergence, matrix norms

2.1 Norms

Definition 2.1. Let $(\mathcal{V}, \mathcal{F})$ be a vector space, a **norm** of $\mathcal{V}$ is a function shown as

$$\|\cdot\| : \mathcal{V} \to \mathbb{R}_{\geq 0}$$

satisfying the following properties:

(1) $\|v\| \geq 0 \quad \forall v \in \mathcal{V}, \quad \|v\| = 0 \iff v = 0_{\mathcal{V}}$

(2) $\|\alpha v\| = |\alpha|\,\|v\| \quad \forall v \in \mathcal{V}\ \&\ \forall \alpha \in \mathcal{F}$ (absolute homogeneity)

(3) $\|v_1 + v_2\| \leq \|v_1\| + \|v_2\| \quad \forall v_1, v_2 \in \mathcal{V}$ (triangle inequality)

The triplet $(\mathcal{V}, \mathcal{F}, \|\cdot\|)$ is called a **normed space**.

$\|v\| = \|v - 0\|$: size of v

$\|v_1 - v_2\|$: distance between v_1 & v_2

We can define a sphere in $\mathcal{V}$:

$$S = \{v \in \mathcal{V} \mid \|v - v_0\| \leq r\} \qquad \text{sphere radius } r, \quad v_0 : \text{center vector}$$

▷ *Strictly, with "*$\leq$*" this is the (closed)* ***ball****; the sphere is its boundary* $\|v - v_0\| = r$*. The "unit spheres" drawn below are the sets* $\|x\| = 1$*.*

Example 2.2. $\mathcal{V} = \mathbb{R}^n$, $\mathcal{F} = \mathbb{R}$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \qquad \|x\|_1 \stackrel{?}{=} \sum_{i=1}^{n} |x_i|$$

(1) $|x_i| \geq 0 \ \forall i \Rightarrow \|x\|_1 \geq 0$ ✓. $0_{\mathcal{V}} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$; if $\|x\|_1 = 0$:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i| = 0 \ \Rightarrow \ |x_i| = 0 \ \forall i \ \Rightarrow \ x = 0_{\mathcal{V}}$$

(2) $\alpha x = \begin{bmatrix} \alpha x_1 & \alpha x_2 & \cdots & \alpha x_n \end{bmatrix}^T$:

$$\|\alpha x\| = \sum_{i=1}^{n} |\alpha x_i| = \sum_{i=1}^{n} |\alpha| \cdot |x_i| = |\alpha| \cdot \underbrace{\sum_{i=1}^{n} |x_i|}_{\|x\|_1} = |\alpha| \, \|x\|_1$$

(3) $x + y = \begin{bmatrix} x_1 + y_1 & x_2 + y_2 & \cdots & x_n + y_n \end{bmatrix}^T$: p. 11

$$\|x + y\|_1 = \sum_{i=1}^{n} |x_i + y_i| \leq \sum_{i=1}^{n} (|x_i| + |y_i|) = \underbrace{\sum_{i=1}^{n} |x_i|}_{\|x\|_1} + \underbrace{\sum_{i=1}^{n} |y_i|}_{\|y\|_1}$$

▷ *The inequality uses the scalar triangle inequality $|x_i + y_i| \leq |x_i| + |y_i|$ term by term. Hence $\|x + y\|_1 \leq \|x\|_1 + \|y\|_1$ and all three properties hold: $\|\cdot\|_1$ is a norm.*

2.2 The ℓ_p Norms and Unit Spheres

► $\|\cdot\|_1$ is called ℓ_1 **norm**.

Example 2.3 (ℓ_2 norm)**.**

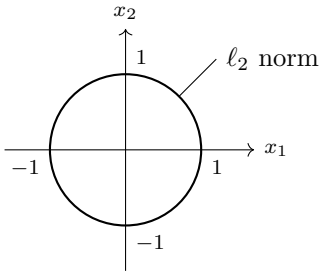
$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} |x_i|^2} = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2}$$

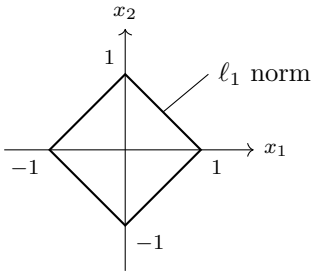
Unit spheres for $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$: $\|x\|_2 = 1$ and $\|x\|_1 = 1$.

For ℓ_1 norm: $|x_1| + |x_2| = 1$

- 1st quadrant $(x_1, x_2 \geq 0)$: $x_1 + x_2 = 1$
- 2nd quadrant $(x_1 \leq 0 \leq x_2)$: $-x_1 + x_2 = 1$
- 3rd quadrant $(x_1, x_2 \leq 0)$: $-x_1 - x_2 = 1$
- 4th quadrant $(x_2 \leq 0 \leq x_1)$: $x_1 - x_2 = 1$

▷ *Four line segments joining $(\pm 1, 0)$ and $(0, \pm 1)$: the ℓ_1 unit sphere is a diamond (a square rotated by $45°$), while the ℓ_2 unit sphere $x_1^2 + x_2^2 = 1$ is the circle.*





ℓ_p norm. The **ℓ_p norm** is

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \quad \text{where } p \geq 1$$

ℓ_∞ norm. The **ℓ_∞ norm** is

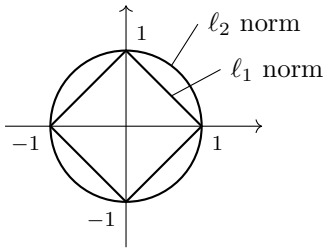
$$\|x\|_\infty = \max_i |x_i| = \lim_{p\to\infty} \|x\|_p$$

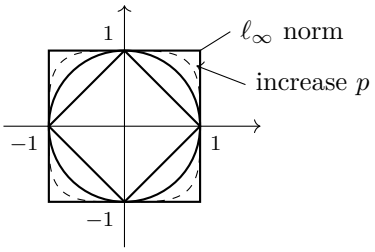
For $\mathcal{V} = \mathbb{R}^2$: $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\|x\|_p = (|x_1|^p + |x_2|^p)^{1/p}$. For $x_1 = 2$ & $x_2 = 3$:

$$\begin{aligned} \left\| \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\|_p &= (2^p + 3^p)^{1/p} \\ &= (3^p)^{1/p} \cdot \left(1 + \left(\tfrac{2}{3}\right)^p\right)^{1/p} \\ &= 3\left(1 + \left(\tfrac{2}{3}\right)^p\right)^{1/p} \\ \lim_{p\to\infty} \left\| \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\|_p &= 3 = \max\left(|x_1|, |x_2|\right) = \left\| \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\|_\infty \end{aligned}$$

▷ $\left(\frac{2}{3}\right)^p \to 0$, *so the factor* $\left(1 + (\frac{2}{3})^p\right)^{1/p} \to 1$. *In general the largest* $|x_i|$ *dominates the sum:* $\max_i |x_i| \leq \|x\|_p \leq n^{1/p} \max_i |x_i|$ *and* $n^{1/p} \to 1$.

► Unit spheres $\|x\|_p = 1$, i.e. $\|x - 0\|_p = 1$. The ℓ_∞ unit sphere is $\max(|x_1|, |x_2|) = 1$.





▷ *Dashed curve:* $|x_1|^p + |x_2|^p = 1$ *for an intermediate* p *(drawn for* $p = 4$*); as* p *increases the unit sphere moves from the* ℓ_2 *circle towards the* ℓ_∞ *square.*

For $p = 1/2$:

$$\|x\|_{1/2} = \left(|x_1|^{1/2} + |x_2|^{1/2}\right)^2$$

▷ *satisfy properties 1 & 2 but might not satisfy triangular inequality*

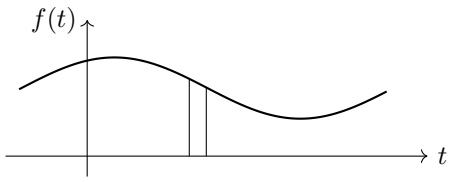
$$x = \begin{bmatrix} 9 \\ 0 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \quad x + y = \begin{bmatrix} 9 \\ 4 \end{bmatrix}$$

$$\|x + y\|_{1/2} = \left(\sqrt{9} + \sqrt{4}\right)^2 = 25$$
$$\|x\|_{1/2} = 9, \qquad \|y\|_{1/2} = 4$$
$$\|x + y\|_{1/2} \not\leq \|x\|_{1/2} + \|y\|_{1/2}$$
$$p < 1 \;\Rightarrow\; \textbf{no norms!}$$

▷ *Here* $25 > 9 + 4 = 13$. *Geometrically, for* $p < 1$ *the "unit sphere"* $|x_1|^p + |x_2|^p = 1$ *is star-shaped (curved inwards) and the unit ball is not convex;* ***a norm's unit ball is always convex.***

2.3 Norms on Function Spaces

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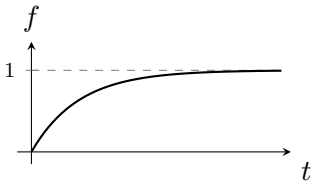


$$\mathcal{L}_p = \left\{ f(\cdot) \;\middle|\; f : \mathbb{R} \to \mathbb{R} \text{ s.t. } \int_{-\infty}^{\infty} |f(t)|^p \, dt < \infty \right\}, \qquad 1 \leq p < \infty$$

$$\|f(t)\|_p = \left(\int_{-\infty}^{\infty} |f(t)|^p \, dt \right)^{1/p} \qquad \mathcal{L}_p \text{ norm}$$
$$\|f(t)\|_1 = \int_{-\infty}^{\infty} |f(t)| \, dt \qquad \mathcal{L}_1 \text{ norm}$$
$$\|f(t)\|_2 = \left(\int_{-\infty}^{\infty} |f(t)|^2 \, dt \right)^{1/2} \qquad \mathcal{L}_2 \text{ norm}$$
$$\|f(t)\|_\infty = \sup_t |f(t)| \qquad \mathcal{L}_\infty \text{ norm}$$

▷ *These are the continuous analogues of the* ℓ_p *norms: the sum over* i *becomes an integral over* t.

$$f(t) = 1 - e^{-t}, \quad t \geq 0$$

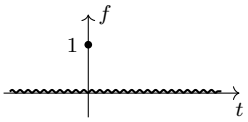


▷ *Value 1 is never reached. A maximum would have to be attained at a finite* t, *so* $\max_t |f(t)|$ *does not exist here.*

$$\sup |f(t)| = 1 \quad \text{(least upper bound)}$$

▷ *Therefore* $\|f\|_\infty$ ***is defined with*** sup *(it always exists for bounded* f*), which equals* max *whenever the maximum is attained.*

Integral norms do not satisfy $\|x\| = 0 \iff x = 0_{\mathcal{V}}$**!**



$$f(t) = \begin{cases} 1, & t = 0 \\ 0, & \text{otherwise} \end{cases} \qquad \Rightarrow \qquad \int_{-\infty}^{\infty} |f(t)|\, dt = 0$$

▷ *they are equivalent to zero functions as they acquire nonzero values only at some finite points.*

How this is fixed

$f \neq 0$ but $\|f\|_1 = 0$, so property (1) fails on the set of all functions. The standard remedy: regard two functions as *equal* in $\mathcal{L}_p$ if they differ only on a set of zero length (e.g. finitely many points), i.e. if $\|f - g\|_p = 0$. With this identification $\|\cdot\|_p$ is a genuine norm on $\mathcal{L}_p$ for $p \geq 1$ (the triangle inequality is Minkowski's inequality).

2.4 Matrix Norms

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}, \qquad \|A\| = ?$$

2.4.1 Entrywise Norms (Frobenius)

1st Method:

$$\mathrm{Vec}(A) \triangleq \begin{bmatrix} 1 \\ 3 \\ -2 \\ 4 \end{bmatrix}, \qquad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\|A\|_p = \|\mathrm{Vec}(A)\|_p = \left\| \begin{bmatrix} 1 \\ 3 \\ -2 \\ 4 \end{bmatrix} \right\|_p = (|1|^p + |-2|^p + |3|^p + |4|^p)^{\frac{1}{p}}$$

$$= \left(\sum_{i,j} |a_{ij}|^p \right)^{1/p}, \qquad p \geq 1$$

Numbers for this A

$\|A\|_1 = 1 + 2 + 3 + 4 = 10$, $\quad \|A\|_2 = \sqrt{1 + 4 + 9 + 16} = \sqrt{30} \approx 5.48$, $\quad \|A\|_\infty = \max\limits_{i,j} |a_{ij}| = 4$

The case $p = 2$ is the **Frobenius norm** $\|A\|_F$. Since $\mathbb{R}^{m \times n}$ is a vector space that is identified with $\mathbb{R}^{mn}$ through Vec, every vector norm gives a matrix norm in this way.

1st method (Frobenius norm).

$$\|A\|_2 = \Big(\sum_{i,j} |a_{ij}|^2\Big)^{1/2} \Rightarrow \textbf{Frobenius norm} = \sqrt{\mathrm{tr}(A^T A)}$$
$$= \sqrt{\mathrm{tr}(A^H A)} \quad \text{for complex}$$

For example, with

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad A^T A = \underbrace{\begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_{A} = \begin{bmatrix} a_{11}^2 + a_{21}^2 & \\ & a_{12}^2 + a_{22}^2 \end{bmatrix}$$

$$\mathrm{tr}(A^T A) = a_{11}^2 + a_{21}^2 + a_{12}^2 + a_{22}^2$$
$$= \sum_{i,j} a_{ij}^2$$

▷ *Only the diagonal of $A^T A$ matters for the trace, so its off-diagonal entries ($a_{11}a_{12} + a_{21}a_{22}$) are left blank.*

Explanation

In general $(A^H A)_{jj} = \sum_i \overline{a_{ij}}\, a_{ij} = \sum_i |a_{ij}|^2$ is the squared length of the j-th column of A, so $\mathrm{tr}(A^H A) = \sum_j \sum_i |a_{ij}|^2$. The Frobenius norm is simply the Euclidean norm of A viewed as a long vector in $\mathcal{F}^{mn}$; it is usually written $\|A\|_F$, to distinguish it from the *induced* 2-norm of the 2nd method below (which is in general different, e.g. $\|I_n\|_F = \sqrt{n}$ but the induced norm of I_n is 1).

2.4.2 Induced Matrix Norms

2nd Method.

Definition 2.4. Let $A \in \mathbb{R}^n \to \mathbb{R}^m$ be an $m \times n$ matrix. Let $\|\cdot\|_{\mathbb{R}^n}$ and $\|\cdot\|_{\mathbb{R}^m}$ denote the norms in $\mathbb{R}^n$ & $\mathbb{R}^m$ respectively. The **induced matrix norm** is defined as

$$\|A\| \triangleq \max_{\substack{x \in \mathbb{R}^n \\ x \neq 0_\mathcal{V}}} \frac{\|Ax\|_{\mathbb{R}^m}}{\|x\|_{\mathbb{R}^n}}$$

$$\underbrace{\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}}_{y}$$

Why second method?

$$|\langle x, y \rangle| \leq \|x\|\,\|y\|$$
$$\|Ax\| \leq \|A\|\,\|x\|$$
$$\|Ax\| = \frac{\|Ax\|}{\|x\|} \cdot \|x\| \leq \left(\max \frac{\|Ax\|}{\|x\|}\right) \cdot \|x\| = \|A\|\,\|x\|$$

▷ *"Cauchy–Schwarz"-like inequality for matrices: $\|Ax\| \leq \|A\|\,\|x\|$ **for every** x.*

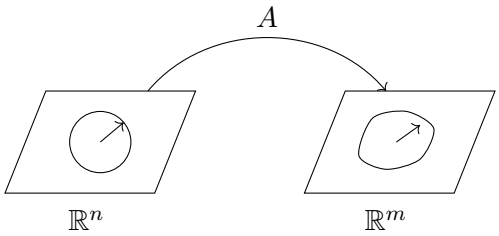
Why the induced norm is useful

The induced norm is the largest "gain" $\|Ax\| / \|x\|$ of A, so $\|Ax\| \le \|A\| \, \|x\|$ holds for every x with the smallest possible constant. It is also submultiplicative: $\|ABx\| \le \|A\| \, \|Bx\| \le \|A\| \, \|B\| \, \|x\|$, hence $\|AB\| \le \|A\| \, \|B\|$, which is what makes it convenient for bounding $\left\|A^k\right\|$ in system analysis.

$$\begin{aligned}\|A\| &= \max_{\substack{x \in \mathbb{R}^n \\ x \neq 0_{\mathcal{V}}}} \left\| A \frac{x}{\|x\|_{\mathbb{R}^n}} \right\|_{\mathbb{R}^m} \\ &= \max_{\|x\|_{\mathbb{R}^n}=1} \|Ax\|_{\mathbb{R}^m}\end{aligned}$$

▷ *First step: absolute homogeneity of the norm,* $\frac{\|Ax\|}{\|x\|} = \left\|A \frac{x}{\|x\|}\right\|$, *and* $\frac{x}{\|x\|}$ *ranges over all unit vectors.*

$$\therefore \ \|A\| = \max_{\|x\|_{\mathbb{R}^n}=1} \|Ax\|_{\mathbb{R}^m}$$

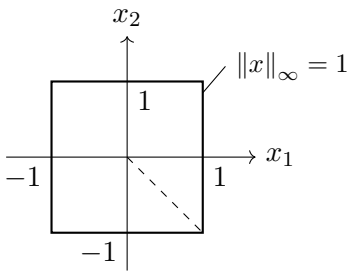


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Example 2.5. Consider an $m \times n$ matrix with real elements. Choose ℓ_∞-norm $\|\cdot\|_\infty$ for both $\mathbb{R}^n$ & $\mathbb{R}^m$.

$$\|A\|_\infty = \max_{\|x\|_\infty=1} \|Ax\|_\infty, \qquad A = \begin{bmatrix} 1 & -2 \\ -3 & 4 \\ -2 & 2 \end{bmatrix}, \qquad x \in \mathbb{R}^2, \quad y = Ax \in \mathbb{R}^3$$

$$\|x\|_\infty = \max(|x_1|, |x_2|) = 1$$



First row only:

$$A_1 = \begin{bmatrix} 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\max_{\|x\|_\infty=1} |A_1 x| = \max_{\|x\|_\infty=1} |x_1 - 2x_2| = 3$$

▷ *attained at* $x_1 = 1$, $x_2 = -1$ *(dashed corner).*

For the whole matrix:

$$Ax = \begin{bmatrix} x_1 - 2x_2 \\ -3x_1 + 4x_2 \\ -2x_1 + 2x_2 \end{bmatrix}$$

$$\|Ax\|_\infty = \max\left(|x_1 - 2x_2|, \ |-3x_1 + 4x_2|, \ |-2x_1 + 2x_2|\right)$$

$$x^* = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{or} \quad x^* = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$Ax^* = \begin{bmatrix} 3 \\ -7 \\ -4 \end{bmatrix} \Rightarrow \|Ax^*\|_\infty = 7$$

$$\underline{\underline{\|A\|_\infty = 7}}$$

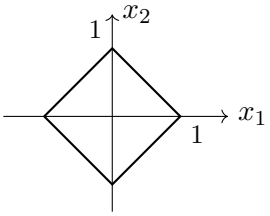
$$\|A\|_\infty = \max_i \sum_{j=1}^{n} |a_{ij}| \qquad \text{(max row sum)}$$

Why the max row sum

If $\|x\|_\infty = 1$ then $|x_j| \le 1$ for all j, so for each row $|(Ax)_i| = |\sum_j a_{ij}x_j| \le \sum_j |a_{ij}||x_j| \le \sum_j |a_{ij}|$. Hence $\|Ax\|_\infty \le \max_i \sum_j |a_{ij}|$. Equality is reached by taking the row i^* with the largest sum and $x_j = \text{sign}(a_{i^*j})$ (a corner of the square). In the example the row sums are $3, 7, 4$ and row $2 = [-3 \;\; 4]$ gives $x^* = [-1 \;\; 1]^T$ (or its negative).

ℓ_1-norm.

$$\|A\|_1 = \max_{\|x\|_1=1} \|Ax\|_1, \qquad \|x\|_1 = |x_1| + |x_2| = 1$$



$$A_1 = \begin{bmatrix} 1 & -2 \end{bmatrix}$$
$$A_1 x = x_1 - 2x_2$$
$$\|A_1 x\| = |x_1 - 2x_2|$$
$$\max_{\|x\|_1=1} |x_1 - 2x_2|$$

▷ *weighted average of* 1 *and* -2*: on* $\|x\|_1 = 1$, $|x_1 - 2x_2| \le |x_1| + 2|x_2| \le 2(|x_1| + |x_2|) = 2$, *so the best is to put all the weight on the entry of largest magnitude.*

$$x = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix} \quad \Rightarrow \quad \tfrac{1}{2} + 1 = \tfrac{3}{2}$$
$$x = \begin{bmatrix} 0.2 \\ -0.8 \end{bmatrix} \quad \Rightarrow \quad \text{larger}$$
$$x = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad \Rightarrow \quad \text{largest:} \quad \underline{\|A_1 x\| = 2}$$

Now the whole matrix: p. 15

$$A = \begin{bmatrix} 1 & -2 \\ -3 & 4 \\ -2 & 2 \end{bmatrix}, \qquad Ax = \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} x_1 + \begin{bmatrix} -2 \\ 4 \\ 2 \end{bmatrix} x_2$$

$$\|Ax\|_1 = \left\| \underbrace{\begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}}_{6} x_1 + \underbrace{\begin{bmatrix} -2 \\ 4 \\ 2 \end{bmatrix}}_{8\ \checkmark} x_2 \right\|_1 = 8 \qquad (x_1 = 0,\ x_2 = 1)$$

▷ *Column sums:* $|1|+|-3|+|-2| = 6$ *and* $|-2|+|4|+|2| = 8$, *so* $\|A\|_1 = 8$ *(compare* $\|A\|_\infty = 7$*: different vector norms give different matrix norms).*

$$\|A\|_1 = \max_j \sum_i |a_{ij}| \qquad \text{sum columns.}$$

Why the max column sum

Write $A = [a_1 \ \cdots \ a_n]$ by columns. If $\|x\|_1 = 1$, $\|Ax\|_1 = \left\|\sum_j x_j a_j\right\|_1 \le \sum_j |x_j| \, \|a_j\|_1 \le \left(\max_j \|a_j\|_1\right) \sum_j |x_j| = \max_j \|a_j\|_1$, with equality for $x = e_{j^*}$ (a corner of the diamond), j^* the column of largest absolute sum.

2.5 Convergence of Sequences

Definition 2.6. Let $(\mathcal{V}, \mathcal{F}, \|\cdot\|)$ be a normed space. Let $\{v_n\}_{n=1}^\infty$ be a sequence of vectors in $\mathcal{V}$, i.e., $v_n \in \mathcal{V}\ \forall n$. The sequence $\{v_n\}_{n=1}^\infty$ is said to **converge** to the **limit** $\boxed{v \in \mathcal{V}}$ iff

$$\boxed{\|v_n - v\| \to 0 \quad \text{as } n \to \infty}$$

Remark. A sequence is **divergent** if it is not convergent.

Example 2.7. $\mathcal{V} = \mathbb{R}$, $\|v\| = |v|$.

Sequence $\{(1/2)^n\}_{n=1}^\infty$, $v_n = (1/2)^n$, $v = 0 \in \mathcal{V}$:

$$\|v_n - v\| = \left|\frac{1}{2^n} - 0\right| = \frac{1}{2^n}$$
$$\lim_{n\to\infty} \|v_n - v\| = 0 \ \Rightarrow\ v_n \to 0 \text{ as } n \to \infty$$

Sequence $\{(-1)^n\}$, $v_n = (-1)^n$.
Try $v = 0$:

$$\|(-1)^n - 0\| = |(-1)^n| = 1, \qquad \lim_{n\to\infty} \overbrace{\|v_n - v\|}^{1} = 1 \neq 0$$

▷ *sequence is not convergent to* $v = 0$.
Try $v = 1$:

$$\|(-1)^n - 1\| = |(-1)^n - 1| = \begin{cases} 0, & n \text{ even} \\ 2, & n \text{ odd} \end{cases} \ \Rightarrow \text{ do not converge}$$

▷ *The distance keeps jumping back to* 2, *so it does not tend to* 0.
∴ **Sequence is not convergent anywhere.**

Explanation

For an arbitrary candidate $v \in \mathbb{R}$: the even terms are 1 and the odd terms are -1, so by the triangle inequality $2 = |1 - (-1)| \le |v_{2k} - v| + |v - v_{2k+1}|$. Thus at least one of two consecutive distances is ≥ 1, and $\|v_n - v\| \not\to 0$ for every v.

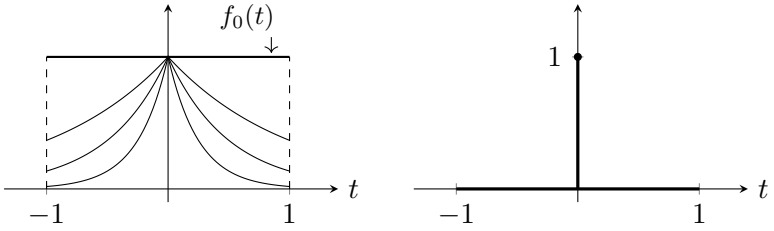
Example 2.8. $\mathcal{V} = \left\{ f : [-1,1] \to \mathbb{R} \;\middle|\; f(\cdot) \text{ is continuous, } \int_{-1}^{1} |f(t)|\, dt < \infty \right\}$

$$\|f(\cdot)\| = \int_{-1}^{1} |f(t)|\, dt \qquad (\mathcal{L}_1 \text{ norm})$$

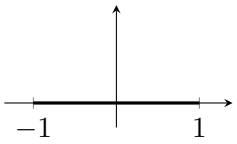
Is f_n convergent?

$$\begin{aligned} f_n(t) &= e^{-n|t|}, \quad n \geq 0 \\ f_0(t) &= e^{0t} = 1 \\ f_1(t) &= e^{-|t|} \\ f_2(t) &= e^{-2|t|} \\ &\vdots \end{aligned}$$

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▷ *Left:* f_0, f_1, f_2, f_4. *Right: pointwise limit as* $n \to \infty$ *(*0 *for* $t \neq 0$, 1 *at* $t = 0$*)* $\notin \mathcal{V}$ *(not continuous).*



$0(t) \in \mathcal{V}$

Candidate limit: the zero function $0(t)$.
$v \in \mathcal{V}$ is satisfied ✓

$$\begin{aligned} \|f_n(t) - 0(t)\| = \|f_n(t)\| &= \int_{-1}^{1} \left| e^{-n|t|} \right| dt \\ &= \int_{-1}^{1} e^{-n|t|}\, dt && \text{(exponential is positive)} \\ &= 2 \cdot \int_{0}^{1} e^{-nt}\, dt && \text{(even function)} \\ &= 2\, \frac{e^{-nt}}{-n}\bigg|_0^1 = -2\left(\frac{e^{-n}}{n} - \frac{1}{n} \right) \\ \|f_n(t) - 0(t)\| &= -\frac{2}{n}\Big(\underbrace{e^{-n} - 1}_{\to -1} \Big) = \frac{2\,(1 - e^{-n})}{n} \leq \frac{2}{n} \\ \lim_{n\to\infty} \|f_n(t) - 0(t)\| = 0 \;&\Rightarrow\; \|v_n - v\| \to 0 \text{ as } n \to \infty \;\checkmark \end{aligned}$$

$\therefore$ **It is convergent to the zero function.**

Convergence depends on the distance definition (the norm):

- for ℓ_1 norm $\Rightarrow$ convergent

- for ℓ_∞ norm $\Rightarrow$ divergent

Explanation

With $\|f\|_\infty = \max_{-1\le t\le 1} |f(t)|$ we get $\|f_n - 0\|_\infty = \max_t e^{-n|t|} = 1$ (at $t = 0$) for every n, so $f_n \not\to 0$. No other continuous g works either: $\|f_n - g\|_\infty \to 0$ would force $g(t) = \lim_n f_n(t)$ pointwise, i.e. $g(0) = 1$ and $g(t) = 0$ for $t \neq 0$, which is not continuous. The $\mathcal{L}_1$ norm only "sees" area, and the area under the narrowing peak vanishes; the ℓ_∞ (sup) norm sees the peak height, which stays 1.

2.6 Cauchy Sequences and Completeness

Definition 2.9. Let $(\mathcal{V}, \mathcal{F}, \|\cdot\|)$ be a normed space. A sequence $\{v_n\}_{n=0}^\infty$ is said to be a **Cauchy sequence** iff

$$\lim_{n,m\to\infty} \|v_n - v_m\| = 0$$

Claim. Every convergent sequence is a Cauchy sequence. The reverse statement, in general, is not correct.

$$\text{Convergent} \underset{\not\Leftarrow}{\Rightarrow} \text{Cauchy}$$

Proof. Suppose that $\{v_n\}_{n=0}^\infty$ is convergent to $v \in \mathcal{V}$, i.e. $\lim_{n\to\infty} \|v_n - v\| = 0$. Then

$$\begin{aligned}\lim_{n,m\to\infty} \|v_n - v_m\| &= \lim_{n,m\to\infty} \|v_n - v + v - v_m\| \\ &\le \lim_{\substack{n\to\infty \\ m\to\infty}} \|v_n - v\| + \|v - v_m\| \\ &= \underbrace{\lim_{n\to\infty} \|v_n - v\|}_{=0} + \underbrace{\lim_{m\to\infty} \|v_m - v\|}_{=0} = 0\end{aligned}$$

▷ *Inequality step: triangle inequality,* $\|a + b\| \le \|a\| + \|b\|$ *with* $a = v_n - v$, $b = v - v_m$.

$$0 \le \lim_{n,m\to\infty} \|v_n - v_m\| \le 0 \;\Rightarrow\; \lim_{n,m\to\infty} \|v_n - v_m\| = 0 \;\Rightarrow\; \{v_n\} \text{ is Cauchy.} \qquad \square$$

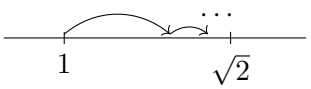
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Example 2.10. $\mathcal{V} = \mathbb{Q}$: set of rational numbers, $\mathcal{F} = \mathbb{Q}$ as well.

$$\sqrt{2} = 1.41421356237310\ldots \;\Rightarrow\; \text{not a rational number}$$

$$\begin{aligned} v_0 &= 1 = \tfrac{1}{1} \\ v_1 &= 1.4 = 14/10 \\ v_2 &= 1.41 = \tfrac{141}{100} \\ v_3 &= 1.414 \\ &\vdots \end{aligned}$$

We need $v \in \mathcal{V}$ s.t. $\|v_n - v\| \to 0$ as $n \to \infty$, where $v_n \in \mathcal{V}\ \forall n$. But v should be $\sqrt{2}$ for $\|v_n - v\| \to 0$ as $n \to \infty$.



$\sqrt{2} \notin \mathcal{V}$. This sequence is not convergent. $\therefore \{v_n\}$ is not convergent.

What about Cauchyness?

$$\|v_n - v_m\| \to 0 \text{ as } n, m \to \infty$$

v_n is actually Cauchy as elements are approaching each other.

▷ *Indeed v_n and v_m agree in the first $\min(n,m)$ decimals, so $|v_n - v_m| < 10^{-\min(n,m)} \to 0$.*

$$\therefore \textbf{ Convergent} \Rightarrow \textbf{Cauchy} \quad \text{but} \quad \text{Cauchy} \not\Rightarrow \text{Convergent}$$

▷ *Margin note (Turkish): "ben bi sequence olsam sana converge ederdim."*

If a set does not have holes $\Rightarrow$ **complete** set.

Explanation

The same sequence viewed in $\mathcal{V} = \mathbb{R}$ converges (to $\sqrt{2} \in \mathbb{R}$): convergence depends on the space, not only on the terms. $\mathbb{R}$ is complete (it "fills the holes" of $\mathbb{Q}$), and so is every finite-dimensional normed space over $\mathbb{R}$ or $\mathbb{C}$, e.g. $(\mathbb{R}^n, \|\cdot\|_p)$. Incompleteness is therefore a phenomenon of infinite-dimensional spaces such as function spaces, as the next example shows.

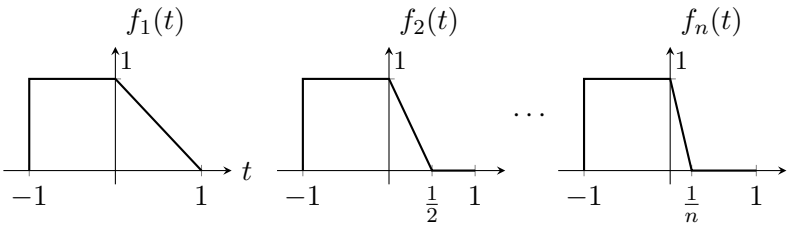
Definition 2.11. A normed space is said to be **complete** iff **every Cauchy sequence is convergent**. A complete normed space is called a **Banach space**.

Example 2.12. A normed space defined as

$$\mathcal{V} = \left\{ f(\cdot) \;\middle|\; f : [-1,1] \to \mathbb{R},\ f \text{ is continuous } \& \int_{-1}^{1} |f(t)|\,dt < \infty \right\}, \qquad \|f\|_1 = \int_{-1}^{1} |f(t)|\,dt$$

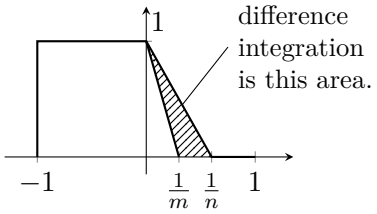
Consider the sequence $\{f_n\}_{n=1}^{\infty}$:

$$f_n(t) = \begin{cases} 1, & -1 \le t \le 0 \\ 1 - nt, & 0 \le t \le \frac{1}{n} \\ 0, & \frac{1}{n} \le t \le 1 \end{cases}$$



Is the sequence $\{f_n\}$ Cauchy? Convergent?

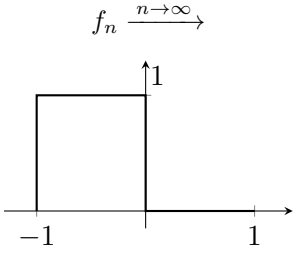
$$\|f_n(t) - f_m(t)\|_1 = \int_{-1}^{1} |f_n(t) - f_m(t)|\,dt$$



$$\text{Area} = \left|\frac{1}{n} - \frac{1}{m}\right| \cdot \frac{1}{2}$$

▷ *triangle with vertices $(0,1)$, $(\frac{1}{m}, 0)$, $(\frac{1}{n}, 0)$: base $|\frac{1}{n} - \frac{1}{m}|$, height 1.*

$$\lim_{n,m \to \infty} \left|\frac{1}{n} - \frac{1}{m}\right| \cdot \frac{1}{2} = 0$$



The limit $\notin \mathcal{V}$ (not continuous).
$\Rightarrow \{f_n\}$ is Cauchy but not convergent.
$\therefore (\mathcal{V}, \mathcal{F}, \|\cdot\|_1)$ **is not complete.**

Explanation

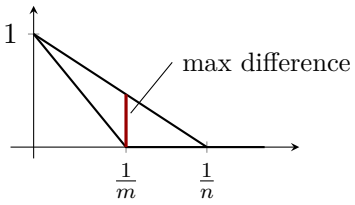
To be precise, "not convergent" means no *continuous* g satisfies $\|f_n - g\|_1 \to 0$. If one did, then on $[-1, 0]$ we would need $\int_{-1}^{0} |1 - g|\, dt = 0$, so $g = 1$ there; and on every $[\delta, 1]$, $\delta > 0$, $f_n = 0$ eventually, so $g = 0$ there. Then g jumps at $t = 0$, contradicting continuity. The "limit" is the step function, which lies outside $\mathcal{V}$.

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Now with $(\mathcal{V}, \mathcal{F}, \|\cdot\|_\infty)$:

$$\|f\|_\infty = \max_{-1\le t\le 1} |f(t)|$$
$$\|f_n(t) - f_m(t)\|_\infty = \max_{-1\le t\le 1} |f_n(t) - f_m(t)|$$

$$f_n(t) = 1 - nt \quad \left(0 \le t \le \tfrac{1}{n}\right)$$
$$f_n\left(\tfrac{1}{m}\right) = 1 - \frac{n}{m} \quad (m \ge n)$$



$$\|f_n(t) - f_m(t)\|_\infty = f_n\left(\tfrac{1}{m}\right) - \underbrace{f_m\left(\tfrac{1}{m}\right)}_{0} = \underbrace{\left|1 - \frac{n}{m}\right|}_{\neq 0 \text{ unless } n=m} \qquad (m \ge n)$$

e.g. $m = 2n$: $\quad \|f_n - f_{2n}\|_\infty = \frac{1}{2}$ for every n

n ve m'in nasıl ∞'a gittiği fark etmez, her zaman limit sonucu sıfır olmalı. Bu senaryoda if $n = m$ o zaman limit sıfır, onun dışında değil.

▷ *It does not matter how n and m go to ∞; the limit must always be zero. Here it is zero only if $n = m$.*

▷ *The maximum is at $t = \frac{1}{m}$: on $[0, \frac{1}{m}]$ the difference $(m - n)t$ increases, and on $[\frac{1}{m}, \frac{1}{n}]$ it equals $1 - nt$, which decreases.*

$\therefore \{f_n\}$ **is not Cauchy** in the sup norm.

3 Inner Product Spaces

Syllabus: Week 4 — inner product spaces, orthogonality, Gram–Schmidt

3.1 Inner Products

Definition 3.1. Let $(\mathcal{V}, \mathbb{C})$ be a vector space. An **inner product** is a function from $\mathcal{V} \times \mathcal{V}$ into $\mathbb{C}$,

$$\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \to \mathbb{C}$$

satisfying the following properties:

1) $\langle x, y \rangle = \overline{\langle y, x \rangle} \quad \forall x, y \in \mathcal{V}$ (conjugate symmetry)

2) Linearity wrt first argument:

   a) $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle \quad \forall x, y, z \in \mathcal{V}$ (additivity)

   b) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle \quad \forall \alpha \in \mathbb{C}$ and $\forall x, y \in \mathcal{V}$ (homogeneity)

3) $\langle x, x \rangle$ is real & nonnegative: $\langle x, x \rangle \geq 0$, equality holds iff $x = 0_{\mathcal{V}}$, i.e. $\langle x, x \rangle = 0 \Leftrightarrow x = 0_{\mathcal{V}}$.

$\therefore$ The triplet $(\mathcal{V}, \mathbb{C}, \langle \cdot, \cdot \rangle)$ is called an **inner product space**.

$$\langle x, y + z \rangle = \overline{\langle y + z, x \rangle} = \overline{\langle y, x \rangle} + \overline{\langle z, x \rangle} = \langle x, y \rangle + \langle x, z \rangle$$

▷ *Additivity wrt second argument* ✓

$$\langle x, \alpha y \rangle = \overline{\langle \alpha y, x \rangle} = \bar{\alpha}\, \overline{\langle y, x \rangle} = \bar{\alpha} \langle x, y \rangle$$

▷ *Homogeneity wrt second argument* × *(α should be real). We have* ***conjugate homogeneity wrt second argument.***

Explanation

So an inner product is linear in the first argument and *conjugate*-linear in the second: $\left\langle \sum_i \alpha_i x_i, \sum_j \beta_j y_j \right\rangle = \sum_{i,j} \alpha_i \bar{\beta}_j \langle x_i, y_j \rangle$. Over $\mathcal{F} = \mathbb{R}$ the conjugates disappear and the inner product is bilinear and symmetric. Also $\langle x, 0_{\mathcal{V}} \rangle = \langle 0_{\mathcal{V}}, x \rangle = 0$ for all x (take $\alpha = 0$ in homogeneity).

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Example 3.2. $\mathcal{V} = \mathbb{C}^n$, $\mathcal{F} = \mathbb{C}$, $\quad x = [x_1\ x_2 \dots x_n]^T$, $\quad y = [y_1\ y_2 \dots y_n]^T$, $\quad x_i, y_i \in \mathbb{C}$.

$$\langle x, y \rangle = \sum_{i=1}^{n} x_i \bar{y}_i = \bar{y}^T x = y^H x \qquad \left(\begin{bmatrix} \bar{y}_1 & \bar{y}_2 & \cdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \end{bmatrix} \right)$$

Check the axioms:

1) $\langle x, y \rangle = \sum_{i=1}^{n} x_i \bar{y}_i = \bar{y}^T x = y^H x$

$$\overline{\langle y, x \rangle} = \overline{x^H y} = \overline{\bar{x}^T y} = \underbrace{x^T \bar{y} = \bar{y}^T x}_{\text{scalar=its transpose}} = y^H x = \langle x, y \rangle$$

2) a) $\langle x+y,z\rangle = z^H(x+y) = z^Hx + z^Hy = \langle x,z\rangle + \langle y,z\rangle$

b) $\langle \alpha x, y\rangle = y^H(\alpha x) = \alpha y^H x = \alpha \langle x,y\rangle$

3) $\langle x,x\rangle = x^Hx = \sum_{i=1}^n |x_i|^2 \geq 0 \Rightarrow$ real value; $\langle x,x\rangle = 0 \Leftrightarrow x = 0$.

Example 3.3. $A, B \in \mathbb{R}^{m\times n}$, $\mathcal{V} = \mathbb{R}^{m\times n}$, $\mathcal{F} = \mathbb{R}$, $A = [a_{ij}]$, $B = [b_{ij}]$:

$$\langle A, B\rangle = \sum_{i,j} a_{ij}b_{ij}$$

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \\ b_{13} & b_{23} \end{bmatrix}}_{A\cdot B^T} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{12} + a_{13}b_{13} & \times \\ \times & a_{21}b_{21} + a_{22}b_{22} + a_{23}b_{23} \end{bmatrix}$$

$$\begin{aligned} \langle A, B\rangle &\triangleq \operatorname{tr}(AB^T) \\ &= \operatorname{tr}(A^TB) \quad \text{(in the real field)} \end{aligned}$$

▷ *$(AB^T)_{ii} = \sum_j a_{ij}b_{ij}$ (row i of A against row i of B), so the trace adds up all $a_{ij}b_{ij}$; the off-diagonal entries ($\times$) are irrelevant. For complex matrices one uses $\langle A, B\rangle = \operatorname{tr}(AB^H) = \operatorname{tr}(B^HA)$, whose induced norm is the Frobenius norm.*

Example 3.4. $\mathcal{V} = \left\{ f(\cdot) \;\middle|\; f : [0,1] \to \mathbb{C} \text{ s.t. } \int_0^1 |f(t)|^2\, dt < \infty \right\}$

$$\langle f_1, f_2\rangle \triangleq \int_0^1 f_1(t)\overline{f_2(t)}\, dt \qquad (L_2 \text{ inner product})$$

3.2 The Cauchy–Schwarz Inequality and the Induced Norm

Theorem 3.5 (Cauchy–Schwarz Inequality)**.** Let $\mathcal{V}$ be an inner product space. Let $x, y \in \mathcal{V}$.

$$|\langle x,y\rangle|^2 \leq \langle x,x\rangle \langle y,y\rangle$$

Equality is satisfied iff (only) $x = cy$ for some $c \in \mathbb{C}$ (or $y = 0_{\mathcal{V}}$), i.e. **equality iff x and y are linearly dependent**.

Equality check for $x = cy$:

$$\begin{aligned} \langle x,y\rangle &= \langle cy,y\rangle = c\langle y,y\rangle \\ |\langle x,y\rangle|^2 &= |c|^2\langle y,y\rangle^2 \end{aligned} \qquad \begin{aligned} \langle x,x\rangle &= \langle cy,cy\rangle = \overbrace{c\bar{c}}^{\text{real value}}\langle y,y\rangle \\ &= |c|^2\langle y,y\rangle \end{aligned}$$

▷ *using the first- and second-argument axioms.*

$$\underbrace{|c|^2\langle y,y\rangle^2}_{|\langle x,y\rangle|^2} = \underbrace{|c|^2\langle y,y\rangle}_{\langle x,x\rangle}\underbrace{\langle y,y\rangle}_{\langle y,y\rangle}$$

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Proof of theorem. **Case 1:** $y = 0_{\mathcal{V}}$.

$$\langle x, y\rangle = \langle x, 0_{\mathcal{V}}\rangle = \langle x, 0 \cdot 0_{\mathcal{V}}\rangle = 0 \cdot \langle x, 0_{\mathcal{V}}\rangle = 0$$
$$\langle y, y\rangle = 0 \quad \text{(by axioms)}$$
$$\underbrace{|\langle x, y\rangle|^2}_{0} \leq \underbrace{\langle x, x\rangle}_{?}\,\underbrace{\langle y, y\rangle}_{0} \;\Rightarrow\; 0 \leq 0 \;\checkmark$$

Case 2: $y \neq 0_{\mathcal{V}} \Rightarrow \langle y, y\rangle \neq 0$ (in fact $\langle y, y\rangle > 0$ by axiom 3, so we may divide by it and multiply inequalities by it). For $\alpha \in \mathbb{C}$ (in field):

$$\begin{aligned} 0 &\leq \langle x - \alpha y, x - \alpha y\rangle \\ &= \langle x, x\rangle - \langle x, \alpha y\rangle - \langle \alpha y, x\rangle + \langle \alpha y, \alpha y\rangle \\ &= \langle x, x\rangle - \bar{\alpha}\langle x, y\rangle - \alpha \underbrace{\langle y, x\rangle}_{\overline{\langle x, y\rangle}} + |\alpha|^2 \langle y, y\rangle \end{aligned}$$

Choose $\alpha = \dfrac{\langle x, y\rangle}{\langle y, y\rangle}$:

$$0 \leq \langle x, x\rangle - \frac{\overline{\langle x, y\rangle}}{\langle y, y\rangle} \cdot \langle x, y\rangle - \frac{\langle x, y\rangle}{\cancel{\langle y, y\rangle}} \cdot \cancel{\overline{\langle x, y\rangle}} + \frac{|\langle x, y\rangle|^2}{\cancel{\langle y, y\rangle^2}} \cdot \cancel{\langle y, y\rangle}$$
$$0 \leq \langle x, x\rangle - \frac{|\langle x, y\rangle|^2}{\langle y, y\rangle}$$
$$\frac{|\langle x, y\rangle|^2}{\langle y, y\rangle} \leq \langle x, x\rangle \;\Rightarrow \therefore\; |\langle x, y\rangle|^2 \leq \langle x, x\rangle \langle y, y\rangle \qquad \square$$

Choice of α and the equality case

αy with $\alpha = \langle x, y\rangle / \langle y, y\rangle$ is the projection of x onto y (see Gram–Schmidt below), so $x - \alpha y \perp y$ and the first line says $\|x - \alpha y\|^2 \geq 0$: the error of the best approximation of x along y has nonnegative length. Equality in the theorem (with $y \neq 0_{\mathcal{V}}$) holds iff $\langle x - \alpha y, x - \alpha y\rangle = 0$, i.e. iff $x = \alpha y$ by axiom 3. This proves the "only if" part of the equality statement; the "if" part is the check for $x = cy$ above.

Remark. An inner product **induces a norm** (the **induced norm**) defined as

$$\|x\|^2 \triangleq \langle x, x\rangle, \qquad \|x\| \triangleq \sqrt{\langle x, x\rangle}$$

▷ *induced norm of inner product.*

Show that this is a norm:

1-) $\|x\| \geq 0$, $\|x\| = 0 \Leftrightarrow x = 0_{\mathcal{V}}$ ✓ (axiom 3 of the inner product)

2-) $\|\alpha x\| \triangleq \sqrt{\langle \alpha x, \alpha x\rangle} = \sqrt{\alpha \cdot \bar{\alpha}\langle x, x\rangle} = |\alpha| \underbrace{\sqrt{\langle x, x\rangle}}_{\|x\|} = |\alpha|\,\|x\|$ ✓

3-) $\|x + y\| = \sqrt{\langle x + y, x + y\rangle}$

$$\|x + y\|^2 = \langle x + y, x + y\rangle = \langle x, x\rangle + \langle x, y\rangle + \underbrace{\langle y, x\rangle}_{\overline{\langle x, y\rangle}} + \langle y, y\rangle$$
$$= \langle x, x\rangle + 2 \cdot \mathrm{Re}\,\langle x, y\rangle + \langle y, y\rangle$$

$$\leq \langle x,x\rangle + 2|\operatorname{Re}\langle x,y\rangle| + \langle y,y\rangle$$
$$\leq \langle x,x\rangle + 2|\langle x,y\rangle| + \langle y,y\rangle$$

By Cauchy–Schwarz, $|\langle x,y\rangle| \leq \sqrt{\langle x,x\rangle\langle y,y\rangle}$: p. 21

$$\leq \langle x,x\rangle + 2\sqrt{\langle x,x\rangle}\sqrt{\langle y,y\rangle} + \langle y,y\rangle$$
$$= \left(\sqrt{\langle x,x\rangle} + \sqrt{\langle y,y\rangle}\right)^2$$
$$= (\|x\| + \|y\|)^2$$
$$\|x+y\|^2 \leq (\|x\| + \|y\|)^2 \;\Rightarrow\; \|x+y\| \leq \|x\| + \|y\| \;\checkmark$$

▷ *Both sides are nonnegative, so taking square roots preserves the inequality: the triangle inequality holds.*

∗ $|\langle x,y\rangle| \leq \|x\|\,\|y\|$; suppose $x \neq 0_{\mathcal{V}}$, $y \neq 0_{\mathcal{V}}$:

$$0 \leq \frac{|\langle x,y\rangle|}{\|x\|\,\|y\|} \leq 1$$
$$\cos\theta \triangleq \frac{|\langle x,y\rangle|}{\|x\|\,\|y\|} \;\Rightarrow\; \theta = \arccos\left(\frac{|\langle x,y\rangle|}{\|x\|\,\|y\|}\right)$$

▷ *Cauchy–Schwarz is exactly what makes this angle well defined. With the absolute value $\theta \in [0, \frac{\pi}{2}]$; in a real space one often drops it, $\cos\theta = \frac{\langle x,y\rangle}{\|x\|\|y\|} \in [-1,1]$, which for $\mathbb{R}^2, \mathbb{R}^3$ is the usual geometric angle. $\theta = \frac{\pi}{2}$ $\Leftrightarrow$ $\langle x,y\rangle = 0$ (orthogonal), $\theta = 0 \Leftrightarrow x = cy$.*

Remark. Every inner product space is also a normed space. The converse/reverse is not correct in general.

Explanation

A norm comes from an inner product **iff it satisfies the parallelogram law** $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$. For example $\|\cdot\|_1$ on $\mathbb{R}^2$ fails it: $x = e_1$, $y = e_2$ give $4 + 4 = 8 \neq 4 = 2 + 2$. So ℓ_1 and ℓ_∞ are norms without an inner product behind them; only ℓ_2 is induced by $\langle x,y\rangle = y^T x$.

Definition 3.6. An inner product space that is complete wrt the induced norm is called a **complete inner product space** / **Hilbert space**.

3.3 Orthogonality and Orthogonal Expansions

Definition 3.7. Two vectors $x, y \in \mathcal{V}$ are said to be **orthogonal** iff $\langle x,y\rangle = 0$. Similarly subsets $S_1, S_2 \subset \mathcal{V}$ are orthogonal iff $\langle x,y\rangle = 0\ \forall x \in S_1, \forall y \in S_2$.

Examples:

1-) $\mathcal{V} = \mathbb{R}^n$, $\mathcal{F} = \mathbb{R}$, $\quad \langle x,y\rangle = \sum_{i=1}^n x_i y_i = y^T x = x^T y$

$$\underbrace{B = \{e_1, e_2, \ldots, e_n\}}_{\text{canonical basis set}} \qquad e_1 = \begin{bmatrix}1\\0\\\vdots\\0\end{bmatrix} \quad e_2 = \begin{bmatrix}0\\1\\0\\\vdots\\0\end{bmatrix} \quad \cdots$$

$$\langle e_i, e_j \rangle = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases} = \delta_{ij}$$

2-) $\mathcal{V} = \left\{ f(\cdot) \;\middle|\; f : [-\pi, \pi] \to \mathbb{C} \text{ s.t. } \int_{-\pi}^{\pi} |f(t)|^2 \, dt < \infty \right\}$

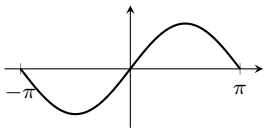
Consider $\{e^{jnt}\}_{n=-\infty}^{\infty}$, $\quad e^{jnt} \in \mathcal{V}$ for all n.

$$\langle f_1(t), f_2(t) \rangle = \int_{-\pi}^{\pi} f_1(t) \, \overline{f_2(t)} \, dt$$

$$\begin{aligned} \left\langle e^{jnt}, e^{jmt} \right\rangle &= \int_{-\pi}^{\pi} e^{jnt} \, e^{-jmt} \, dt = \int_{-\pi}^{\pi} e^{j(n-m)t} \, dt \\ &= \int_{-\pi}^{\pi} \left[\cos((n-m)t) + j \sin((n-m)t) \right] dt \\ &= \int_{-\pi}^{\pi} \cos((n-m)t) \, dt + j \underbrace{\int_{-\pi}^{\pi} \sin((n-m)t) \, dt}_{0} \end{aligned}$$

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$\left.\begin{matrix} \sin t \\ \sin 2t \\ \sin 3t \\ \vdots \end{matrix}\right\}$ all zero due to integral over a period



$$\left\langle e^{jnt}, e^{jmt} \right\rangle = \begin{cases} 2\pi, & n = m \\ 0, & n \neq m \end{cases}$$
$$= 2\pi \, \delta_{nm} \;\; //$$

▷ *For $n = m$: $\int_{-\pi}^{\pi} \cos 0 \, dt = 2\pi$. For $k = n - m \neq 0$: $\int_{-\pi}^{\pi} \cos kt \, dt = \frac{2 \sin k\pi}{k} = 0$, and $\int_{-\pi}^{\pi} \sin kt \, dt = 0$ since $\sin kt$ is odd.*

3-) $S_1 = \{\sin nt\}_{n=1}^{\infty}$, $S_2 = \{\cos mt\}_{m=1}^{\infty}$:

$$\begin{aligned} \langle \sin nt, \cos mt \rangle &= \int_{-\pi}^{\pi} \sin nt \cdot \cos mt \, dt \\ &= \int_{-\pi}^{\pi} \frac{1}{2} \Big(\underbrace{\sin((n+m)t)}_{0} + \underbrace{\sin((n-m)t)}_{0} \Big) \, dt = 0 \end{aligned}$$

▷ *Product-to-sum: $\sin a \cos b = \frac{1}{2}[\sin(a+b) + \sin(a-b)]$; both terms are odd functions (or zero when $n = m$), so they integrate to 0 over the symmetric interval $[-\pi, \pi]$.*

$$\therefore \;\; S_1 \perp S_2$$

Expansion in an orthogonal set.

- $z = \alpha_1 z_1 + \alpha_2 z_2$ if $z_1 \perp z_2$:

$$\begin{aligned}\langle z, z_1\rangle &= \langle \alpha_1 z_1 + \alpha_2 z_2, z_1\rangle \\ &= \alpha_1 \langle z_1, z_1\rangle + \alpha_2 \underbrace{\langle z_2, z_1\rangle}_{0} \Rightarrow \alpha_1 = \frac{\langle z, z_1\rangle}{\langle z_1, z_1\rangle}\end{aligned}$$

In general,

$$z = \sum_{i=1}^{n} \alpha_i z_i, \qquad \alpha_i = \frac{\langle z, z_i\rangle}{\langle z_i, z_i\rangle} \quad \forall i$$

▷ *Valid when $z \in \mathrm{sp}\{z_1, \ldots, z_n\}$ and the $z_i \neq 0_{\mathcal{V}}$ are mutually orthogonal: take $\langle \cdot, z_k\rangle$ of both sides and all terms with $i \neq k$ vanish.* ***No linear system has to be solved; each coefficient is computed independently.***

Example (as $z_1 \perp z_2$):

$$\underbrace{\begin{bmatrix}2\\3\end{bmatrix}}_{z} = \beta_1 \underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{z_1} + \beta_2 \underbrace{\begin{bmatrix}1\\-1\end{bmatrix}}_{z_2}, \qquad \beta_1 = \frac{\langle z, z_1\rangle}{\langle z_1, z_1\rangle} = \frac{5}{2}, \qquad \beta_2 = \frac{\langle z, z_2\rangle}{\langle z_2, z_2\rangle} = \frac{2-3}{2} = -\frac{1}{2}$$

▷ *Check:* $\frac{5}{2}\begin{bmatrix}1\\1\end{bmatrix} - \frac{1}{2}\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$ ✓

- $f(t) = \sum_{n=-\infty}^{\infty} \alpha_n e^{jnt}$:

$$\alpha_n = \frac{\langle f(t), e^{jnt}\rangle}{\langle e^{jnt}, e^{jnt}\rangle} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t) e^{-jnt}\, dt \qquad \text{(Fourier Series Coefficients)}$$

▷ *with* $\langle f_1, f_2\rangle = \int f_1(t) \cdot \overline{f_2(t)}\, dt$.

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3.4 Gram–Schmidt Orthogonalization

Question: Let $\mathcal{V}$ be an inner product space. Let $S_1 = \{v_1, v_2, \ldots, v_n\}$ be a linearly indep. set. Can we obtain another set $S_2 = \{w_1, w_2, \ldots, w_n\}$ s.t.

1- $w_i \perp w_j$ when $i \neq j$

2- $\mathrm{sp}\{S_1\} = \mathrm{sp}\{S_2\}$?

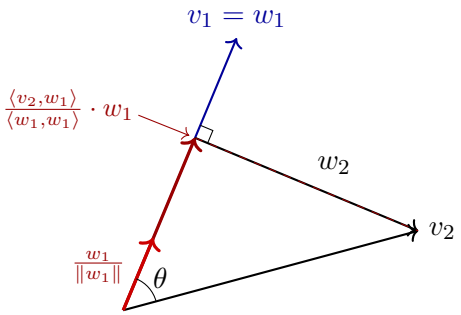
Yes! (constructively, by the following algorithm)

Gram–Schmidt Orthogonalization Algorithm.

Step 1: $w_1 = v_1$

Step 2: $w_2 = v_2 - \dfrac{\langle v_2, w_1\rangle}{\langle w_1, w_1\rangle}\cdot w_1$

$$\langle w_2, w_1\rangle = \left\langle v_2 - \frac{\langle v_2, w_1\rangle}{\langle w_1, w_1\rangle}\cdot w_1, w_1 \right\rangle$$
$$= \cancel{\langle v_2, w_1\rangle} - \underbrace{\frac{\cancel{\langle v_2, w_1\rangle}}{\cancel{\langle w_1, w_1\rangle}}}_{\text{scalar}} \cancel{\langle w_1, w_1\rangle} = 0$$



$\langle x, y\rangle = \|x\|\,\|y\|\cos\theta$

$$\underbrace{\frac{\langle v_2, w_1\rangle}{\langle w_1, w_1\rangle}}_{\|w_1\|^2}\cdot w_1 = \frac{\langle v_2, w_1\rangle}{\|w_1\|}\cdot\frac{w_1}{\|w_1\|} = \underbrace{\left\langle v_2, \frac{w_1}{\|w_1\|}\right\rangle}_{\|v_2\|\cos\theta}\frac{w_1}{\|w_1\|}$$

Step 3: $w_3 = v_3 - \dfrac{\langle v_3, w_1\rangle}{\langle w_1, w_1\rangle}\cdot w_1 - \dfrac{\langle v_3, w_2\rangle}{\langle w_2, w_2\rangle}\cdot w_2$

$\dfrac{\langle v_i, w_j\rangle}{\langle w_j, w_j\rangle}\cdot w_j$: **projection** of v_i onto w_j

Step n: $w_n = v_n - \displaystyle\sum_{i=1}^{n-1}\frac{\langle v_n, w_i\rangle}{\langle w_i, w_i\rangle}\cdot w_i$

Why Gram–Schmidt works

Orthogonality (by induction): if $w_1, \dots, w_{k-1}$ are mutually orthogonal, then for $j < k$ $\langle w_k, w_j\rangle = \langle v_k, w_j\rangle - \sum_{i<k}\frac{\langle v_k, w_i\rangle}{\langle w_i, w_i\rangle}\langle w_i, w_j\rangle = \langle v_k, w_j\rangle - \langle v_k, w_j\rangle = 0$, since only the $i = j$ term survives. *Same span*: each w_k is v_k minus a combination of $w_1, \dots, w_{k-1} \in \mathrm{sp}\{v_1, \dots, v_{k-1}\}$, so $\mathrm{sp}\{w_1, \dots, w_k\} = \mathrm{sp}\{v_1, \dots, v_k\}$ for every k. *No division by zero*: $w_k = 0_{\mathcal{V}}$ would make $v_k \in \mathrm{sp}\{v_1, \dots, v_{k-1}\}$, impossible for a linearly independent S_1; so $\langle w_k, w_k\rangle > 0$. In words: w_k **is the component of v_k orthogonal to the span of the previous vectors.**

Example 3.8. $\mathcal{V} = \left\{ f \;\middle|\; f : [-1,1] \to \mathbb{R},\ \int_{-1}^{1}(f(t))^2\,dt < \infty \right\}$, $\mathcal{F} = \mathbb{R}$,

$$\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)\,dt$$

$S_1 = \{v_1, v_2, v_3\}$, $v_1 = 1(t)$, $v_2(t) = t$, $v_3(t) = t^2$, i.e. $S_1 = \{1, t, t^2\}$

p. 24

$w_1 = v_1 = 1(t)$

$$w_2 = t - \frac{\overset{0}{\cancel{\langle t, 1\rangle}}}{\langle 1, 1\rangle}\cdot 1 = t, \qquad \langle t, 1\rangle = \int_{-1}^{1} t\,dt = 0$$

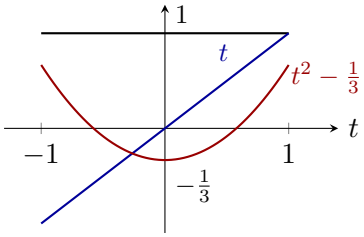
$$w_3 = t^2 - \frac{\langle t^2, 1\rangle}{\langle 1,1\rangle}\cdot 1 - \cancelto{0}{\frac{\langle t^2, t\rangle}{\langle t, t\rangle}}\cdot t = t^2 - \frac{2/3}{2} = t^2 - \frac{1}{3}$$

using

$$\begin{aligned}\left\langle t^2, 1\right\rangle &= \int_{-1}^{1} t^2\,dt = 2\int_{0}^{1} t^2\,dt = \frac{2}{3}\\ \langle 1,1\rangle &= \int_{-1}^{1} 1\,dt = 2\\ \left\langle t^2, t\right\rangle &= \int_{-1}^{1} t^3\,dt = 0\\ \langle t, t\rangle &= \int_{-1}^{1} t^2\,dt = \frac{2}{3}\end{aligned}$$

$$S_2 = \left\{1,\ t,\ t^2 - \tfrac{1}{3}\right\}$$

Legendre polynomials



▷ *Check:* $\left\langle t^2 - \frac{1}{3}, 1\right\rangle = \frac{2}{3} - \frac{1}{3}\cdot 2 = 0$ *and* $\left\langle t^2 - \frac{1}{3}, t\right\rangle = \int_{-1}^{1}(t^3 - \frac{t}{3})\,dt = 0$ *(odd integrand)* ✓. *These are the Legendre polynomials up to scaling: the standard ones are* $P_0 = 1$, $P_1 = t$, $P_2 = \frac{1}{2}(3t^2 - 1) = \frac{3}{2}\left(t^2 - \frac{1}{3}\right)$, *normalised so that* $P_k(1) = 1$.

If vectors are linearly dependent, e.g. $S_1 = \{v_1, v_1\}$:

$$\begin{aligned}w_1 &= v_1\\ w_2 &= v_1 - \cancelto{1}{\frac{\langle v_1, v_1\rangle}{\langle v_1, v_1\rangle}}\cdot v_1 = 0\end{aligned}$$

▷ *When this is the case,* ***remove the redundant vector and resume the process****. In general* $w_k = 0_{\mathcal{V}}$ *exactly when* v_k *depends linearly on* $v_1, \ldots, v_{k-1}$*; discarding such vectors, Gram–Schmidt returns an orthogonal basis of* $\mathrm{sp}\{S_1\}$ *(and its size is* $\dim \mathrm{sp}\{S_1\}$*).*

Gram–Schmidt Orthonormalization. Require $\|w_i\| = 1$, i.e. $\langle w_i, w_i\rangle = 1$:

$$\begin{aligned}\bar{w}_1 &= v_1, & w_1 &= \frac{\bar{w}_1}{\|\bar{w}_1\|}\\ \bar{w}_2 &= v_2 - \langle v_2, w_1\rangle\, w_1, & w_2 &= \frac{\bar{w}_2}{\|\bar{w}_2\|}\\ &\vdots\\ \bar{w}_n &= v_n - \sum_{i=1}^{n-1}\langle v_n, w_i\rangle\, w_i, & w_n &= \frac{\bar{w}_n}{\|\bar{w}_n\|}\end{aligned}$$

▷ *Since* $\langle w_i, w_i\rangle = 1$, *the projection coefficients* $\frac{\langle v_k, w_i\rangle}{\langle w_i, w_i\rangle}$ *simplify to* $\langle v_k, w_i\rangle$. *The resulting set is* ***orthonormal****:* $\langle w_i, w_j\rangle = \delta_{ij}$.

4 Linear Transformations

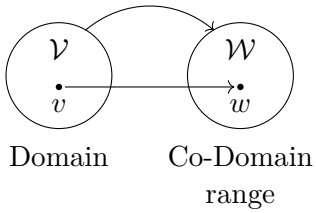
Syllabus: Week 5 — linear transformations, null space, range space, matrix representations

4.1 Definition and Examples

Definition 4.1. Let $\mathcal{V}$ & $\mathcal{W}$ be two vector spaces over the same field $\mathcal{F}$. A **linear transformation** $\mathcal{A}$ is a mapping from $\mathcal{V}$ into $\mathcal{W}$ shown as $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ s.t.

$$\mathcal{A}(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 \mathcal{A} v_1 + \alpha_2 \mathcal{A} v_2$$

for all $v_1, v_2 \in \mathcal{V}$ and all $\alpha_1, \alpha_2 \in \mathcal{F}$.



▷ *$\mathcal{V}$ is the **domain** and $\mathcal{W}$ the **co-domain**; the **range** $\mathcal{R}(\mathcal{A}) = \{\mathcal{A}v : v \in \mathcal{V}\}$ is the subset of $\mathcal{W}$ actually reached (in general $\mathcal{R}(\mathcal{A}) \subseteq \mathcal{W}$).*

Explanation

The single condition is equivalent to additivity $\mathcal{A}(v_1 + v_2) = \mathcal{A}v_1 + \mathcal{A}v_2$ together with homogeneity $\mathcal{A}(\alpha v) = \alpha \mathcal{A} v$. Taking $\alpha_1 = \alpha_2 = 0$ gives $\mathcal{A}0_{\mathcal{V}} = 0_{\mathcal{W}}$. Standard examples: $x \mapsto Ax$ for $A \in \mathcal{F}^{m \times n}$ ($\mathcal{V} = \mathcal{F}^n$, $\mathcal{W} = \mathcal{F}^m$, as in the induced-norm section), differentiation $f \mapsto f'$ on polynomials, and $f \mapsto \int_0^1 f(t)\,dt$.

Example 4.2. $\mathcal{V} = \mathcal{W} \triangleq \{\text{polynomials with degree} \le n\}$, $\mathcal{A} = \dfrac{d}{dt}$, p. 25

$$v_1 = \sum_{i=0}^{n} \alpha_i t^i, \qquad v_2 = \sum_{i=0}^{n} \beta_i t^i.$$

$$\begin{aligned} a_1 v_1 + a_2 v_2 &= \sum_{i=0}^{n} (a_1 \alpha_i + a_2 \beta_i)\, t^i \\ \mathcal{A}(a_1 v_1 + a_2 v_2) &= \sum_{i=1}^{n} (a_1 \alpha_i + a_2 \beta_i) \cdot i\, t^{i-1} \\ &= a_1 \sum_{i=1}^{n} i \alpha_i t^{i-1} + a_2 \sum_{i=1}^{n} i \beta_i t^{i-1} \\ &= a_1 \mathcal{A} v_1 + a_2 \mathcal{A} v_2 \end{aligned}$$

▷ *The constant terms ($i = 0$) differentiate to zero, which is why the sums after differentiation start at $i = 1$. Hence $\frac{d}{dt}$ is linear.*

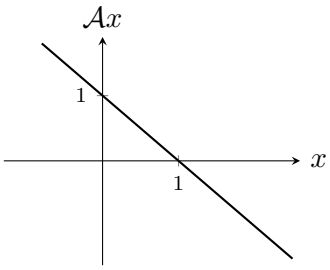
Example 4.3. $\mathcal{V} = \mathcal{W} = \mathbb{R}^2, \quad x = \begin{bmatrix}\alpha_1 \\ \alpha_2\end{bmatrix}, \quad y = \begin{bmatrix}\beta_1 \\ \beta_2\end{bmatrix}.$

$$\mathcal{A}x = \begin{bmatrix}\alpha_1 \\ \alpha_1 + \alpha_2\end{bmatrix}, \qquad ax + by = \begin{bmatrix}a\alpha_1 + b\beta_1 \\ a\alpha_2 + b\beta_2\end{bmatrix}$$

$$\mathcal{A}(ax + by) = \begin{bmatrix}a\alpha_1 + b\beta_1 \\ a\alpha_1 + b\beta_1 + a\alpha_2 + b\beta_2\end{bmatrix}$$

$$= a\begin{bmatrix}\alpha_1 \\ \alpha_1 + \alpha_2\end{bmatrix} + b\begin{bmatrix}\beta_1 \\ \beta_1 + \beta_2\end{bmatrix} = a\mathcal{A}x + b\mathcal{A}y \quad //$$

Example 4.4. $\mathcal{V} = \mathcal{W} = \mathbb{R}, \quad \mathcal{A}x = 1 - x.$



Take $x_1 = 0$, $x_2 = 3$, $\alpha_1 = 1$, $\alpha_2 = 1$:

$$\alpha_1 x_1 + \alpha_2 x_2 = 1 \cdot 0 + 1 \cdot 3 = 3$$
$$\mathcal{A}(\alpha_1 x_1 + \alpha_2 x_2) = \mathcal{A}3 = 1 - 3 = -2$$
$$\mathcal{A}x_1 = 1, \qquad \mathcal{A}x_2 = -2$$
$$\alpha_1 \mathcal{A}x_1 + \alpha_2 \mathcal{A}x_2 = -1$$

▷ -2 *and* -1 *are not equal* $\Rightarrow \therefore$ ***Not a linear transformation.***

It should cross from zero: $\mathcal{A}0_V = 0_W$ **should satisfy this.**

$$\mathcal{A}(\alpha_1 x_1 + \cancel{\alpha_2 x_2}^{0}) = \alpha_1 \mathcal{A}x_1 + \cancel{\alpha_2 \mathcal{A}x_2}^{0} \qquad \text{(suppose } \alpha_2 = 0\text{)}$$
$$\mathcal{A}(\alpha_1 x_1) = \alpha_1 \mathcal{A}(x_1) \qquad \text{(suppose } \alpha_1 = 0\text{)}$$
$$\mathcal{A}(0_V) = 0_W$$

Quick test for linearity

Every linear transformation maps 0_V to 0_W (take $\alpha_1 = 0$ in $\mathcal{A}(\alpha_1 x_1) = \alpha_1 \mathcal{A}x_1$). Here $\mathcal{A}0 = 1 - 0 = 1 \neq 0$, so $\mathcal{A}x = 1 - x$ cannot be linear: the graph is a straight line, but it does not pass through the origin (an **affine** map). The converse is false: $\mathcal{A}0 = 0$ alone does not guarantee linearity (e.g. $\mathcal{A}x = x^2$).

Example 4.5. p. 26

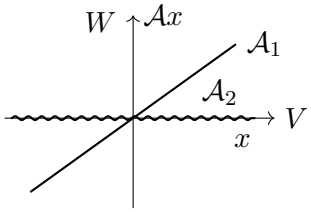
$$\mathcal{V} = \{f \mid f : [0,1] \to \mathbb{R},\ f \text{ is integrable}\}, \qquad \mathcal{W} = \mathbb{R}$$
$$\mathcal{A}f = \int_0^1 f(t)\,dt$$

▷ ✓ *A linear transformation:* $\int_0^1 (af + bg)\,dt = a\int_0^1 f\,dt + b\int_0^1 g\,dt$ *by linearity of the integral.*

4.2 Null Space

Definition 4.6. Let $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ be a linear transformation. The **null space** of $\mathcal{A}$ is defined as

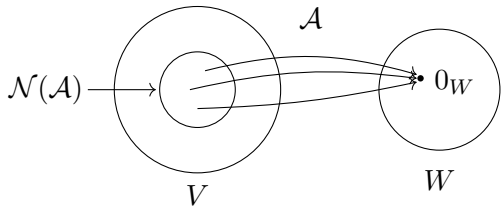
$$\mathcal{N}(\mathcal{A}) \triangleq \{v \in \mathcal{V} \mid \mathcal{A}v = 0_W\}$$



$$\mathcal{N}(\mathcal{A}_1) = \{0_V\}$$
$$\mathcal{N}(\mathcal{A}_2) = V = \mathbb{R}$$

▷ *$\mathcal{A}_1$: a line through the origin with nonzero slope (only $x = 0$ maps to 0); $\mathcal{A}_2$: the zero map (every x maps to 0).*

Claim. $\mathcal{N}(\mathcal{A})$ is a subspace of $\mathcal{V}$.



Let $x_1 \,\&\, x_2 \in \mathcal{N}(\mathcal{A})$. Show

$$\left.\begin{aligned} x_1 + x_2 &\overset{?}{\in} \mathcal{N}(\mathcal{A}) \\ \alpha x_1 &\overset{?}{\in} \mathcal{N}(\mathcal{A}) \end{aligned}\right\} \Rightarrow \mathcal{N}(\mathcal{A}) \text{ is a subspace.}$$

We have $\mathcal{A}x_1 = 0_W$, $\mathcal{A}x_2 = 0_W$. Then

$$\mathcal{A}(x_1 + x_2) = \underbrace{\mathcal{A}x_1}_{0_W} + \underbrace{\mathcal{A}x_2}_{0_W} = 0_W \qquad \Rightarrow\ x_1 + x_2 \in \mathcal{N}(\mathcal{A})$$
$$\mathcal{A}(\alpha x_1) = \alpha \mathcal{A}x_1 = \alpha 0_W = 0_W \qquad \Rightarrow\ \alpha x_1 \in \mathcal{N}(\mathcal{A})$$

$\therefore \mathcal{N}(\mathcal{A})$ is a subspace.

Example 4.7. $\mathcal{V} = \{\text{set of polynomials degree} \le 2 \text{ with interval } [0, 1]\}, \quad \mathcal{W} = \mathbb{R},$

$$\mathcal{A}f = \int_0^1 f(t)\,dt. \qquad \text{Find a basis for } \mathcal{N}(\mathcal{A}).$$

Let $v = at^2 + bt + c$. Then

$$\begin{aligned} \mathcal{A}v = 0_W \ &\Rightarrow\ \int_0^1 (at^2 + bt + c)\,dt = 0 \\ &\Rightarrow\ \frac{at^3}{3} + \frac{bt^2}{2} + ct\Big|_0^1 = \frac{a}{3} + \frac{b}{2} + c = 0 \\ &\Rightarrow\ c = -\frac{a}{3} - \frac{b}{2}, \qquad \dim \mathcal{N}(\mathcal{A}) = 2 \end{aligned}$$

Hence p. 27

$$\begin{aligned} at^2 + bt + c &= at^2 + bt - \frac{a}{3} - \frac{b}{2} \\ &= a\left(t^2 - \frac{1}{3}\right) + b\left(t - \frac{1}{2}\right) \end{aligned}$$

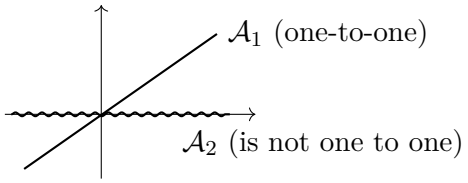
A basis for $\mathcal{N}(\mathcal{A})$: $\left\{t^2 - \frac{1}{3},\ t - \frac{1}{2}\right\}$.

▷ *Check:* $\int_0^1 (t^2 - \frac{1}{3})\,dt = \frac{1}{3} - \frac{1}{3} = 0$, $\int_0^1 (t - \frac{1}{2})\,dt = \frac{1}{2} - \frac{1}{2} = 0$*; the two polynomials have different degrees, so they are linearly independent. Consistent with rank–nullity:* $\dim \mathcal{V} = 3 = \dim \mathcal{N}(\mathcal{A}) + \dim \mathcal{R}(\mathcal{A}) = 2 + 1$.

4.3 One-to-One Transformations

Definition 4.8. A linear transformation is called **one to one** iff

$$\mathcal{A}v_1 = \mathcal{A}v_2 \quad \text{implies} \quad v_1 = v_2.$$



Proposition 4.9. A linear transformation $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ is **one to one iff** $\mathcal{N}(\mathcal{A}) = \{0_V\}$:

$$\text{one to one} \iff \mathcal{N}(\mathcal{A}) = \{0_V\} \iff \dim \mathcal{N}(\mathcal{A}) = \underline{0}.$$

Proof techniques.

$p \Rightarrow q$	(constructive proof)
$\bar{q} \Rightarrow \bar{p}$	(contraposition proof)
$p \wedge \bar{q} = \text{False}$	(contradiction proof)

Proof. **One-to-one** $\Rightarrow \mathcal{N}(\mathcal{A}) = \{0_V\}$**.** We use contraposition ($p \Rightarrow q \ \equiv\ \bar{q} \Rightarrow \bar{p}$), i.e. we show

$$\mathcal{N}(\mathcal{A}) \neq \{0_V\} \ \Rightarrow\ \text{not one-to-one.}$$

$\exists x \neq 0_V$ s.t. $x \in \mathcal{N}(\mathcal{A})$, i.e. $\mathcal{A}x = 0_W$.
Let $v_1 \in \mathcal{V}$ and $v_2 = v_1 + x \in \mathcal{V}$. As $x \neq 0 \Rightarrow v_2 \neq v_1$.
$\mathcal{A}v_2 = \mathcal{A}v_1 + \underbrace{\mathcal{A}x}_{0_W} = \mathcal{A}v_1 \ \Rightarrow\ \mathcal{A}$ is not one to one.

$\mathcal{N}(\mathcal{A}) = \{0_V\} \Rightarrow$ **one-to-one.** Again by contraposition: not-one-to-one $\Rightarrow \mathcal{N}(\mathcal{A}) \neq \{0_V\}$.

$$\begin{aligned} &\exists v_1 \neq v_2 \text{ s.t. } \mathcal{A}v_1 = \mathcal{A}v_2 \\ &\Rightarrow\ \mathcal{A}v_1 - \mathcal{A}v_2 = 0_W \\ &\Rightarrow\ \mathcal{A}(v_1 - v_2) = 0_W \\ &\Rightarrow\ 0_V \neq v_1 - v_2 \in \mathcal{N}(\mathcal{A}) \\ &\Rightarrow\ \mathcal{N}(\mathcal{A}) \neq \{0_V\} \end{aligned}$$

□

Why the null space decides injectivity

For a linear map, $\mathcal{A}v_1 = \mathcal{A}v_2 \iff \mathcal{A}(v_1 - v_2) = 0_W \iff v_1 - v_2 \in \mathcal{N}(\mathcal{A})$. So the set of vectors with the same image as v_1 is the shifted copy $v_1 + \mathcal{N}(\mathcal{A})$; images are unique exactly when this copy is the single point v_1, i.e. when $\mathcal{N}(\mathcal{A}) = \{0_V\}$.

Example 4.10. $\mathcal{V} = \mathcal{W} = \{\text{polynomials with degree} \le 2\}$, $\mathcal{A} = \dfrac{d}{dt}$. Is $\mathcal{A}$ one-to-one? p. 28

$$\mathcal{A}(at^2 + bt + c) = 2at + b = 0_W = 0(t) \;\Rightarrow\; a = 0,\; b = 0$$
$$\mathcal{N}(\mathcal{A}) = \{c\} = \{\text{constant polynomials}\}, \qquad \dim(\mathcal{N}(\mathcal{A})) = 1$$
$$\mathcal{N}(\mathcal{A}) \neq \{0_V\} \;\Rightarrow\; \textbf{Not one to one}$$

▷ *Side note in the notes:* $\mathcal{R}(\mathcal{A}) = \{\text{poly. degree} \le 1\} = \mathrm{sp}\{1, t\}$ *(2 elements,* $\dim \mathcal{R}(\mathcal{A}) = 2$*).*

Example 4.11. $\mathcal{V} = \{f \mid f \text{ is integrable},\ f : [0,1] \to \mathbb{R}\}$, $\mathcal{W} = \mathbb{R}$, $\mathcal{A}f = \displaystyle\int_0^1 f(t)\,dt$. Is $\mathcal{A}$ one-to-one?

$$f(t) = 2t - 1 \neq 0_V, \qquad \mathcal{A}f = \left(t^2 - t\right)\Big|_0^1 = 0$$

$\therefore$ Not one-to-one. $(\dim(\mathcal{N}(\mathcal{A})) = \infty)$

▷ *E.g.* $t^k - \frac{1}{k+1}$, $k = 1, 2, 3, \ldots$ *all integrate to* 0 *and are linearly independent, so the null space is infinite dimensional.*

Example 4.12. $\mathcal{V} = \mathcal{W} = \mathbb{R}^2$, $x = \begin{bmatrix}\alpha_1 \\ \alpha_2\end{bmatrix}$, $\mathcal{A}x = \begin{bmatrix}\alpha_1 \\ \alpha_1 + \alpha_2\end{bmatrix}$. Is $\mathcal{A}$ one-to-one?

$$\mathcal{A}x = \begin{bmatrix}0 \\ 0\end{bmatrix} = \begin{bmatrix}\alpha_1 \\ \alpha_1 + \alpha_2\end{bmatrix} \;\Rightarrow\; \alpha_1 = \alpha_2 = 0$$
$$\mathcal{N}(\mathcal{A}) = \left\{\begin{bmatrix}0 \\ 0\end{bmatrix}\right\} \checkmark \qquad \cancel{\mathcal{N}(\mathcal{A}) = \begin{bmatrix}0 \\ 0\end{bmatrix}}$$

▷ *Crossed out:* $\mathcal{N}(\mathcal{A})$ *is a subspace (a set), whereas* $\begin{bmatrix}0 \\ 0\end{bmatrix}$ *alone is not a set/subset — write it inside braces.*

4.4 Range Space, Onto Transformations and the Rank–Nullity Theorem

Definition 4.13. Let $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ be a linear transformation. The **range space** of $\mathcal{A}$ is defined as

$$\mathcal{R}(\mathcal{A}) \triangleq \{w \in \mathcal{W} \mid \exists v \in \mathcal{V} \text{ s.t. } \mathcal{A}v = w\}$$

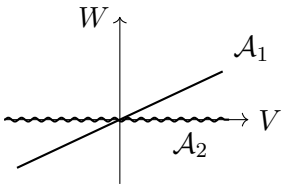
Claim. $\mathcal{R}(\mathcal{A})$ **is a subspace of** $\mathcal{W}$**.**

Let $w_1, w_2 \in \mathcal{R}(\mathcal{A})$. By the definition of range space,

$$\exists v_1, v_2 \text{ s.t. } \mathcal{A}v_1 = w_1, \quad \mathcal{A}v_2 = w_2$$
$$\mathcal{A}(v_1 + v_2) = \mathcal{A}v_1 + \mathcal{A}v_2 = w_1 + w_2 \;\Rightarrow\; w_1 + w_2 \in \mathcal{R}(\mathcal{A})$$
$$\mathcal{A}(\alpha v_1) = \alpha \mathcal{A}v_1 = \alpha w_1 \;\Rightarrow\; \alpha w_1 \in \mathcal{R}(\mathcal{A})$$

$\therefore$ $\mathcal{R}(\mathcal{A})$ is closed under addition and scalar multiplication, hence a subspace of $\mathcal{W}$.

Example 4.14. $\mathcal{V} = \mathcal{W} = \mathbb{R}$ p. 29



$$\begin{aligned} \mathcal{N}(\mathcal{A}_1) &= \{0\}, & \dim \mathcal{N}(\mathcal{A}_1) &= 0 \\ \mathcal{N}(\mathcal{A}_2) &= V = \mathbb{R}, & \dim \mathcal{N}(\mathcal{A}_2) &= 1 \\ \mathcal{R}(\mathcal{A}_1) &= W = \mathbb{R}, & \dim \mathcal{R}(\mathcal{A}_1) &= 1 \\ \mathcal{R}(\mathcal{A}_2) &= \{0\}, & \dim \mathcal{R}(\mathcal{A}_2) &= 0 \end{aligned}$$

▷ $\dim \mathcal{N}(\mathcal{A}_1) + \dim \mathcal{R}(\mathcal{A}_1) = 0 + 1 = 1, \quad \dim \mathcal{N}(\mathcal{A}_2) + \dim \mathcal{R}(\mathcal{A}_2) = 1 + 0 = 1.$

Definition 4.15. A linear transformation $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ is called **onto** iff $\mathcal{R}(\mathcal{A}) = \mathcal{W}$; otherwise, if $\mathcal{R}(\mathcal{A}) \neq \mathcal{W}$ (i.e. $\mathcal{R}(\mathcal{A}) \subsetneq \mathcal{W}$), $\mathcal{A}$ is called **into**.

Example 4.16. $\mathcal{V} = \mathcal{W} = \{\text{polyn. with degree} \leq 2\}$ $(\to \dim 3)$,

$$\mathcal{A} = \frac{d}{dt}, \qquad \mathcal{A}(at^2 + bt + c) = 2at + b.$$

$$\begin{aligned} \mathcal{N}(\mathcal{A}) &= \{\text{constant polynomials}\}, & \dim \mathcal{N}(\mathcal{A}) &= 1 \\ \mathcal{R}(\mathcal{A}) &= \{\text{polynomials with degree} \leq 1\}, & \dim \mathcal{R}(\mathcal{A}) &= 2 \end{aligned}$$

$\mathcal{R}(\mathcal{A}) \neq \mathcal{W} \;\Rightarrow \therefore \mathcal{A}$ is not onto.
▷ *Sum:* $1 + 2 = 3$, *the same as* $\dim \mathcal{V}$.

Example 4.17. $\mathcal{V} = \{f : [0,1] \to \mathbb{R},\ f \text{ is integrable}\}, \quad \mathcal{W} = \mathbb{R}, \quad \mathcal{A}f = \int_0^1 f(t)\,dt.$

$$\mathcal{R}(\mathcal{A}) = \mathbb{R}$$

▷ *For any* $c \in \mathbb{R}$, *the constant function* $f(t) = c$ *gives* $\mathcal{A}f = c$, *so* $\mathcal{A}$ *is onto (but not one-to-one, see p. 28).*

$$\left.\begin{aligned} \dim \mathcal{N}(\mathcal{A}) &\triangleq \textbf{nullity} \\ \dim \mathcal{R}(\mathcal{A}) &\triangleq \textbf{rank} \end{aligned}\right\} \xrightarrow{+} \text{constant}$$

Theorem 4.18 (Rank–Nullity Theorem)**.** If $\mathcal{V}$ is finite dimensional and $\mathcal{A}$ is a linear transformation from $\mathcal{V}$ into $\mathcal{W}$, then $\mathcal{R}(\mathcal{A})$ is finite dimensional and **nullity** + **rank** $= \dim \mathcal{V}$:

$$\dim \mathcal{N}(\mathcal{A}) + \dim \mathcal{R}(\mathcal{A}) = \dim \mathcal{V}.$$

Proof sketch and consequences

Take a basis $\{u_1, \ldots, u_r\}$ of $\mathcal{N}(\mathcal{A})$ and extend it to a basis $\{u_1, \ldots, u_r, v_{r+1}, \ldots, v_n\}$ of $\mathcal{V}$. Then $\mathcal{R}(\mathcal{A}) = \text{sp}\{\mathcal{A}v_{r+1}, \ldots, \mathcal{A}v_n\}$ (since $\mathcal{A}u_i = 0_W$), and these $n - r$ images are linearly independent: $\sum \beta_j \mathcal{A}v_j = 0_W \Rightarrow \sum \beta_j v_j \in \mathcal{N}(\mathcal{A}) \Rightarrow$ all $\beta_j = 0$. Hence $\dim \mathcal{R}(\mathcal{A}) = n - r$. Consequences: $\mathcal{A}$ one-to-one $\iff$ $\dim \mathcal{R}(\mathcal{A}) = \dim \mathcal{V}$; and if $\dim \mathcal{V} = \dim \mathcal{W} < \infty$, then one-to-one $\iff$ onto. In the examples above: $\frac{d}{dt}$ on polynomials of degree ≤ 2: nullity + rank $= 1 + 2 = 3 = \dim \mathcal{V}$, so it is neither one-to-one nor onto.

4.5 Matrix Representations

Definition 4.19. Let $\mathcal{V}$ (dim $= n$) and $\mathcal{W}$ (dim $= m$) be two linear spaces over the same field $\mathcal{F}$. Let $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ be a linear transformation. Given the basis $B = \{v_1, v_2, \ldots, v_n\}$ and $C = \{w_1, w_2, \ldots, w_m\}$ for $\mathcal{V}$ and $\mathcal{W}$, the transformation can be represented by a matrix $A = [a_{ij}]$ ($a_{ij} \in \mathcal{F}$), its **matrix representation**, s.t.

$$[w]_C = A\,[v]_B$$

for any w, v satisfying $w = \mathcal{A}v$.

▷ *A is $m \times n$; its jth column is the coordinate vector $[\mathcal{A}v_j]_C$ (procedure below). The matrix depends on the chosen bases B and C.*

Example 4.20. p. 30

$$\mathcal{V} = \{\text{polynomials with degree} \le 3\}, \qquad \mathcal{W} = \{\text{polynomials with degree} \le 2\}$$
$$\mathcal{A} : \mathcal{V} \to \mathcal{W}, \qquad \mathcal{A}f = \frac{df}{dt}$$
$$B = \{\underbrace{1}_{v_1}, \underbrace{1+t}_{v_2}, \underbrace{1+t+t^2}_{v_3}, \underbrace{1+t+t^2+t^3}_{v_4}\}$$
$$C = \{\underbrace{1}_{w_1}, \underbrace{1+t}_{w_2}, \underbrace{1+t+t^2}_{w_3}\}$$
$$[w]_C = A\,[v]_B \qquad \text{e.g. } [v]_B = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

Procedure.

* Choose $v = v_j$ for $j = 1, \ldots, n$:

$$[v]_B = e_j = \begin{bmatrix}0\\0\\\vdots\\1\\\vdots\\0\end{bmatrix} \leftarrow j\text{th element}$$

▷ *$A\,e_j$ picks out the jth column of A: this gives the first column for $j = 1$, and in general the jth column.*

* Find $w = \mathcal{A}v_j$.
* **jth column of A is $[\mathcal{A}v_j]_C$.**

Columns of A.

$$v = v_1 : \quad [v_1]_B = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \quad \mathcal{A}v_1 = 0 = 0w_1 + 0w_2 + 0w_3 \qquad \Rightarrow\ 1^{\text{st}} \text{ column of } A \text{ is } \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$v = v_2: \quad [v_2]_B = \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \quad \mathcal{A}v_2 = 1 = 1w_1 + 0w_2 + 0w_3 \qquad \Rightarrow \ 2^{\text{nd}} \text{ column of } A \text{ is } \begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$v = v_3: \quad [v_3]_B = \begin{bmatrix}0\\0\\1\\0\end{bmatrix}, \quad \mathcal{A}v_3 = 2t + 1 = -w_1 + 2w_2 + 0w_3 \quad \Rightarrow \ 3^{\text{rd}} \text{ column of } A \text{ is } \begin{bmatrix}-1\\2\\0\end{bmatrix}$$

$$v = v_4: \quad [v_4]_B = \begin{bmatrix}0\\0\\0\\1\end{bmatrix}, \quad \mathcal{A}v_4 = 3t^2 + 2t + 1$$

$$= -1w_1 - 1w_2 + 3w_3 \qquad \Rightarrow \ 4^{\text{th}} \text{ column of } A \text{ is } \begin{bmatrix}-1\\-1\\3\end{bmatrix}$$

$$\boxed{A = \begin{bmatrix}0 & 1 & -1 & -1\\0 & 0 & 2 & -1\\0 & 0 & 0 & 3\end{bmatrix}}$$

▷ *Coordinates in C:* $2t + 1 = -1 \cdot 1 + 2(1 + t)$; $3t^2 + 2t + 1 = 3(1 + t + t^2) - 1 \cdot (1 + t) - 1 \cdot 1$ *(match the t^2, then t, then constant coefficients).*

---

p. 31

Check. Let $v = t^3 + 2t^2 + 3t + 4$. Then

$$[v]_B = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \qquad [w]_C = A\,[v]_B = \begin{bmatrix}-1\\1\\3\end{bmatrix}$$

$$\begin{aligned} w &= -w_1 + w_2 + 3w_3 \\ &= -1 + 1 + t + 3 + 3t + 3t^2 = 3t^2 + 4t + 3 \quad // \end{aligned}$$

▷ *Indeed* $\frac{d}{dt}(t^3 + 2t^2 + 3t + 4) = 3t^2 + 4t + 3$. *($[v]_B$: $v = 1 \cdot v_1 + 1 \cdot v_2 + 1 \cdot v_3 + 1 \cdot v_4$, since $v_1 + v_2 + v_3 + v_4 = 4 + 3t + 2t^2 + t^3$.)*

4.6 Change of Basis and Similar Matrices

1) Basis change in the domain. Let $[v]_B = P\,[v]_{\bar{B}}$.

$$\begin{aligned} [w]_C &= A\,[v]_B \\ [w]_C &= \bar{A}\,[v]_{\bar{B}} \\ [w]_C &= A \cdot P\,[v]_{\bar{B}} \end{aligned}$$

$$\therefore \boxed{\bar{A} = A \cdot P}$$

2) Basis change in the co-domain. Let $[w]_C = Q\,[w]_{\bar{C}}$.

$$\begin{aligned} [w]_C &= A\,[v]_B \\ [w]_{\bar{C}} &= \bar{A}\,[v]_B \\ Q\,[w]_{\bar{C}} &= A\,[v]_B \\ [w]_{\bar{C}} &= Q^{-1}A\,[v]_B \end{aligned}$$

$$\therefore \boxed{\bar{A} = Q^{-1} \cdot A}$$

3) Basis change in both domain and co-domain. With $[v]_B = P\,[v]_{\bar{B}}$ and $[w]_C = Q\,[w]_{\bar{C}}$:

$$\begin{aligned} [w]_C &= A\,[v]_B \\ [w]_{\bar{C}} &= \bar{A}\,[v]_{\bar{B}} \\ Q\,[w]_{\bar{C}} &= A\,P\,[v]_{\bar{B}} \\ [w]_{\bar{C}} &= Q^{-1}AP\,[v]_{\bar{B}} \end{aligned}$$

$$\therefore \boxed{\bar{A} = Q^{-1}AP}$$

Statement: Special Case. If $\mathcal{V} = \mathcal{W}$, $\mathcal{A}$ is a linear transformation from $\mathcal{V}$ into $\mathcal{V}$. Then the basis B can be taken to be the same as C. Then a basis change in both the domain and co-domain would give

$$\therefore \boxed{\bar{A} = P^{-1}AP}$$

▷ *Here $Q = P$ since the same basis change is applied to domain and co-domain; A and $\bar{A}$ are then called* ***similar*** *matrices.*

Example 4.21.

$$\begin{aligned} \mathcal{V} &= \{\text{polynomial with degree} \le 3\}, & B &= \{1,\ 1+t,\ 1+t+t^2,\ 1+t+t^2+t^3\} \\ \mathcal{W} &= \{\text{polynomial with degree} \le 2\}, & C &= \{1,\ 1+t,\ 1+t+t^2\} \end{aligned}$$

$$A = \begin{bmatrix} 0 & 1 & -1 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

∗ Change the basis in the domain to $\bar{B} = \{1, t, t^2, t^3\}$: p. 32

$$\bar{A} = AP, \qquad [v]_B = P\,[v]_{\bar{B}}$$

$$P = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \bar{A} = AP = \begin{bmatrix} 0 & 1 & -2 & 0 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

▷ *Column j of P is $[\bar{v}_j]_B$: $1 = v_1$, $t = v_2 - v_1$, $t^2 = v_3 - v_2$, $t^3 = v_4 - v_3$. Check of $\bar{A}$: $\frac{d}{dt}t^2 = 2t = -2 \cdot 1 + 2(1+t)$ and $\frac{d}{dt}t^3 = 3t^2 = 3(1+t+t^2) - 3(1+t)$, matching columns 3 and 4.*

∗ Change the basis in the co-domain to canonical basis while keeping the basis of the domain as the canonical form: $\bar{C} = \{1, t, t^2\}$,

$$\begin{aligned} [w]_{\bar{C}} &= \bar{\bar{A}}\,[v]_{\bar{B}} \\ \bar{\bar{A}} &= Q^{-1}\bar{A}, & [w]_C &= Q\,[w]_{\bar{C}} \\ \bar{\bar{A}} &= Q^{-1}\bar{A}, & R\,[w]_C &= [w]_{\bar{C}} \\ \bar{\bar{A}} &= R\bar{A}, & [w]_{\bar{C}} &= R\,[w]_C \end{aligned}$$

$$R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = Q^{-1}, \qquad \bar{\bar{A}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

▷ *$Q^{-1}\bar{A} \Rightarrow$ linear combinations of rows of $\bar{A}$ (row 1 = rows $1+2+3$, row 2 = rows $2+3$, row 3 unchanged). Column j of R is $[w_j]_{\bar{C}}$: 1, $1+t$, $1+t+t^2$.*

Check

$\bar{\bar{A}}$ is the matrix of $\frac{d}{dt}$ in the canonical bases: $1 \mapsto 0$, $t \mapsto 1$, $t^2 \mapsto 2t$, $t^3 \mapsto 3t^2$, which gives exactly the columns $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$. Altogether $\bar{\bar{A}} = Q^{-1}AP = RAP$.

5 Direct Sums and Orthogonal Projections

Syllabus: Week 6 — direct sum decompositions, projection theorem, orthogonal projections

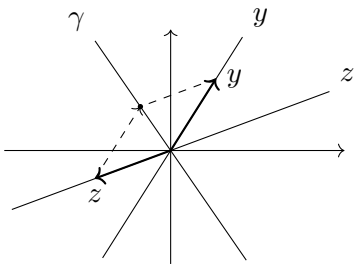
5.1 Direct Sums

Definition 5.1. Let $\mathcal{V}$ be a vector space and let $M_1, \dots, M_k$ be subspaces of $\mathcal{V}$. The **sum** of these subspaces is defined as

$$M = M_1 + M_2 + \cdots + M_k = \{m \in \mathcal{V} \mid m = m_1 + m_2 + \cdots + m_k,\ m_i \in M_i\ \forall i\}$$

The subspaces are called **linearly independent** iff

$$m_1 + m_2 + \cdots + m_k = 0_V \quad \text{where } m_i \in M_i\ \forall i \quad \Rightarrow \quad m_1 = m_2 = \cdots = m_k = 0_V.$$



$y + z - \gamma = 0$
$\Rightarrow$ not linearly independent

Remark. Let $\mathcal{V} = M_1 + M_2 + \cdots + M_k$ and let $M_1, \dots, M_k$ be lin. indep. Then for $x \in \mathcal{V}$, p. 33

$$x = m_1 + m_2 + \cdots + m_k, \qquad m_i \in M_i\ \forall i,$$

$m_1, m_2, \dots, m_k$ **are unique.**

Proof. Assume $m_1, \dots, m_k$ are not unique, i.e. $\exists i$ s.t. $m_i \neq \bar{m}_i$:

$$\begin{aligned} x &= m_1 + \cdots + m_k \\ x &= \bar{m}_1 + \cdots + \bar{m}_k \\ 0_V &= \underbrace{(m_1 - \bar{m}_1)}_{\in M_1} + \underbrace{(m_2 - \bar{m}_2)}_{\in M_2} + \cdots + \underbrace{(m_k - \bar{m}_k)}_{\in M_k} \end{aligned}$$

$\Rightarrow M_1, \dots, M_k$ are not linearly independent. □

Definition 5.2. Let $M_1, \dots, M_k$ be subspaces of $\mathcal{V}$. Let

i-) $M = M_1 + M_2 + \cdots + M_k$

ii-) $M_1, M_2, \dots, M_k$ are linearly indep.

Then M is said to be the **direct sum** of $M_1, \dots, M_k$. This is shown as

$$M = M_1 \oplus M_2 \oplus \cdots \oplus M_k$$

Example 5.3. $\mathcal{V} = \mathbb{R}^4, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4.$

$$M_1 \triangleq \{x \in \mathbb{R}^4 \mid x_4 = x_3 = 0\} : \quad \begin{bmatrix} x_1 \\ x_2 \\ 0 \\ 0 \end{bmatrix}, \quad \dim = 2$$

$$M_2 \triangleq \{x \in \mathbb{R}^4 \mid x_1 = x_2 = 0\} : \quad \begin{bmatrix} 0 \\ 0 \\ x_3 \\ x_4 \end{bmatrix}, \quad \dim = 2$$

$$M_3 \triangleq \{x \in \mathbb{R}^4 \mid x_1 = 0\} : \quad \begin{bmatrix} 0 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad \dim = 3$$

$M_1 + M_2 = \mathbb{R}^4$ $\Rightarrow$ direct sum: $M_1 \,\&\, M_2$ linearly indep $\Rightarrow M_1 \oplus M_2 = \mathbb{R}^4$
$M_1 + M_3 = \mathbb{R}^4$ $\Rightarrow$ not direct sum
$M_2 + M_3 = M_3$ $M_2 \subset M_3 \Rightarrow$ not direct sum

---

For M_1 and M_3:

$$\begin{bmatrix} x_1 \\ x_2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2' \\ x_3' \\ x_4' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \text{e.g.} \quad \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\in M_1} + \underbrace{\begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}}_{\in M_3} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{0_V}$$

As the vectors are not equal to zero, $M_1 \,\&\, M_3$ are linearly dependent. $\therefore$ Not a direct sum. $M_2 \,\&\, M_3$ are linearly dependent (e.g. $m_2 = (0,0,1,0)^T \in M_2$, $m_3 = -m_2 \in M_3$). $\therefore$ Not a direct sum.

Dimension test

For two subspaces, $M_1 + M_2$ is direct $\iff$ $M_1 \cap M_2 = \{0_V\}$ $\iff$ $\dim(M_1 + M_2) = \dim M_1 + \dim M_2$. Here: $M_1 \cap M_2 = \{0\}$ and $2+2=4$ (direct); $M_1 \cap M_3 = \{(0, x_2, 0, 0)^T\}$, $2+3=5 \neq 4$ (not direct); $M_2 \cap M_3 = M_2$, $2+3 \neq 3$ (not direct).

5.2 Orthogonal Direct Sums and Orthogonal Complements

Definition 5.4. Let $\mathcal{V}$ be an inner product space. Two subspaces $M_1 \,\&\, M_2$ are said to be **orthogonal** iff p. 34

$$\langle m_1, m_2 \rangle = 0 \qquad \forall m_1 \in M_1,\ \forall m_2 \in M_2.$$

We denote this $M_1 \perp M_2$.

Definition 5.5. Let M be equal to $M_1 + M_2 + \cdots + M_k$ and $M_i \perp M_j$ for $i \neq j$. Then M is said to be an **orthogonal direct sum** of $M_1, \ldots, M_k$. We show this as

$$M = M_1 \overset{\perp}{\oplus} M_2 \overset{\perp}{\oplus} \ldots \overset{\perp}{\oplus} M_k.$$

If $M = \mathcal{V}$ then

$$\mathcal{V} = M_1 \overset{\perp}{\oplus} M_2 \overset{\perp}{\oplus} \ldots \overset{\perp}{\oplus} M_k$$

is called an **orthogonal direct sum decomposition** of $\mathcal{V}$.

Orthogonality implies linear independence. Let $\sum_{i=1}^{k} m_i = 0_V, \;\; m_i \in M_i$. Then

$$\left\langle \sum_{i=1}^{k} m_i,\, m_j \right\rangle = \langle 0_V, m_j \rangle = 0$$
$$\sum_{i=1}^{k} \langle m_i, m_j \rangle = 0$$

Since $\langle m_i, m_j \rangle = 0$ when $i \neq j$:

$$\langle m_j, m_j \rangle = 0 \;\Rightarrow\; m_j = 0_V.$$

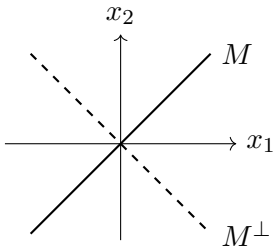
▷ *The last step uses positive definiteness of the inner product. Since j was arbitrary, all $m_j = 0_V$: orthogonality implies linear independence.*

Orthogonal complement.

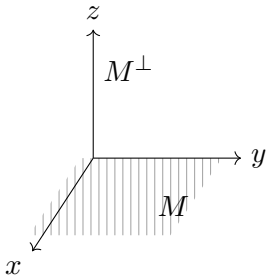
Definition 5.6. Let M be a subspace of an inner product space $\mathcal{V}$. The **orthogonal complement** $M^\perp$ of the subspace is defined as

$$M^\perp = \{x \in \mathcal{V} \mid \langle x, m \rangle = 0 \;\forall m \in M\}$$

Zero vector is orthogonal to any vector.



$$M = \operatorname{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \begin{bmatrix}-1\\1\end{bmatrix} \in M^\perp, \quad \begin{bmatrix}2\\-2\end{bmatrix} \in M^\perp$$
$$M^\perp = \operatorname{sp}\left\{\begin{bmatrix}-1\\1\end{bmatrix}\right\}, \quad M + M^\perp = M \overset{\perp}{\oplus} M^\perp$$



$$M = \{x\text{-}y \text{ plane}\}$$
$$M^\perp = \{z \text{ axis}\}$$
$$\underset{\dim=2}{M} \overset{\perp}{\oplus} \underset{\dim=1}{M^\perp} = \underset{\dim 3}{\mathbb{R}^3}$$

Orthogonality implies linear independency, thus this is a direct sum. Also orthogonal, so orthogonal direct sum.

Claim. $M^\perp$ **is a subspace of** $\mathcal{V}$**.**

Let $x, y \in M^\perp$, i.e. $\langle x, m\rangle = 0$, $\langle y, m\rangle = 0\ \forall m \in M$. Then

$$\begin{aligned} \langle x+y, m\rangle &= \underbrace{\langle x, m\rangle + \langle y, m\rangle}_{0} \quad (m \in M) \\ &= 0 \quad \forall m \in M && \Rightarrow\ x+y \in M^\perp \\ \langle \alpha x, m\rangle &= \alpha \langle x, m\rangle = 0 \quad \forall m \in M && \Rightarrow\ \alpha x \in M^\perp \end{aligned}$$

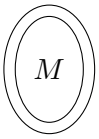
Theorem 5.7 (Orthogonal decomposition)**.** Let $\mathcal{V}$ be an inner product space and M be a subspace of $\mathcal{V}$, then $\mathcal{V}$ **is the orthogonal direct sum of** M **and** $M^\perp$: p. 35

$$\boxed{\mathcal{V} = M \overset{\perp}{\oplus} M^\perp}$$

To prove this we check:
- $\mathcal{V} = M + M^\perp \rightarrow$ show (proof below);
- $M \,\&\, M^\perp$ are linearly indep ✓;
- $M \perp M^\perp$ ✓.

Proof. **Part I:** $\dim M = n \triangleq \dim \mathcal{V} \Rightarrow M = \mathcal{V}$ ⍰

▷ *Answer to ⍰: a basis of M consists of n linearly independent vectors of $\mathcal{V}$; in an n-dimensional space any n independent vectors form a basis, so they span $\mathcal{V}$ and $M = \mathcal{V}$.*



$V = n$

$$\begin{aligned} M^\perp = \mathcal{V}^\perp &= \{0_V\} \\ \mathcal{V} &= M + M^\perp \\ x = \underset{\in M}{x_1} + \underset{\in M^\perp}{x_2} \ \Rightarrow\ x &= \underset{\in M}{x} + \underset{\in M^\perp}{0_V} \quad \checkmark \end{aligned}$$

Part II: $\dim M < n$, with $\dim M = k$. Let

$$\begin{aligned} B &= \{m_1, m_2, m_3, \ldots, m_k\} && \text{(basis for } M\text{)} \\ C &= \{m_1, m_2, \ldots, m_k, m_{k+1}, \ldots, m_n\} && \text{(basis for } \mathcal{V}\text{)} \end{aligned}$$

Apply Gram–Schmidt orthogonalization to C:

$$\bar{C} = \{ \underbrace{\underset{\uparrow m_1}{e_1}, e_2, \ldots, e_k}_{\text{form a basis for } M}, \overbrace{e_{k+1}, \ldots, e_n}^{\text{basis for } M^\perp}\}, \qquad \operatorname{sp}\{e_1, \ldots, e_k\} = \operatorname{sp}\{m_1, \ldots, m_k\}.$$

$\bar{C}$ is a basis for $\mathcal{V}$. For $x \in \mathcal{V}$:

$$x = \sum_{i=1}^{n} \alpha_i e_i = \underbrace{\sum_{i=1}^{k} \alpha_i e_i}_{x_1 \in M} + \underbrace{\sum_{j=k+1}^{n} \alpha_j e_j}_{x_2 \in M^\perp}$$

▷ *$x_2 \in M^\perp$ by construction: Gram–Schmidt makes each e_j $(j > k)$ orthogonal to $e_1, \ldots, e_k$, which span M; hence $e_j \perp m$ for all $m \in M$, and so is every combination of them.*

$$\Rightarrow\ x = \underset{\in M}{x_1} + \underset{\in M^\perp}{x_2} \ \Rightarrow\ \mathcal{V} = M + M^\perp \qquad \square$$

Explanation

The other two items are immediate: $M \perp M^\perp$ by the definition of $M^\perp$, and orthogonality implies linear independence (shown above), equivalently $M \cap M^\perp = \{0_V\}$ since $x \in M \cap M^\perp \Rightarrow \langle x, x\rangle = 0$. Consequently $\dim M^\perp = \dim \mathcal{V} - \dim M$, and $\{e_{k+1}, \dots, e_n\}$ is in fact a basis of $M^\perp$. (Finite dimension of $\mathcal{V}$ is assumed in the proof.)

Example 5.8. $\mathcal{V} = \mathbb{R}^4$,

$$M = \operatorname{sp}\left\{ \begin{bmatrix}1\\2\\3\\4\end{bmatrix}, \begin{bmatrix}4\\3\\2\\1\end{bmatrix} \right\}. \qquad \text{Find a basis for } M^\perp.$$

Complete the basis of M with two canonical vectors to a basis of $\mathbb{R}^4$ and orthogonalize:

$$C = \Big\{ \underbrace{\begin{bmatrix}1\\2\\3\\4\end{bmatrix}}_{m_1}, \underbrace{\begin{bmatrix}4\\3\\2\\1\end{bmatrix}}_{m_2}, \underbrace{\begin{bmatrix}1\\0\\0\\0\end{bmatrix}}_{m_3}, \underbrace{\begin{bmatrix}0\\1\\0\\0\end{bmatrix}}_{m_4} \Big\} \xrightarrow{\text{Gram–Schmidt}} \bar{C} = \{e_1, e_2, e_3, e_4\}$$

$$e_1 = m_1 = \begin{bmatrix}1\\2\\3\\4\end{bmatrix}, \qquad \langle e_1, e_1\rangle = 30$$

$$e_2 = m_2 - \frac{\langle m_2, e_1\rangle}{\langle e_1, e_1\rangle} e_1 = \begin{bmatrix}4\\3\\2\\1\end{bmatrix} - \frac{20}{30}\begin{bmatrix}1\\2\\3\\4\end{bmatrix} = \frac{5}{3}\begin{bmatrix}2\\1\\0\\-1\end{bmatrix} \;\to\; \begin{bmatrix}2\\1\\0\\-1\end{bmatrix}$$

$$e_3 = m_3 - \frac{\langle m_3, e_1\rangle}{\langle e_1, e_1\rangle} e_1 - \frac{\langle m_3, e_2\rangle}{\langle e_2, e_2\rangle} e_2 = \begin{bmatrix}1\\0\\0\\0\end{bmatrix} - \frac{1}{30}\begin{bmatrix}1\\2\\3\\4\end{bmatrix} - \frac{2}{6}\begin{bmatrix}2\\1\\0\\-1\end{bmatrix} = \frac{1}{10}\begin{bmatrix}3\\-4\\-1\\2\end{bmatrix} \;\to\; \begin{bmatrix}3\\-4\\-1\\2\end{bmatrix}$$

$$e_4 = m_4 - \frac{\langle m_4, e_1\rangle}{\langle e_1, e_1\rangle} e_1 - \frac{\langle m_4, e_2\rangle}{\langle e_2, e_2\rangle} e_2 - \frac{\langle m_4, e_3\rangle}{\langle e_3, e_3\rangle} e_3$$

$$= \begin{bmatrix}0\\1\\0\\0\end{bmatrix} - \frac{2}{30}\begin{bmatrix}1\\2\\3\\4\end{bmatrix} - \frac{1}{6}\begin{bmatrix}2\\1\\0\\-1\end{bmatrix} + \frac{4}{30}\begin{bmatrix}3\\-4\\-1\\2\end{bmatrix} = \frac{1}{6}\begin{bmatrix}0\\1\\-2\\1\end{bmatrix} \;\to\; \begin{bmatrix}0\\1\\-2\\1\end{bmatrix}$$

▷ *Scaling a vector does not affect orthogonality, so after each step the vector is rescaled to integer entries ($\to$) and the scaled version is used in the following steps.*

$$\boxed{M^\perp = \operatorname{sp}\{e_3, e_4\} = \operatorname{sp}\left\{ \begin{bmatrix}3\\-4\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\-2\\1\end{bmatrix} \right\}}$$

Verification

$\langle e_3, m_1\rangle = 3 - 8 - 3 + 8 = 0$, $\langle e_3, m_2\rangle = 12 - 12 - 2 + 2 = 0$, $\langle e_4, m_1\rangle = 2 - 6 + 4 = 0$, $\langle e_4, m_2\rangle = 3 - 4 + 1 = 0$, and $\langle e_3, e_4\rangle = -4 + 2 + 2 = 0$. Since $e_3, e_4 \neq 0$, the set C was indeed a basis of $\mathbb{R}^4$.

Alternatively, $x \in M^\perp \iff \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix} x = 0$, i.e. $M^\perp = \mathcal{N}\left(\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}\right)$, which has dimension $4 - 2 = 2$ and is spanned e.g. by $\begin{bmatrix} 1 & -2 & 1 & 0 \end{bmatrix}^T$ and $\begin{bmatrix} 0 & 1 & -2 & 1 \end{bmatrix}^T$; note $\begin{bmatrix} 3 & -4 & -1 & 2 \end{bmatrix}^T = 3\begin{bmatrix} 1 & -2 & 1 & 0 \end{bmatrix}^T + 2\begin{bmatrix} 0 & 1 & -2 & 1 \end{bmatrix}^T$, so both answers describe the same subspace. Gram–Schmidt additionally gives an *orthogonal* basis.

5.3 The Projection Theorem

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Theorem 5.9 (Hilbert's Projection Theorem)**.** Let H be a Hilbert space and let M be a finite dimensional subspace of H. For any $x \in H$ the following minimization problem has a solution:

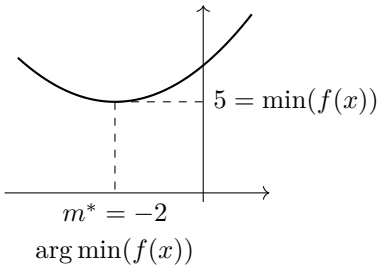
$$m^* = \arg\min_{m \in M} \|x - m\|^2$$

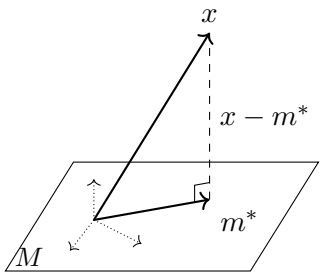
- the exponent 2 does not matter, the norm is positive;
- $\|\cdot\|$ is the norm induced by the inner product.

In other words, we can find the closest vector m^* to x lying in the subspace M. If m^* is a solution, then $x - m^*$ **is orthogonal to** M. Furthermore, **the solution** m^* **is unique**:

$$m^* = \mathrm{Proj}_M x : \text{ “}\textbf{Orthogonal Projection}\text{ of } x \text{ onto } M\text{”}$$

▷ *Reminder on* arg min *vs.* min *(left), and the geometry of the theorem (right).*





$\Rightarrow$ solution exists
$\Rightarrow$ solution is unique
$\Rightarrow$ $x - m^* \perp m^*$

$$H = M \overset{\perp}{\oplus} M^\perp, \qquad x = \underbrace{x_1}_{\in M} + \underbrace{x_2}_{\in M^\perp}$$

$$\|x - m\|^2 = \langle x - m, x - m\rangle = \langle x_1 + x_2 - m, x_1 + x_2 - m\rangle$$

$$= \langle \underbrace{(\underbrace{x_1}_{\in M} - \underbrace{m}_{\in M})}_{\in M} + \underbrace{x_2}_{\in M^\perp},\ (x_1 - m) + x_2 \rangle$$

$$= \langle x_1 - m, x_1 - m \rangle + \underbrace{\left\langle \underbrace{x_1 - m}_{\in M},\ \underbrace{x_2}_{\in M^\perp} \right\rangle}_{0} + \underbrace{\langle x_2, x_1 - m \rangle}_{0} + \langle x_2, x_2 \rangle$$

$$= \|x_1 - m\|^2 + \|x_2\|^2$$

$\boxed{m^* = x_1}$ to minimize this $\quad\Longrightarrow\quad$ $\boxed{\therefore\ \min \|x - m\| = \|x_2\|}$

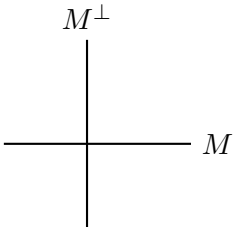
- x_1 exists
- x_1 is unique as $M \oplus M^\perp$ is a direct sum property
- $x - m^* = x - x_1 = x_2 \in M^\perp$

Why the proof works

The key identity is the Pythagorean theorem $\|x - m\|^2 = \|x_1 - m\|^2 + \|x_2\|^2$, valid because $x_1 - m \in M$ and $x_2 \in M^\perp$ are orthogonal. The second term does not depend on m, and the first one is ≥ 0 with equality *only* for $m = x_1$; this gives existence and uniqueness at once. The decomposition $H = M \oplus M^\perp$ itself exists because M is finite dimensional: take an orthonormal basis $\{e_1, \ldots, e_k\}$ of M (Gram–Schmidt) and set $x_1 = \sum_i \langle x, e_i \rangle e_i$, $x_2 = x - x_1$; then $\langle x_2, e_j \rangle = \langle x, e_j \rangle - \langle x, e_j \rangle = 0$ for every j, so $x_2 \in M^\perp$.

Note. $\mathcal{V} = M^\perp \overset{\perp}{\oplus} M, \quad x \in \mathcal{V}, \quad x \notin M.$

$$\mathbb{R}^2 = x \text{ axis} \overset{\perp}{\oplus} y \text{ axis}, \qquad \mathbb{R}^2 = \underbrace{\operatorname{sp}\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}}_{M} \overset{\perp}{\oplus} \underbrace{\operatorname{sp}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}}_{M^\perp}$$



$$\begin{bmatrix}1\\2\end{bmatrix} = \underbrace{\begin{bmatrix}1\\0\end{bmatrix}}_{\in M} + \underbrace{\begin{bmatrix}0\\2\end{bmatrix}}_{\in M^\perp}$$

If a vector is not in M, it does not directly mean that it should be inside $M^\perp$. The vector can be written using M and $M^\perp$.

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$$x \in M \ \Rightarrow\ m^* = \operatorname{Proj}_M x = x, \qquad x = \underbrace{x_1}_{x} + \underbrace{x_2}_{0}: \quad \text{if } x \in M \ \Rightarrow\ \operatorname{Proj}_{M^\perp} x = 0$$

Example 5.10. $\mathcal{V} = \mathbb{R}^2, \quad M = \operatorname{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \quad \langle x, y \rangle = y^T x, \quad x = \begin{bmatrix}1\\2\end{bmatrix}, \quad m^* = \operatorname{Proj}_M x.$

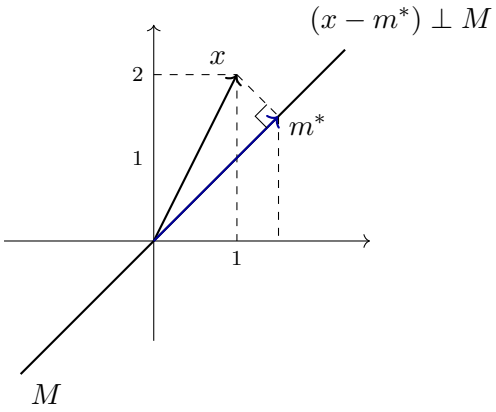
$$m = \alpha \begin{bmatrix}1\\1\end{bmatrix} \ \Rightarrow\ m^* = \alpha^* \begin{bmatrix}1\\1\end{bmatrix}, \qquad (x - m^*) \perp M$$

$$x - m^* = \left(\begin{bmatrix}1\\2\end{bmatrix} - \alpha^* \begin{bmatrix}1\\1\end{bmatrix}\right) \perp \begin{bmatrix}1\\1\end{bmatrix}$$

$$\begin{bmatrix} 1-\alpha^* \\ 2-\alpha^* \end{bmatrix} \perp \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$1-\alpha^*+2-\alpha^*=0 \;\Rightarrow\; \alpha^*=\tfrac{3}{2}$$

$$\boxed{m^*=\frac{3}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

▷ *Check:* $x-m^*=\begin{bmatrix} -1/2 \\ 1/2 \end{bmatrix}$ *and* $\begin{bmatrix} -1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}=0$*; the distance from* x *to* M *is* $\|x-m^*\|=1/\sqrt{2}$.



Example 5.11. **(1)** $\mathcal{V}$, $\quad M=\mathrm{sp}\{v_1\}$.

$$m^*=\mathrm{Proj}_M\, x=? \qquad m^*=\alpha^* v_1, \qquad x-m^*=(x-\alpha^* v_1)\perp v_1$$

$$\langle x-\alpha^* v_1, v_1\rangle = 0$$
$$\langle x, v_1\rangle - \alpha^*\langle v_1, v_1\rangle = 0$$

$$\Rightarrow \quad \alpha^*=\frac{\langle x, v_1\rangle}{\langle v_1, v_1\rangle}, \qquad m^*=\frac{\langle x, v_1\rangle}{\langle v_1, v_1\rangle}\cdot v_1$$

(2) $\mathcal{V}$, $\quad M=\mathrm{sp}\{v_1, v_2\}$.

$$m^*=\mathrm{Proj}_M\, x=?$$
$$m^*=\alpha_1^* v_1+\alpha_2^* v_2$$
$$(x-\alpha_1^* v_1-\alpha_2^* v_2)\perp M \;\Rightarrow\; (x-\alpha_1^* v_1-\alpha_2^* v_2)\perp v_1 \text{ and } v_2$$

$$\langle x-\alpha_1^* v_1-\alpha_2^* v_2, v_1\rangle = 0 \;\Rightarrow\; \langle v_1, v_1\rangle\alpha_1^*+\langle v_2, v_1\rangle\alpha_2^*=\langle x, v_1\rangle$$
$$\langle x-\alpha_1^* v_1-\alpha_2^* v_2, v_2\rangle = 0 \;\Rightarrow\; \langle v_1, v_2\rangle\alpha_1^*+\langle v_2, v_2\rangle\alpha_2^*=\langle x, v_2\rangle$$

$$\underbrace{\begin{bmatrix} \langle v_1, v_1\rangle & \langle v_2, v_1\rangle \\ \langle v_1, v_2\rangle & \langle v_2, v_2\rangle \end{bmatrix}}_{G}\underbrace{\begin{bmatrix} \alpha_1^* \\ \alpha_2^* \end{bmatrix}}_{\alpha^*}=\underbrace{\begin{bmatrix} \langle x, v_1\rangle \\ \langle x, v_2\rangle \end{bmatrix}}_{b} \qquad \Longrightarrow \qquad G\cdot\alpha^*=b, \quad \boxed{\alpha^*=G^{-1}\cdot b}$$

Gramian Matrix

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$$\det G=\langle v_1, v_1\rangle\langle v_2, v_2\rangle-\langle v_1, v_2\rangle\underbrace{\langle v_2, v_1\rangle}_{\overline{\langle v_1, v_2\rangle}}$$

$$= \langle v_1, v_1\rangle \langle v_2, v_2\rangle - |\langle v_1, v_2\rangle|^2$$

▷ *Cauchy–Schwarz: equality satisfied if $v_1 = \alpha v_2$; then only in this case* $\det G = 0$.

$$\det G > 0 \quad \text{if} \quad \underbrace{v_1 \neq \alpha v_2}_{\text{linearly indep}}, \qquad M = \operatorname{sp}\{v_1, v_2\}, \quad m^* = \alpha_1^* v_1 + \alpha_2^* v_2$$

Explanation

By the Cauchy–Schwarz inequality $|\langle v_1, v_2\rangle|^2 \le \langle v_1, v_1\rangle \langle v_2, v_2\rangle$, so always $\det G \ge 0$, with equality iff v_1, v_2 are linearly dependent. Hence for a basis $\{v_1, v_2\}$ of M the system $G\alpha^* = b$ has exactly one solution, which is the unique m^* promised by the projection theorem. Note also that G is Hermitian: its (j,i) entry is the complex conjugate of its (i,j) entry.

Example 5.12. $\mathcal{V} = \{f : [0,1] \to \mathbb{R}\}$, $\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt$, $M = \operatorname{sp}\{1, t\}$, $f(t) = t^2$. Find $m^* = \operatorname{Proj}_M f$.

$$v_1(t) = 1, \qquad v_2(t) = t$$

$$\underbrace{\begin{bmatrix}\langle 1,1\rangle & \langle t,1\rangle \\ \langle 1,t\rangle & \langle t,t\rangle\end{bmatrix}}_{G}\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} = b = \begin{bmatrix}\langle t^2,1\rangle \\ \langle t^2,t\rangle\end{bmatrix}$$

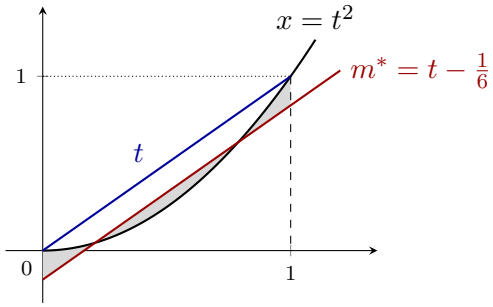
$$\Rightarrow \begin{bmatrix}\int_0^1 dt & \int_0^1 t\,dt \\ \int_0^1 t\,dt & \int_0^1 t^2\,dt\end{bmatrix}\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} = \begin{bmatrix}\int_0^1 t^2\,dt \\ \int_0^1 t^3\,dt\end{bmatrix}$$

$$\Rightarrow \underbrace{\begin{bmatrix}1 & 1/2 \\ 1/2 & 1/3\end{bmatrix}}_{G}\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} = \begin{bmatrix}1/3 \\ 1/4\end{bmatrix}, \qquad \det G = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} = \frac{1}{\frac{1}{12}}\begin{bmatrix}1/3 & -1/2 \\ -1/2 & 1\end{bmatrix}\begin{bmatrix}1/3 \\ 1/4\end{bmatrix} = \begin{bmatrix}-1/6 \\ 1\end{bmatrix}$$

$$\boxed{m^* = \operatorname{Proj}_M f = -\frac{1}{6} + t}$$

$$\|f(t)\| = \sqrt{\int_0^1 |f(t)|^2\,dt}, \qquad \min_{m\in M}\left\|\underbrace{t^2}_{x} - m(t)\right\| = \min_{m(t)\in M}\sqrt{\int_0^1 |t^2 - m(t)|^2\,dt}$$



▷ *Shaded: the gap between t^2 and its best approximation m^* in* $\operatorname{sp}\{1, t\}$.

Checking the answer

The error $e(t) = t^2 - t + \frac{1}{6}$ must be orthogonal to 1 and t:

$$\langle e, 1\rangle = \int_0^1 (t^2 - t + \tfrac{1}{6})\,dt = \tfrac{1}{3} - \tfrac{1}{2} + \tfrac{1}{6} = 0,$$
$$\langle e, t\rangle = \int_0^1 (t^3 - t^2 + \tfrac{1}{6}t)\,dt = \tfrac{1}{4} - \tfrac{1}{3} + \tfrac{1}{12} = 0. \quad \checkmark$$

Since $e \perp m^*$, the minimum error is $\|e\|^2 = \langle e, t^2\rangle = \frac{1}{5} - \frac{1}{4} + \frac{1}{18} = \frac{1}{180}$, i.e. $\min_{m\in M}\|t^2 - m\| = \frac{1}{6\sqrt{5}} \approx 0.075$. So $t - \frac{1}{6}$ is the best straight-line (least-squares) fit to t^2 on $[0,1]$; it crosses t^2 at $t = \frac{1}{2} \pm \frac{1}{2\sqrt{3}}$.

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5.4 Computing Projections: the Gram Matrix and Normal Equations

General space. $\mathcal{V}$, n dimensional, $\quad M = \mathrm{sp}\{\underbrace{v_1, v_2, \ldots, v_k}_{\text{linearly indep.}}\}, \quad k < n$

$$m^* = \sum_{i=1}^{k} \alpha_i^* v_i, \qquad x - m^* = x - \sum_{i=1}^{k} \alpha_i^* v_i \;\perp\; v_1, v_2, \ldots, v_k$$

$$\Big\langle x - \sum_{i=1}^{k} \alpha_i^* v_i,\, v_j \Big\rangle = 0, \qquad j = 1, 2, \ldots, k$$
$$\langle x, v_j\rangle - \sum_{i=1}^{k} \alpha_i^* \langle v_i, v_j\rangle = 0$$

$$\underbrace{\begin{bmatrix} \langle v_1, v_1\rangle & \cdots & \langle v_k, v_1\rangle \\ \vdots & \ddots & \vdots \\ \langle v_1, v_k\rangle & \cdots & \langle v_k, v_k\rangle \end{bmatrix}}_{G} \underbrace{\begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_k^* \end{bmatrix}}_{\alpha^*} = \underbrace{\begin{bmatrix} \langle x, v_1\rangle \\ \vdots \\ \langle x, v_k\rangle \end{bmatrix}}_{b}$$

G is invertible iff $v_1, \ldots, v_k$ are linearly indep. $\qquad \alpha^* = G^{-1}b$

▷ *These k equations are the **normal equations**; G is the **Gram matrix**. G is Hermitian and positive definite for independent v_i: $\beta^H G\beta = \|\sum_i \beta_i v_i\|^2 > 0$ for $\beta \neq 0$ (written for $\mathcal{F} = \mathbb{C}$ with $G_{ji} = \langle v_i, v_j\rangle$).*

Specific case: orthogonal basis. $v_i \perp v_j$ when $i \neq j$: $\quad \{v_1, \ldots, v_k\}$ are orthogonal.

$$\begin{bmatrix} \langle v_1, v_1\rangle & & & 0 \\ & \langle v_2, v_2\rangle & & \\ & & \ddots & \\ 0 & & & \langle v_k, v_k\rangle \end{bmatrix} \begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_k^* \end{bmatrix} = \begin{bmatrix} \langle x, v_1\rangle \\ \vdots \\ \langle x, v_k\rangle \end{bmatrix}$$

$$G^{-1}b = \begin{bmatrix} \dfrac{1}{\langle v_1, v_1\rangle} & & & 0 \\ & & \ddots & \\ 0 & & & \dfrac{1}{\langle v_k, v_k\rangle} \end{bmatrix} \begin{bmatrix} \langle x, v_1\rangle \\ \vdots \\ \langle x, v_k\rangle \end{bmatrix} = \begin{bmatrix} \dfrac{\langle x, v_1\rangle}{\langle v_1, v_1\rangle} \\ \vdots \\ \dfrac{\langle x, v_k\rangle}{\langle v_k, v_k\rangle} \end{bmatrix}$$

$$\alpha_i^* = \frac{\langle x, v_i\rangle}{\langle v_i, v_i\rangle}, \qquad \boxed{m^* = \sum_{i=1}^{k} \frac{\langle x, v_i\rangle}{\langle v_i, v_i\rangle} \cdot v_i}$$

For an orthonormal set ($v_i \perp v_j$, $i \neq j$, and $\|v_i\| = 1$):

$$\boxed{m^* = \sum_{i=1}^{k} \langle x, v_i\rangle \; v_i}$$

Example 5.13. $\mathcal{V} = \mathbb{R}^{2\times 2}$, $\quad M = \{\text{set of symmetric matrices}\}$

$$x = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad m^* = \operatorname{Proj}_M x, \qquad \langle A, B\rangle = \operatorname{tr}(A^T B)$$

$$\operatorname{tr}\left(\begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}\right) = \operatorname{tr}\left(\begin{bmatrix} a_{11}b_{11} + a_{21}b_{21} & \times \\ \times & a_{12}b_{12} + a_{22}b_{22} \end{bmatrix}\right)$$
$$= a_{11}b_{11} + a_{21}b_{21} + a_{12}b_{12} + a_{22}b_{22}$$

$$\|A\| = \sqrt{\operatorname{tr}(A^T A)} = \sqrt{\langle A, A\rangle}$$

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$$\min_{S \text{ sym.}} \left(\operatorname{tr}\left[(x-S)^T(x-S)\right]\right)^{1/2}$$

$$M = \operatorname{sp}\left\{\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{v_2}, \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{v_3}\right\}$$

▷ *v_1, v_2, v_3 are mutually orthogonal (no common nonzero entries), so the orthogonal-basis formula $\alpha_i^* = \langle x, v_i\rangle \,/\, \langle v_i, v_i\rangle$ applies.*

$$\begin{aligned} \langle x, v_1\rangle &= a, & \langle v_1, v_1\rangle &= 1 \\ \langle x, v_2\rangle &= b + c, & \langle v_2, v_2\rangle &= 2 \\ \langle x, v_3\rangle &= d, & \langle v_3, v_3\rangle &= 1 \end{aligned}$$

$$m^* = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{b+c}{2}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & \frac{b+c}{2} \\ \frac{b+c}{2} & d \end{bmatrix} = \frac{x + x^T}{2}$$

$$\operatorname{Proj}_{M^\perp} x = x - m^* = x - \left(\frac{x + x^T}{2}\right) = \frac{x - x^T}{2}$$

$$\left.\begin{aligned} M &\to 3 \text{ dim.} \\ M^\perp &\to 1 \text{ dim} \end{aligned}\right\} \; \mathcal{V} \Rightarrow 4 \text{ dim}$$

Explanation

$M^\perp$ is the set of **skew-symmetric** matrices, $M^\perp = \operatorname{sp}\left\{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right\}$: for $S = S^T$ and $K = -K^T$, $\langle K, S\rangle = \operatorname{tr}(S^T K) = \operatorname{tr}(SK) = \operatorname{tr}\left((SK)^T\right) = \operatorname{tr}(K^T S) = -\operatorname{tr}(KS) = -\operatorname{tr}(SK)$, so $\langle K, S\rangle = 0$. Thus every square matrix splits uniquely as $x = \underbrace{\frac{x+x^T}{2}}_{\text{symmetric}} + \underbrace{\frac{x-x^T}{2}}_{\text{skew-symmetric}}$, and $\frac{x+x^T}{2}$ is the symmetric matrix closest to x in the Frobenius norm $\sqrt{\operatorname{tr}(A^T A)}$.

5.5 Orthogonal Projections in $\mathbb{C}^n$: Projection Matrices

$\mathcal{V} = \mathbb{C}^n, \quad M = \text{sp}\{m_1, \dots, m_k\}, \quad m_i = [\ \ ] \in \mathbb{C}^n$ (column vectors), $\quad x \in \mathbb{C}^n$.

$$m^* = \text{Proj}_M x = ?$$
$$m^* = P_M \cdot x \quad \Rightarrow \quad P_M\text{: } \textbf{projection matrix} \text{ onto } M.$$

$$m^* = \alpha_1^* m_1 + \alpha_2^* m_2 + \cdots + \alpha_k^* m_k = \underbrace{[\, m_1 \vdots m_2 \vdots \cdots \vdots m_k \,]}_{B} \underbrace{\begin{bmatrix} \alpha_1^* \\ \alpha_2^* \\ \vdots \\ \alpha_k^* \end{bmatrix}}_{\alpha^*}$$

$$m^* = B\alpha^*, \qquad x - B\alpha^* \perp m_1, m_2, \dots, m_k, \qquad \langle x, y \rangle = y^H x$$

$$\begin{aligned} \langle x - B\alpha^*, m_j \rangle = 0 \quad \Leftrightarrow \quad & m_j^H (x - B\alpha^*) = 0 \quad \forall j \\ & m_1^H (x - B\alpha^*) = 0 \\ & \vdots \\ & m_k^H (x - B\alpha^*) = 0 \end{aligned}$$

$$\underbrace{\begin{bmatrix} m_1^H \\ \hline m_2^H \\ \hline \vdots \\ \hline m_k^H \end{bmatrix}}_{B^H} (x - B\alpha^*) = 0 \qquad \Longrightarrow \qquad \begin{aligned} B^H (x - B\alpha^*) &= 0 \\ B^H x - B^H B\alpha^* &= 0 \\ \underbrace{B^H B}_{G}\, \alpha^* &= \underbrace{B^H x}_{b} \end{aligned}$$

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$$\underbrace{\begin{bmatrix} m_1^H \\ \vdots \\ m_k^H \end{bmatrix}}_{B^H} \underbrace{[\, m_1 \vdots m_2 \vdots \cdots \vdots m_k \,]}_{B} = \begin{bmatrix} m_1^H m_1 & \cdots & m_1^H m_k \\ \vdots & \ddots & \vdots \\ m_k^H m_1 & \cdots & m_k^H m_k \end{bmatrix} = \underbrace{\begin{bmatrix} \langle m_1, m_1 \rangle & \cdots & \langle m_k, m_1 \rangle \\ \vdots & \ddots & \vdots \\ \langle m_1, m_k \rangle & \cdots & \langle m_k, m_k \rangle \end{bmatrix}}_{G}$$

$$B^H x = \begin{bmatrix} m_1^H x \\ m_2^H x \\ \vdots \\ m_k^H x \end{bmatrix} = \begin{bmatrix} \langle x, m_1 \rangle \\ \langle x, m_2 \rangle \\ \vdots \\ \langle x, m_k \rangle \end{bmatrix} = b$$

▷ *So $B^H B\, \alpha^* = B^H x$ is exactly the Gram system $G\alpha^* = b$ of the general case, with $\langle x, y \rangle = y^H x$.*

$$\alpha^* = (B^H B)^{-1} B^H x$$
$$m^* = B\alpha^*$$
$$m^* = \text{Proj}_M x = B(B^H B)^{-1} B^H x$$

$$\boxed{P_M = B(B^H B)^{-1} B^H}, \qquad m^* = P_M x$$

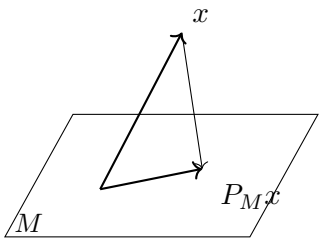
$$P_{M^\perp} = I - P_M$$
$$\text{Proj}_{M^\perp} x = x - \text{Proj}_M x$$

$$= x - P_M x = (I - P_M)x$$

$$P_M^H = \left(B(B^HB)^{-1}B^H\right)^H = B\left((B^HB)^{-1}\right)^H B^H = B(B^HB)^{-1}B^H = P_M$$
$$\Rightarrow \; P_M^H = P_M$$
$$P_M^2 = P_M \cdot P_M$$
$$P_M^2 = B(B^HB)^{-1}\cancel{B^HB}\cancel{(B^HB)^{-1}}B^H = B(B^HB)^{-1}B^H = P_M$$

▷ $(B^HB)^{-1}$ *is Hermitian because* B^HB *is:* $\left((B^HB)^{-1}\right)^H = \left((B^HB)^H\right)^{-1} = (B^HB)^{-1}$.

$$\boxed{P_M^2 = P_M}$$



$$P_M \cdot P_M x = P_M x$$
$$\underline{P_M^2 x = P_M x}$$

Remark. An **orthogonal projection matrix** $P \in \mathbb{C}^{n\times n}$ satisfies

i-) $P^H = P$
ii-) $P^2 = P \quad (P^n = P)$

Also,

$$(I-P)^2 = I - P$$
$$(I-P)(I-P) = I - \cancel{P} - P + \overbrace{P^2}^{\cancel{P}}$$
$$= I - P$$

Remarks on P_M

- Conversely, **any P with $P^H = P$ and $P^2 = P$ is the orthogonal projection onto** $\mathcal{R}(P)$: $x = Px + (I-P)x$ with $Px \in \mathcal{R}(P)$, and $(I-P)x \perp \mathcal{R}(P)$ because $(Py)^H(I-P)x = y^H(P-P^2)x = 0$ for all y.
- P_M **depends only on** M, not on the chosen basis: if $B' = BT$ with T invertible, then $B'(B'^HB')^{-1}B'^H = BT\,T^{-1}(B^HB)^{-1}T^{-H}\,T^HB^H = P_M$.
- If the columns of B are orthonormal, $B^HB = I$ and $P_M = BB^H = \sum_i m_i m_i^H$, which is the matrix form of $m^* = \sum_i \langle x, m_i\rangle\, m_i$.
- $P_M m = m$ for $m \in M$ and $P_M y = 0$ for $y \in M^\perp$ (since $B^Hy = 0$); hence $\mathcal{R}(P_M) = M$, $\mathcal{N}(P_M) = M^\perp$.

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Example 5.14. $\mathcal{V} = \mathbb{R}^3$, $\quad M = \operatorname{sp}\left\{ \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\\1\end{bmatrix} \right\}$, $\quad x = \begin{bmatrix}1\\0\\0\end{bmatrix}$. $\quad$ Find $\operatorname{Proj}_M x = ?$

$$B = \begin{bmatrix}1 & 1\\1 & -1\\1 & 1\end{bmatrix}, \qquad B^T B = \begin{bmatrix}1 & 1 & 1\\1 & -1 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\1 & -1\\1 & 1\end{bmatrix},$$

$$B^H B = \begin{bmatrix}3 & 1\\1 & 3\end{bmatrix}, \qquad (B^H B)^{-1} = \begin{bmatrix}3 & -1\\-1 & 3\end{bmatrix}\cdot\frac{1}{8}$$

$$\begin{aligned}\operatorname{Proj}_M x &= B(B^H B)^{-1}B^H x = P_M x\\ &= \frac{1}{8}\begin{bmatrix}1 & 1\\1 & -1\\1 & 1\end{bmatrix}\begin{bmatrix}3 & -1\\-1 & 3\end{bmatrix}\underbrace{\begin{bmatrix}1 & 1 & 1\\1 & -1 & 1\end{bmatrix}\begin{bmatrix}1\\0\\0\end{bmatrix}}_{\begin{bmatrix}1\\1\end{bmatrix}}\\ &= \frac{1}{8}\begin{bmatrix}1 & 1\\1 & -1\\1 & 1\end{bmatrix}\underbrace{\begin{bmatrix}3 & -1\\-1 & 3\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}}_{\begin{bmatrix}2\\2\end{bmatrix}}\\ &= \frac{1}{8}\cdot\begin{bmatrix}4\\0\\4\end{bmatrix} = \begin{bmatrix}1/2\\0\\1/2\end{bmatrix}\end{aligned}$$

$$\begin{bmatrix}1\\0\\0\end{bmatrix} - \begin{bmatrix}1/2\\0\\1/2\end{bmatrix} = \begin{bmatrix}1/2\\0\\-1/2\end{bmatrix}$$

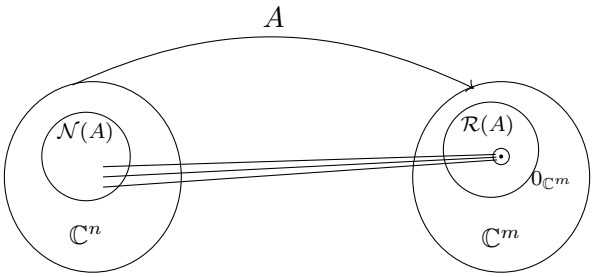
▷ *should be orthogonal to basis vectors (sağlaması için kullanılabilir — can be used to check the result).*
▷ *Check:* $\begin{bmatrix}1 & 1 & 1\end{bmatrix}\begin{bmatrix}1/2\\0\\-1/2\end{bmatrix} = 0$ *and* $\begin{bmatrix}1 & -1 & 1\end{bmatrix}\begin{bmatrix}1/2\\0\\-1/2\end{bmatrix} = 0$ ✓. *The full matrix is* $P_M = \frac{1}{8}B\begin{bmatrix}3 & -1\\-1 & 3\end{bmatrix}B^T = \frac{1}{2}\begin{bmatrix}1 & 0 & 1\\0 & 2 & 0\\1 & 0 & 1\end{bmatrix}$, *whose first column is indeed* $\begin{bmatrix}1/2\\0\\1/2\end{bmatrix}$.

6 Systems of Linear Equations $Ax = b$

Syllabus: Week 7 — systems of linear equations of the form $Ax = b$

6.1 Existence and Uniqueness of Solutions

$$Ax = b, \qquad A \in \mathbb{C}^{m\times n}, \quad x \in \mathbb{C}^n, \quad b \in \mathbb{C}^m$$



▷ *A maps every vector of $\mathcal{N}(A)$ to $0 \in \mathbb{C}^m$.*

range space: $\mathcal{R}(A) = \{y \in \mathbb{C}^m \mid \text{s.t. } \exists x \in \mathbb{C}^n \text{ satisfying } y = Ax\}$

null space: $\mathcal{N}(A) = \{x \in \mathbb{C}^n \mid Ax = 0_{\mathbb{C}^m}\}$

$$\underbrace{\dim(\mathcal{R}(A))}_{\text{rank}} + \underbrace{\dim(\mathcal{N}(A))}_{\text{nullity}} = n = \#\text{ of columns of } A$$

$$\mathcal{V} = M \overset{\perp}{\oplus} M^{\perp}$$
$$v = v_1 + v_2, \qquad v_1 \in M, \quad v_2 \in M^{\perp}, \qquad v_1, v_2 \text{ are unique}$$
$$v_1 \perp v_2, \qquad \dim M + \dim M^{\perp} = \dim \mathcal{V}$$

1-) Is there an exact solution? $(\mathcal{R}(A))$
2-) If there is, is it unique? $(\mathcal{N}(A))$

▷ *Question 1 is answered by $\mathcal{R}(A) \subseteq \mathbb{C}^m$ (the codomain side), question 2 by $\mathcal{N}(A) \subseteq \mathbb{C}^n$ (the domain side).*

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★ $\underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{A} \underset{\in\mathbb{R}}{x} = \underbrace{\begin{bmatrix}2\\2\end{bmatrix}}_{b}, \qquad b \in \mathcal{R}(A)$

$\Rightarrow \exists$ an exact solution. $\Rightarrow$ The solution is unique.

★ $\underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{A} x = \underbrace{\begin{bmatrix}2\\1\end{bmatrix}}_{b}$ $\Rightarrow$ No exact solution! $b \notin \mathcal{R}(A)$

$$y = Ax = \left[\, a_1 \vdots a_2 \vdots a_3 \vdots \cdots \vdots a_n \,\right] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

$$\mathcal{R}(A) = \text{sp}\{\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n\} \triangleq \textbf{column space of } A$$

$$\mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \begin{bmatrix}2\\1\end{bmatrix} \notin \mathcal{R}(A)$$

$\therefore$ 1-) **There exists an exact solution iff** $b \in \mathcal{R}(A)$**.**

★ $\underbrace{\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix}}_{A} \underset{\in\mathbb{R}^2}{x} = \underbrace{\begin{bmatrix}2\\2\end{bmatrix}}_{b}$: $\quad \mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$ & $b \in \mathcal{R}(A) \Rightarrow \therefore$ $\exists$ an exact solution

$$x_1 = \begin{bmatrix}2\\0\end{bmatrix} \quad \text{or} \quad x_2 = \begin{bmatrix}0\\1\end{bmatrix} \quad \text{or} \quad x_3 = \begin{bmatrix}4\\-1\end{bmatrix} \quad \ldots$$

$$\left.\begin{array}{l} A = \begin{bmatrix}1\\1\end{bmatrix}, \qquad \dim\mathbb{C} = 1 \\ \mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \dim\mathcal{R}(A) = 1 \end{array}\right\} \quad \begin{array}{l} \underbrace{\dim(\mathcal{R}(A))}_{1} + \underbrace{\dim(\mathcal{N}(A))}_{0} = 1 \leftarrow \#\text{ of columns} \\ \mathcal{N}(A) = \{0\}, \qquad \dim(\mathcal{N}(A)) = 0 \end{array}$$

$$A = \begin{bmatrix}1 & 2\\1 & 2\end{bmatrix}, \qquad \mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \dim\mathcal{R}(A) = 1$$

$$\underbrace{\dim\mathcal{R}(A)}_{1} + \underbrace{\dim\mathcal{N}(A)}_{1} = 2, \qquad \mathcal{N}(A) = \text{sp}\left\{\begin{bmatrix}2\\-1\end{bmatrix}\right\}$$

$$x_1 - x_2 = \begin{bmatrix}2\\-1\end{bmatrix}, \qquad x_1 - x_3 = \begin{bmatrix}-2\\1\end{bmatrix}, \qquad x_2 - x_3 = \begin{bmatrix}-4\\2\end{bmatrix} \qquad \text{all inside null space.}$$

$A\hat{x} = b \qquad \hat{x}$ is exact

$$\underset{\in\mathcal{N}(A)}{\hat{x} + m} \;\Rightarrow\; A(\hat{x}+m) = A\hat{x} + \underbrace{\overset{\in\mathcal{N}(A)}{Am}}_{0} = b + 0 = b$$

- if $\hat{x}$ satisfies $A\hat{x} = b$ exactly $\Rightarrow \hat{x} + m$ satisfies $A(\hat{x}+m) = b$ exactly $\quad (m \in \mathcal{N}(A))$;
- if $m \neq 0 \Rightarrow \hat{x} \neq \hat{x} + m$;
- if $\mathcal{N}(A)$ includes nonzero vectors $\Rightarrow$ **infinitely many exact solutions**.

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$\therefore$ 2-) If an exact solution $\hat{x}$ exists, then **it is unique** $\iff \mathcal{N}(A) = \{0\}$ or $\dim\mathcal{N}(A) = 0$.

Explanation

Conversely, if $A\hat{x} = b$ and $A\tilde{x} = b$, then $A(\hat{x} - \tilde{x}) = 0$, i.e. any two exact solutions differ by a vector of $\mathcal{N}(A)$. Hence **the full solution set is the affine set** $\{\hat{x} + m : m \in \mathcal{N}(A)\}$. By rank–nullity, $\mathcal{N}(A) = \{0\}$ iff $\operatorname{rank} A = n$ (the columns of A are linearly independent); this requires $m \geq n$.

6.2 Least-Squares Solutions

★ $\begin{bmatrix}1\\1\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad Ax = b, \qquad b \notin \mathcal{R}(A)$

Least squares problem:

$$\min_{x\in\mathbb{C}^n} \underbrace{\|Ax - b\|^2}_{\text{error}}$$

▷ *The exponent 2 doesn't matter; $\|\cdot\|$ is the induced norm of the inner product.*

$$\hat{x} = \arg\min_{x\in\mathbb{C}^n} \|Ax - b\|$$

▷ $\hat{x}$: ***minimum error solution*** / ***least square solution***.

$$b \in \mathbb{C}^m: \qquad \mathbb{C}^m = M \overset{\perp}{\oplus} M^\perp$$

$$\mathbb{C}^m = \mathcal{R}(A) \overset{\perp}{\oplus} \mathcal{R}(A)^\perp$$

$$b = \underset{\in\mathcal{R}(A)}{b_1} + \underset{\in\mathcal{R}^\perp(A)}{b_2} \;\Rightarrow\; b_1 = \operatorname{Proj}_{\mathcal{R}(A)} b, \quad b_2 = \operatorname{Proj}_{\mathcal{R}^\perp(A)} b \;\Rightarrow\; b_1 \perp b_2$$

$$\begin{aligned}
\|Ax - b\|^2 &= \langle Ax - b, Ax - b\rangle \\
&= \langle (\underset{\in\mathcal{R}(A)}{Ax} - \underset{\in\mathcal{R}(A)}{b_1}) - \underset{\in\mathcal{R}^\perp(A)}{b_2},\ (Ax - b_1) - b_2\rangle \qquad (\mathcal{R}(A) \text{ is a subspace}) \\
&= \langle Ax - b_1, Ax - b_1\rangle \\
&\quad \cancel{- \langle Ax - b_1, b_2\rangle} \\
&\quad \cancel{- \langle b_2, Ax - b_1\rangle} \qquad (\to 0 \text{ because of orthogonality}) \\
&\quad + \langle b_2, b_2\rangle \\
&= \langle Ax - b_1, Ax - b_1\rangle + \langle b_2, b_2\rangle \\
&= \overbrace{\left\| A \underset{\text{find an } x}{\underset{\uparrow}{x}} - \underset{\in\mathcal{R}(A)}{b_1} \right\|^2}^{\text{zero}} + \|b_2\|^2 \geq \|b_2\|^2
\end{aligned}$$

▷ ***The first term can be made zero:*** *$b_1 \in \mathcal{R}(A)$, so some x satisfies $Ax = b_1$ exactly.*

$$\boxed{\min_{x\in\mathbb{C}^n} \|Ax - b\| = \|b_2\| = \left\|\operatorname{Proj}_{\mathcal{R}^\perp(A)} b\right\|}$$

$$A\hat{x} = b_1, \qquad \hat{x} = \arg\min_{x\in\mathbb{C}^n} \|Ax - b\|, \qquad b_1 = \operatorname{Proj}_{\mathcal{R}(A)} b$$

$$\underbrace{\hat{x}}_{\substack{\text{min error /}\\ \text{least squares solution}}} = \arg\min_{x\in\mathbb{C}^n} \|Ax - b\|$$

Normal equations

$A\hat{x} = b_1$ is equivalent to $b - A\hat{x} = b_2 \perp \mathcal{R}(A)$, i.e. $a_j^H(b - A\hat{x}) = 0$ for every column a_j of A:

$$A^H(b - A\hat{x}) = 0 \quad \Longleftrightarrow \quad \boxed{A^H A\,\hat{x} = A^H b}$$

(the same equations as $B^H B\alpha^* = B^H x$ before, with $B = A$). **These normal equations always have a solution.** If the columns of A are linearly independent, $A^H A$ is invertible and $\hat{x} = (A^H A)^{-1}A^H b$ is unique; otherwise there are infinitely many least squares solutions, differing by elements of $\mathcal{N}(A^H A) = \mathcal{N}(A)$.

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$$\begin{bmatrix}1\\1\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad b \notin \mathcal{R}(A), \qquad \mathcal{R}(A) = \operatorname{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$$

$$P_{\mathcal{R}(A)} = B(B^H B)^{-1}B^H, \qquad B = \begin{bmatrix}1\\1\end{bmatrix}, \qquad B^H B = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = 2$$

$$P_{\mathcal{R}(A)} = \begin{bmatrix}1\\1\end{bmatrix}\frac{1}{2}\begin{bmatrix}1 & 1\end{bmatrix} = \begin{bmatrix}1/2 & 1/2\\1/2 & 1/2\end{bmatrix}$$

$$b_1 = P_{\mathcal{R}(A)}b = \begin{bmatrix}1/2 & 1/2\\1/2 & 1/2\end{bmatrix}\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}3/2\\3/2\end{bmatrix}$$

$$\underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{A}\hat{x} = \underbrace{\begin{bmatrix}3/2\\3/2\end{bmatrix}}_{b_1}, \qquad \hat{x} = \frac{3}{2}$$

▷ $\hat{x} = \frac{3}{2}$ *is the average of* $b = \begin{bmatrix}2\\1\end{bmatrix}$.

▷ *Same result from the normal equations:* $A^H A = 2$, $A^H b = 3 \Rightarrow \hat{x} = \frac{3}{2}$; *the minimum error is* $\|b - b_1\| = \left\|\begin{bmatrix}1/2\\-1/2\end{bmatrix}\right\| = \frac{1}{\sqrt{2}}$.

★ $\begin{bmatrix}1\\1\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad \mathcal{R}(A) = \operatorname{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$

$$b_1 = P_{\mathcal{R}(A)}b = \begin{bmatrix}1/2 & 1/2\\1/2 & 1/2\end{bmatrix}\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}3/2\\3/2\end{bmatrix}$$

$$A\hat{x} = b_1, \qquad \hat{x} = 3/2$$
$$\mathcal{N}(A) = \{0\},\ \dim\mathcal{N}(A) = 0 \qquad \Rightarrow\ \hat{x} \textbf{ is unique.}$$

★ $\begin{bmatrix}1 & 2\\1 & 2\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad \mathcal{R}(A) = \operatorname{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad b \notin \mathcal{R}(A)$

$$b_1 = P_{\mathcal{R}(A)}b = \begin{bmatrix}3/2\\3/2\end{bmatrix}$$

$$\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix}\hat{x} = \begin{bmatrix}3/2\\ 3/2\end{bmatrix} \quad \Rightarrow \quad \hat{x} = \begin{bmatrix}3/2\\ 0\end{bmatrix}, \quad \hat{x} = \begin{bmatrix}0\\ 3/4\end{bmatrix}, \quad \hat{x} = \begin{bmatrix}1/2\\ 1/2\end{bmatrix}$$

$$\mathcal{N}(A) = \operatorname{sp}\left\{\begin{bmatrix}2\\ -1\end{bmatrix}\right\} \quad \Rightarrow \quad \hat{x} \textbf{ is not unique!}$$

▷ *All least squares solutions:* $\hat{x} = \begin{bmatrix}3/2\\ 0\end{bmatrix} + t\begin{bmatrix}2\\ -1\end{bmatrix}$, $t \in \mathbb{R}$ *(e.g.* $t = -\frac{3}{4}$ *gives* $\begin{bmatrix}0\\ 3/4\end{bmatrix}$, $t = -\frac{1}{2}$ *gives* $\begin{bmatrix}1/2\\ 1/2\end{bmatrix}$*). Each leaves the same error* $\|b_2\| = \frac{1}{\sqrt{2}}$.

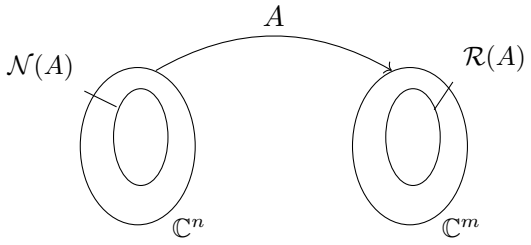
6.3 Minimum-Norm Solutions

Minimum norm solution.

$$b \in \mathcal{R}(A) \;\Rightarrow\; \min_{Ax=b} \|x\|, \qquad \hat{x}_1 = \arg\min_{Ax=b} \|x\|, \qquad x = \hat{x} + \overset{\in\mathcal{N}(A)}{m}$$

$$\min_{m\in\mathcal{N}(A)} \|\hat{x} + m\|, \qquad A\hat{x} = b$$

▷ *When there are infinitely many solutions, pick the one with the smallest norm. Every solution has the form* $\hat{x} + m$ *for one fixed solution* $\hat{x}$*, so we minimize over* $m \in \mathcal{N}(A)$.



$$\mathbb{C}^m = \mathcal{R}(A) \overset{\perp}{\oplus} \mathcal{R}^\perp(A)$$

$$\mathbb{C}^n = \mathcal{N}^\perp(A) \overset{\perp}{\oplus} \mathcal{N}(A)$$

$$\hat{x} = \underset{\in\mathcal{N}^\perp(A)}{\hat{x}_1} + \underset{\in\mathcal{N}(A)}{\hat{x}_2}, \qquad \hat{x}_1 = \operatorname{Proj}_{\mathcal{N}^\perp(A)} \hat{x}, \quad \hat{x}_2 = \operatorname{Proj}_{\mathcal{N}(A)} \hat{x}, \qquad \hat{x}_1 \perp \hat{x}_2$$

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$$\begin{aligned}\|\hat{x} + m\|^2 &= \langle \hat{x} + m, \hat{x} + m\rangle \\ &= \big\langle \underset{\in\mathcal{N}^\perp}{\hat{x}_1} + (\underset{\in\mathcal{N}}{\hat{x}_2} + \underset{\in\mathcal{N}}{m}),\ \underset{\in\mathcal{N}^\perp}{\hat{x}_1} + (\underset{\in\mathcal{N}}{\hat{x}_2} + \underset{\in\mathcal{N}}{m})\big\rangle \\ &= \langle \hat{x}_1, \hat{x}_1\rangle + \langle \hat{x}_2 + m, \hat{x}_2 + m\rangle \\ &= \|\hat{x}_1\|^2 + \|\hat{x}_2 + m\|^2 \\ &\geq \|\hat{x}_1\|^2\end{aligned}$$

▷ *Equality for* $m^* = -\hat{x}_2 \;\Rightarrow\; \|\hat{x}_2 + m\| = 0$.

$$\min_{\substack{Ax=b\\ m\in\mathcal{N}(A)}} \|\hat{x} + m\| = \|\hat{x} - \hat{x}_2\| = \|\hat{x}_1\|$$

$$A\hat{x} = b$$

$$A(\hat{x}_1 + \hat{x}_2) = b$$
$$A\hat{x}_1 + \underbrace{A\hat{x}_2}_{0} = b$$
$$\Rightarrow \quad A\hat{x}_1 = b$$

$$\boxed{\hat{x}_1 = \text{Proj}_{\mathcal{N}^\perp(A)}\,\hat{x} = \text{minimum norm solution}}$$

Summary of the method

- $\hat{x}_1$ does not depend on which solution $\hat{x}$ we start from: two solutions differ by an element of $\mathcal{N}(A)$, which the projection onto $\mathcal{N}^\perp(A)$ removes. So **the minimum norm solution is unique**.
- $\mathcal{N}^\perp(A) = \mathcal{R}(A^H)$ (since $Ax = 0 \iff x \perp$ every row of A, i.e. every column of A^H). Hence the minimum norm solution can be written $\hat{x}_1 = A^H z$, and $AA^H z = b$; if A has linearly independent rows, $\hat{x}_1 = A^H(AA^H)^{-1}b$.
- If $b \notin \mathcal{R}(A)$, combine both steps: replace b by $b_1 = \text{Proj}_{\mathcal{R}(A)}\,b$, take any least squares solution and project it onto $\mathcal{N}^\perp(A)$. The result is the **minimum norm least squares solution** (this is what the pseudo-inverse A^+b computes); the example below does exactly this.

Example 6.1.

$$\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix} x = \begin{bmatrix}2\\ 1\end{bmatrix}, \qquad b_1 = \text{Proj}_{\mathcal{R}(A)}\,b = \begin{bmatrix}3/2\\ 3/2\end{bmatrix}$$

$$\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix}\hat{x} = \begin{bmatrix}3/2\\ 3/2\end{bmatrix} \quad \Rightarrow \quad \hat{x} = \begin{bmatrix}3/2\\ 0\end{bmatrix}$$

$$\mathcal{N}(A) = \text{sp}\left\{\begin{bmatrix}2\\ -1\end{bmatrix}\right\}, \qquad \underbrace{\dim \mathcal{N}(A)}_{1} + \underbrace{\dim \mathcal{N}^\perp(A)}_{=1} = 2$$

$$\mathcal{N}^\perp(A) = \text{sp}\left\{\underbrace{\begin{bmatrix}1\\ 2\end{bmatrix}}_{v}\right\} = \mathcal{R}(A^H)$$

$$P_{\mathcal{N}^\perp(A)} = B(B^HB)^{-1}B^H = \frac{vv^H}{v^Hv} = \frac{\begin{bmatrix}1\\ 2\end{bmatrix}\begin{bmatrix}1 & 2\end{bmatrix}}{\begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}1\\ 2\end{bmatrix}} = \frac{1}{5}\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$$

$$\hat{x}_1 = \frac{1}{5}\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}\begin{bmatrix}3/2\\ 0\end{bmatrix} = \frac{1}{5}\begin{bmatrix}3/2\\ 3\end{bmatrix} = \begin{bmatrix}3/10\\ 3/5\end{bmatrix}$$

Checking the answer

- $A\hat{x}_1 = \begin{bmatrix} 3/10+6/5 \\ 3/10+6/5 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix} = b_1$, so $\hat{x}_1$ is a least squares solution.
- $\hat{x}_1 = \frac{3}{10}\begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \mathcal{N}^\perp(A)$, i.e. $\hat{x}_1 \perp \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.
- Starting from $\begin{bmatrix} 0 \\ 3/4 \end{bmatrix}$ or $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ instead gives the same $\hat{x}_1$, e.g. $\frac{1}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 0 \\ 3/4 \end{bmatrix} = \begin{bmatrix} 3/10 \\ 3/5 \end{bmatrix}$.
- Norms: $\|\hat{x}_1\|^2 = \frac{9}{20} = 0.45$, smaller than $\left\|\begin{bmatrix} 3/2 \\ 0 \end{bmatrix}\right\|^2 = 2.25$, $\left\|\begin{bmatrix} 0 \\ 3/4 \end{bmatrix}\right\|^2 = 0.5625$, $\left\|\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}\right\|^2 = 0.5$.

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6.4 General Procedure and the Fundamental Theorem

- If $b \in \mathcal{R}(A)$: $\exists$ **an exact solution** to $A\hat{x} = b$.
- If $b \notin \mathcal{R}(A)$:

$$b_1 = \text{Proj}_{\mathcal{R}(A)}\, b, \qquad A\hat{x} = b_1$$

  - If $\mathcal{N}(A) = \{0\} \Rightarrow \hat{x}$ **is unique**.
  - If $\mathcal{N}(A) \neq \{0\} \Rightarrow \hat{x}_1 = \text{Proj}_{\mathcal{N}^\perp(A)}\, \hat{x}$

$\hat{x}$: **minimum error solution** (least squares)
$\hat{x}_1$: **minimum error-norm solution**

Why the procedure works

- The error $b - A\hat{x}$ is smallest when $A\hat{x}$ is the point of $\mathcal{R}(A)$ closest to b, i.e. $A\hat{x} = b_1 = \text{Proj}_{\mathcal{R}(A)}\, b$ (orthogonality principle: $b - b_1 \perp \mathcal{R}(A)$). Writing this as $A^H(b - A\hat{x}) = 0$ gives the **normal equations** $A^H A\hat{x} = A^H b$.
- If $\mathcal{N}(A) \neq \{0\}$, every $\hat{x} + n$ with $n \in \mathcal{N}(A)$ gives the same $A\hat{x} = b_1$, so the least-squares solution is not unique. Splitting $\hat{x} = \hat{x}_1 + \hat{x}_2$ with $\hat{x}_1 \in \mathcal{N}^\perp(A)$, $\hat{x}_2 \in \mathcal{N}(A)$ gives $A\hat{x}_1 = A\hat{x} = b_1$ and, by Pythagoras, $\|\hat{x} + n\|^2 = \|\hat{x}_1\|^2 + \|\hat{x}_2 + n\|^2 \geq \|\hat{x}_1\|^2$. Hence $\hat{x}_1$ **is the least-squares solution of minimum norm**, and it does not depend on which $\hat{x}$ we started from.

Example 6.2. $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$, $\quad n = 3,\ m = 2 \quad$ ($A \in \mathbb{C}^{m\times n}$: n columns, m rows).

$$\mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}, \qquad \dim(\mathcal{R}(A)) = 1$$

$$\mathcal{N}(A) = \text{sp}\left\{\begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}\right\}$$

$$\underbrace{\dim(\mathcal{R}(A))}_{1} + \underbrace{\dim(\mathcal{N}(A))}_{2} = 3, \qquad \underbrace{\dim(\mathcal{N}(A))}_{2} + \underbrace{\dim(\mathcal{N}^\perp(A))}_{1} = 3$$

For $\mathcal{N}^\perp(A)$, a vector $[a\ b\ c]^T$ orthogonal to both basis vectors of $\mathcal{N}(A)$:

$$\begin{matrix} 3a-c=0 \\ 2a-b=0 \end{matrix} \quad \Rightarrow \quad \begin{matrix} c=3a \\ b=2a \end{matrix} \quad \Rightarrow \quad \begin{bmatrix} a \\ b \\ c \end{bmatrix} = a \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{\text{row of } A} \quad \Rightarrow \quad \mathcal{N}^\perp(A) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$$

$$\mathcal{N}^\perp(A) = \mathcal{R}(A^H)$$

▷ $\therefore$ ***Fundamental theorem of linear algebra.***

$$\hat{x}_1 = \text{Proj}_{\mathcal{N}^\perp(A)}\, \hat{x} = \text{Proj}_{\mathcal{R}(A^H)}\, \hat{x}$$

▷ $\mathcal{R}(A^H)$*: row space of* A.

Dimensions and rank. For $A \in \mathbb{C}^{m\times n}$:

$$\dim(\underbrace{\mathcal{R}(A)}_{k}) + \dim(\underbrace{\mathcal{N}(A)}_{n-k}) = \#\text{ of columns} = n$$

$$\dim(\underbrace{\mathcal{N}(A)}_{n-k}) + \dim(\underbrace{\mathcal{R}(A^H)}_{k}) = n \qquad \big(\dim(\mathcal{N}(A)) + \dim(\mathcal{N}^\perp(A)) = n\big)$$

$$\left.\begin{matrix} \dim(\mathcal{R}(A)) = \#\text{ of lin. indep. columns of } A = \text{column rank} = k \\ \dim(\mathcal{R}(A^H)) = \#\text{ of lin. indep. rows of } A = \text{row rank} = k \end{matrix}\right\} \; k = \textbf{rank} \text{ of a matrix}$$

$$\#\text{ linearly indep. rows} = \#\text{ linearly indep. columns} \quad \text{as} \quad \dim \mathcal{R}(A) = k = \dim \mathcal{R}(A^H)$$

k: **# of linearly indep. rows = # of linearly indep. columns** (column rank = row rank).

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Theorem 6.3 (Fundamental theorem of linear algebra)**.** $\mathcal{R}(A^H) = \mathcal{N}^\perp(A)$

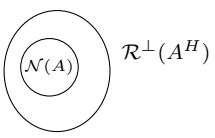
Proof. $\langle x, y\rangle \triangleq y^H x$.

$$\left\langle \underset{\mathbb{C}^m}{Ax}, \underset{\mathbb{C}^m}{y} \right\rangle_{\mathbb{C}^m} = y^H A x = (A^H y)^H x = \left\langle \underset{\mathbb{C}^n}{x}, \underset{\mathbb{C}^n}{A^H y} \right\rangle_{\mathbb{C}^n}$$

Prove $\mathcal{N}(A) = \mathcal{R}^\perp(A^H)$, which is easier to prove.

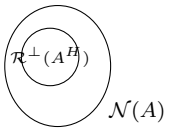
$$\left.\begin{matrix} \mathcal{N}(A) \subset \mathcal{R}^\perp(A^H) \\ \mathcal{R}^\perp(A^H) \subset \mathcal{N}(A) \end{matrix}\right\} \text{ prove both } \Rightarrow \; \mathcal{N}(A) = \mathcal{R}^\perp(A^H)$$

Part 1: $x \in \mathcal{N}(A) \Rightarrow Ax = 0$.



$$\begin{aligned} &\langle Ax, y\rangle \quad \text{where } \forall y \in \mathbb{C}^m \\ &\langle Ax, y\rangle = 0 \\ &\left\langle x, A^H y\right\rangle = 0 \quad \forall y \in \mathbb{C}^m \\ \Rightarrow\ & x \perp A^H y \quad \forall y \in \mathbb{C}^m \\ \Rightarrow\ & x \perp \mathcal{R}(A^H) \Rightarrow x \in \mathcal{R}^\perp(A^H) \end{aligned}$$

Part 2: $x \in \mathcal{R}^\perp(A^H)$.

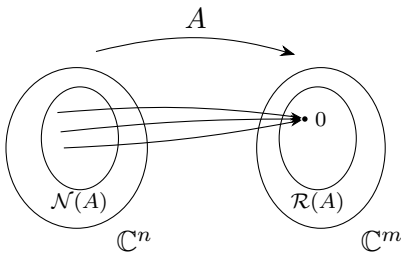


$$
\begin{aligned}
& x \in \mathcal{R}^\perp(A^H) \\
\Rightarrow\ & x \perp A^H y \quad \forall y \in \mathbb{C}^m \\
& \left\langle x, A^H y \right\rangle = 0 \quad \forall y \in \mathbb{C}^m \\
& \langle Ax, y \rangle = 0 \quad \forall y \in \mathbb{C}^m
\end{aligned}
$$

▷ *$Ax \in \mathbb{C}^m$, so choose $y = Ax$:*

$$\langle Ax, Ax \rangle = \|Ax\|^2 = 0 \Rightarrow Ax = 0 \Rightarrow x \in \mathcal{N}(A)$$

Both inclusions hold, so $\mathcal{N}(A) = \mathcal{R}^\perp(A^H)$. Taking orthogonal complements of both sides (in finite dimensions $(S^\perp)^\perp = S$ for a subspace S) gives $\mathcal{N}^\perp(A) = \mathcal{R}(A^H)$. □



$$
\begin{aligned}
& \overset{k}{\dim(\mathcal{R}(A))} + \overset{n-k}{\dim(\mathcal{N}(A))} = n \\
& \dim(\mathcal{N}(A)) + \dim(\mathcal{N}^\perp(A)) = n \\
& \underbrace{\dim(\mathcal{R}(A^H))}_{k} + \underbrace{\dim(\mathcal{N}(A))}_{n-k} = n \\
& \underbrace{\dim(\mathcal{R}(A))}_{\text{column rank}} = \underbrace{\dim(\mathcal{R}(A^H))}_{\text{row rank}} = k
\end{aligned}
$$

▷ *k: rank.*

6.5 Special Cases by Rank

6.5.1 Case I: Full Column Rank

A is/has **full column rank**: columns of A are linearly independent.

$$m\Big\{ \underbrace{\overset{n}{\begin{bmatrix} & \\ & \end{bmatrix}}}_{\text{tall matrix}} \overset{n}{\begin{bmatrix} \\ \end{bmatrix}} = \begin{bmatrix} \\ \end{bmatrix} \Big\} m \qquad m \geq n$$

- $n = \#$ of unknowns, $\quad m = \#$ of equations
- $m > n$: **overdetermined** system of eqn.

Assume

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

3 vectors cannot be linearly indep. for $\mathbb{R}^2$.

▷ *Under the columns: "2 dim", "2 dimension". Each column lies in $\mathbb{R}^2$, so at most 2 of them can be linearly independent; full column rank therefore forces $m \geq n$.*

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$$
\begin{aligned}
& \dim(\mathcal{R}(A)) = n \\
& \dim(\mathcal{N}(A)) = 0 \Rightarrow \mathcal{N}(A) = \{0\} \qquad A \text{ is one-to-one}
\end{aligned}
$$

If we have an exact solution, then the solution is unique.

If $b \notin \mathcal{R}(A)$:

$$P_{\mathcal{R}(A)} = B(B^H B)^{-1} B^H$$

with $B = A$:

$$\begin{aligned} P_{\mathcal{R}(A)} &= A(A^H A)^{-1} A^H \\ b_1 = P_{\mathcal{R}(A)} b &= A(A^H A)^{-1} A^H b \qquad (A\hat{x} = b_1) \\ \Rightarrow \quad A(\underbrace{\hat{x} - (A^H A)^{-1} A^H b}_{\in \mathcal{N}(A)}) &= 0 \\ \Rightarrow \quad \hat{x} - (A^H A)^{-1} A^H b = 0 \;\Rightarrow\; \boxed{\hat{x} = (A^H A)^{-1} A^H b} \end{aligned}$$

▷ *Unique solution as* $\mathcal{N}(A) = \{0\}$.

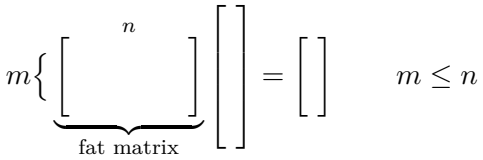
$(A^H A)^{-1} A^H$: **left pseudo inverse** of A.

Explanation

- $A^H A$ $(n \times n)$ **is invertible exactly when** A **has full column rank**: $A^H Ax = 0 \Rightarrow x^H A^H Ax = \|Ax\|^2 = 0 \Rightarrow Ax = 0 \Rightarrow x = 0$.
- The same formula comes directly from the normal equations $A^H A\hat{x} = A^H b$.
- "Left" inverse: $((A^H A)^{-1} A^H)A = I_n$. If $b \in \mathcal{R}(A)$, it returns the exact solution.

6.5.2 Case II: Full Row Rank

A is/has **full row rank**: all rows of A are linearly indep.



$m \leq n$

- m: # of equations, n: # of unknowns
- $m < n$: **underdetermined** system of eqn.

$$\begin{aligned} \dim(\mathcal{R}(A^H)) &= m \\ \dim(\mathcal{R}(A)) = m \;&\Rightarrow\; \mathcal{R}(A) = \mathbb{C}^m \end{aligned}$$

$b \in \mathcal{R}(A) \quad \forall b \in \mathbb{C}^m \Rightarrow$ **Always has an exact solution** (at least one): A is onto.

$$\hat{x}_1 = \text{Proj}_{\mathcal{N}^\perp(A)}\, \hat{x} = \text{Proj}_{\mathcal{R}(A^H)}\, \hat{x}$$

with $B = A^H$:

$$\begin{aligned} P_{\mathcal{R}(A^H)} &= A^H(AA^H)^{-1} A \\ \hat{x}_1 = A^H(AA^H)^{-1} \underbrace{A\hat{x}}_{b} \;&\Rightarrow\; \hat{x}_1 = A^H(AA^H)^{-1} b \end{aligned}$$

$A^H(AA^H)^{-1}$: **right pseudo inverse** of A.

Explanation

- AA^H $(m \times m)$ is invertible since the rows of A are independent (same argument as for $A^H A$, applied to A^H).
- "Right" inverse: $A(A^H(AA^H)^{-1}) = I_m$, so $A\hat{x}_1 = b$ is indeed an exact solution.
- $\hat{x}_1 = A^H((AA^H)^{-1}b) \in \mathcal{R}(A^H) = \mathcal{N}^\perp(A)$, so **it is the minimum-norm one** among the infinitely many exact solutions $\hat{x}_1 + \mathcal{N}(A)$.

6.5.3 Case III: Full Column and Full Row Rank

A is/has both full column and row rank.

$$\left.\begin{matrix} m \geq n \\ n \geq m \end{matrix}\right\} \; m = n: \quad \left.\begin{matrix} A \text{ is a square matrix} \\ A \text{ is one-to-one} \\ A \text{ is onto} \end{matrix}\right\} \; \begin{matrix} A \text{ has an inverse} \\ A \text{ is invertible.} \end{matrix}$$

▷ *Both pseudo inverses reduce to A^{-1} here: $(A^HA)^{-1}A^H = A^{-1}(A^H)^{-1}A^H = A^{-1}$ and $A^H(AA^H)^{-1} = A^{-1}$.*

Exact solution exists. The solution is unique.

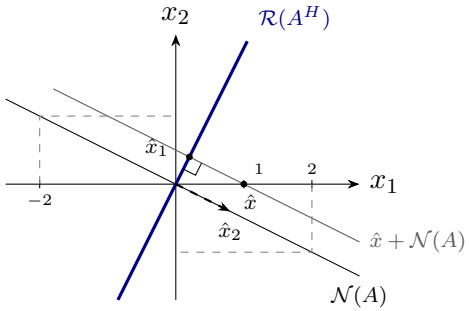
$$Ax = b, \qquad x_1 = \hat{x} = x = A^{-1}b$$

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Example 6.4.

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{b}, \qquad \mathcal{R}(A) = \operatorname{sp}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}, \quad b \in \mathcal{R}(A) \;\Rightarrow\; \text{has exact solution}, \quad \hat{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\dim \mathcal{R}(A) = 1$$
$$\dim \mathcal{N}(A) = 1$$
$$\mathcal{N}(A) = \operatorname{sp}\left\{\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right\}$$
$$\hat{x} + n = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad n \in \mathcal{N}(A)$$



▷ *In the sketch: $\mathcal{R}(A^H) = \operatorname{sp}\{[1\ 2]^T\}$; the solution set $\hat{x} + \mathcal{N}(A)$ is the line through $\hat{x} = [1\ 0]^T$ parallel to $\mathcal{N}(A)$; $\hat{x}_1 = \frac{1}{5}[1\ 2]^T$ is its point on $\mathcal{R}(A^H)$, and $\hat{x}_2 = \hat{x} - \hat{x}_1 \in \mathcal{N}(A)$ is the part to "get rid of".*

▷ *Check: $\hat{x}_1 = \frac{1}{5}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ gives $A\hat{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = b$, and $\|\hat{x}_1\|^2 = \frac{1}{5} < 1 = \|\hat{x}\|^2$; the right pseudo inverse cannot be used here since $AA^H = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$ is singular (A has rank 1, neither full row nor full column rank), so the projection route is used.*

6.5.4 Case IV: Neither Full Column nor Full Row Rank

A is/has neither full column rank nor full row rank.

Example 6.5.

$$\underbrace{\begin{bmatrix}1&0&1\\1&1&2\\0&1&1\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}}_{x}=\underbrace{\begin{bmatrix}1\\0\\3\end{bmatrix}}_{b}$$

2 linearly indep. columns
2 linearly indep. rows
$\dim(\mathcal{R}(A))=\dim(\mathcal{R}(A^H))=2$
$\dim(\mathcal{N}(A))=1$
$\dim(\mathcal{N}^\perp(A))=2$

$$\begin{bmatrix}1\\0\\3\end{bmatrix}=\underset{\alpha}{1}\begin{bmatrix}1\\1\\0\end{bmatrix}+\underset{\beta}{3}\begin{bmatrix}0\\1\\1\end{bmatrix}\quad\times\qquad\Rightarrow\qquad b\notin\mathcal{R}(A)\Rightarrow\textbf{No exact solution.}$$

▷ *Column 3 = column 1 + column 2, so* $\mathcal{R}(A)=\mathrm{sp}\{[1\ 1\ 0]^T,[0\ 1\ 1]^T\}$. *Matching the 1st and 3rd entries of b forces* $\alpha=1$, $\beta=3$, *but then the 2nd entry would be* $\alpha+\beta=4\neq 0$.

Method I.

$$P_{\mathcal{R}(A)}=B(B^HB)^{-1}B^H$$

$$B=\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix},\qquad B^HB=\begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}=\begin{bmatrix}2&1\\1&2\end{bmatrix}$$

$$(B^HB)^{-1}=\frac{1}{3}\begin{bmatrix}2&-1\\-1&2\end{bmatrix}$$

$$P_{\mathcal{R}(A)}=\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}\frac{1}{3}\begin{bmatrix}2&-1\\-1&2\end{bmatrix}\begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}=\frac{1}{3}\begin{bmatrix}2&1&-1\\1&2&1\\-1&1&2\end{bmatrix}$$

$$b_1=P_{\mathcal{R}(A)}b=\frac{1}{3}\begin{bmatrix}-1\\4\\5\end{bmatrix}$$

$$\underbrace{\begin{bmatrix}1&0&1\\1&1&2\\0&1&1\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}}_{\hat{x}}=\underbrace{\frac{1}{3}\begin{bmatrix}-1\\4\\5\end{bmatrix}}_{b_1}\qquad\Rightarrow\qquad\underset{\text{minimum error solution.}}{\hat{x}=\frac{1}{3}\begin{bmatrix}-1\\5\\0\end{bmatrix}}$$

▷ *This was method I. With* $x_3=0$: $x_1=-\frac{1}{3}$, $x_1+x_2=\frac{4}{3}\Rightarrow x_2=\frac{5}{3}$, *and the 3rd row* $x_2+x_3=\frac{5}{3}$ *checks. The error is* $b-b_1=\frac{4}{3}[1\ -1\ 1]^T$, *which is orthogonal to both columns of B, as it must be.*

Method II.

$$\underbrace{\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}}_{A_1}\underbrace{\begin{bmatrix}x_1\\x_2\end{bmatrix}}_{z}=\begin{bmatrix}1\\0\\3\end{bmatrix}$$

$\Rightarrow\mathcal{N}(A_1)=\{0\}$
$\Rightarrow$ full column rank
$\Rightarrow$ Exact solution is known

▷ *Idea: drop the dependent third column (set* $x_3=0$*); the remaining* A_1 *has full column rank and* $\mathcal{R}(A_1)=\mathcal{R}(A)$, *so Case I applies.*

$$\hat{z}=(A_1^HA_1)^{-1}A_1^Hb=\frac{1}{3}\begin{bmatrix}2&-1\\-1&2\end{bmatrix}\begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}1\\0\\3\end{bmatrix}=\underbrace{\frac{1}{3}\begin{bmatrix}-1\\5\end{bmatrix}}_{\hat{z}\ \to\ \text{unique}}\qquad\therefore\ \hat{x}=\frac{1}{3}\begin{bmatrix}-1\\5\\0\end{bmatrix}$$

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Minimum norm solution. $A^H = A^T$ as matrix is real.

$$A^H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

$$P_{\mathcal{R}(A^H)} = B(B^H B)^{-1} B^H, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$(B^H B)^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$P_{\mathcal{R}(A^H)} = \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\hat{x}_1 = P_{\mathcal{R}(A^H)}\hat{x} = \frac{1}{9}\begin{bmatrix} -7 \\ 11 \\ 4 \end{bmatrix}$$

▷ $\therefore \hat{x}_1$***: minimum error, minimum norm solution.***

Check

$A\hat{x}_1 = \frac{1}{9}[-3 \quad 12 \quad 15]^T = \frac{1}{3}[-1 \quad 4 \quad 5]^T = b_1$, so $\hat{x}_1$ has the same (minimum) error as $\hat{x}$. Moreover $\hat{x} - \hat{x}_1 = \frac{4}{9}[1 \quad 1 \quad -1]^T$ spans $\mathcal{N}(A)$ (indeed $A[1\ 1\ -1]^T = 0$), and $\|\hat{x}_1\|^2 = \frac{186}{81} \approx 2.30 < \frac{26}{9} \approx 2.89 = \|\hat{x}\|^2$.

7 Spectral Analysis of Linear Transformations

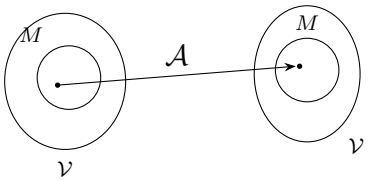
Syllabus: Weeks 8–9 — eigenvalues and eigenvectors, invariance under operators, block diagonal forms

7.1 Invariant Subspaces

$A \qquad \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$: **spectrum** of a matrix

Definition 7.1. Let $\mathcal{A} : \mathcal{V} \to \mathcal{V}$ be a linear transformation defined over a vector space $\mathcal{V}$. A subspace M of $\mathcal{V}$ is said to be **invariant under** $\mathcal{A}$ iff

$$\mathcal{A}x \in M \quad \forall x \in M$$



Example 7.2. $\mathcal{V} = \{f \mid f$ functions defined over $\mathbb{R}$ which are infinitely differentiable$\}$

$$\mathcal{A} = \frac{d}{dt} : \ \mathcal{V} \to \mathcal{V}, \qquad M_1 \triangleq \text{sp}\{\sin t\} \subset \mathcal{V}$$

$$\mathcal{A}(\alpha \sin t) = \alpha \mathcal{A} \sin t = \alpha \cos t \notin M_1$$

$\therefore M_1$ **is not invariant under** $\mathcal{A}$**.**

$$M_2 \triangleq \text{sp}\{\cos t, \sin t\}$$
$$x = \alpha_1 \cos t + \alpha_2 \sin t$$
$$\mathcal{A}x = -\alpha_1 \sin t + \alpha_2 \cos t \in \text{sp}\{\cos t, \sin t\} = M_2$$

$\therefore M_2$ **is invariant under** $\mathcal{A}$**.**

$$M_3 = \{0\}, \qquad \mathcal{A}0 = 0 \in M_3$$
$$M_4 = \text{sp}\{e^{2t}\}, \qquad x = \alpha e^{2t}, \qquad \mathcal{A}x = 2\alpha e^{2t} \in M_4$$

$\therefore M_3$ and M_4 are invariant under $\mathcal{A}$ (e^{2t} is mapped to a multiple of itself: it is an "eigenfunction" of $\frac{d}{dt}$ with eigenvalue 2).

Explanation

The trivial subspaces $\{0\}$ and $\mathcal{V}$ are always invariant. **A one-dimensional subspace** $\text{sp}\{e\}$ **is invariant iff** $\mathcal{A}e = \lambda e$ for some scalar λ, i.e. iff e is an eigenvector — this is the link between invariant subspaces and the eigenvalues defined below.

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Example 7.3. $\mathcal{N}(\mathcal{A})$ & $\mathcal{R}(\mathcal{A})$ are invariant under $\mathcal{A}$.

$\underline{\mathcal{N}(\mathcal{A})}$: $x \in \mathcal{N}(\mathcal{A})$

$$\mathcal{A}x = 0_{\mathcal{V}} \in \mathcal{N}(\mathcal{A})$$

$\mathcal{N}(\mathcal{A})$ is invariant under $\mathcal{A}$.

$\underline{\mathcal{R}(\mathcal{A})}$: $x \in \mathcal{R}(\mathcal{A})$

$$\mathcal{A}x \in \mathcal{R}(\mathcal{A})$$

$\mathcal{R}(\mathcal{A})$ is invariant under $\mathcal{A}$.

7.2 Powers and Polynomials of a Linear Transformation

Definition 7.4. **Powers** of a linear transformation $\mathcal{A} : \mathcal{V} \to \mathcal{V}$ are defined as

$$\mathcal{A}^k x = \underbrace{\mathcal{A}(\mathcal{A} \cdots \mathcal{A}(\mathcal{A}x) \dots)}_{\mathcal{A} \text{ is applied } k \text{ times}}$$

When domain and co-domains are the same, then we can define the power of a linear transformation.

- $\mathbb{R}^n \to \mathbb{R}^m$, $m \neq n \Rightarrow$ not defined.
- $\mathcal{A} : \mathcal{V} \to \mathcal{W}$: $\mathcal{A}(\underbrace{\mathcal{A}x}_{\in \mathcal{W}})$ $\times$ not defined, since $\mathcal{A}$ acts on elements $\in \mathcal{V}$.

$\star$ **Polynomials of** $\mathcal{A}$ can be constructed using linear combinations of powers of $\mathcal{A}$.

$$p(s) = \alpha_0 s^n + \alpha_1 s^{n-1} + \cdots + \alpha_{n-1} s + \alpha_n \, (s^0)$$
$$p(\mathcal{A}) = \alpha_0 \mathcal{A}^n + \alpha_1 \mathcal{A}^{n-1} + \cdots + \alpha_{n-1} \mathcal{A}^1 + \alpha_n \mathcal{A}^0$$
$$p(\mathcal{A}) = \alpha_0 \mathcal{A}^n + \cdots + \alpha_{n-1} \mathcal{A} + \alpha_n I$$

$$\mathcal{A}^0 x = x = Ix$$
$$\mathcal{A}^0 = I$$

For example, with $p(\mathcal{A}) = 3\mathcal{A}^3 + 2\mathcal{A}^2 + \mathcal{A} + I$:

$$p(\mathcal{A})\mathcal{A}x = (3\mathcal{A}^3 + 2\mathcal{A}^2 + \mathcal{A} + I)\mathcal{A}x = 3\mathcal{A}^4 x + 2\mathcal{A}^3 x + \mathcal{A}^2 x + \mathcal{A}x$$

$$= \mathcal{A}(3\mathcal{A}^3x + 2\mathcal{A}^2x + \mathcal{A}x + x)$$
$$= \mathcal{A}\underbrace{(3\mathcal{A}^3 + 2\mathcal{A}^2 + \mathcal{A} + I)}_{p(\mathcal{A})}x$$

$$\boxed{p(\mathcal{A})\mathcal{A}x = \mathcal{A}p(\mathcal{A})x \quad \forall x \in \mathcal{V}} \qquad \Rightarrow \qquad p(\mathcal{A})\mathcal{A} = \mathcal{A}p(\mathcal{A})$$

▷ *$\mathcal{A}$ commutes with $p(\mathcal{A})$.*

Example 7.5. $\mathcal{N}(p(\mathcal{A}))$ & $\mathcal{R}(p(\mathcal{A}))$ are invariant under $\mathcal{A}$.
$\underline{\mathcal{N}(p(\mathcal{A}))}$: $x \in \mathcal{N}(p(\mathcal{A}))$. Is $\mathcal{A}x \in \mathcal{N}(p(\mathcal{A}))$?

$$p(\mathcal{A})\mathcal{A}x = \mathcal{A}\underbrace{p(\mathcal{A})x}_{0_\mathcal{V}} = \mathcal{A}0_\mathcal{V} = 0_\mathcal{V}$$
$$\Rightarrow \mathcal{A}x \in \mathcal{N}(p(\mathcal{A}))$$

$\underline{\mathcal{R}(p(\mathcal{A}))}$: $x \in \mathcal{R}(p(\mathcal{A}))$, $x = p(\mathcal{A})y$.

$$\mathcal{A}x = \mathcal{A}p(\mathcal{A})y = p(\mathcal{A})\underbrace{\mathcal{A}y}_{y'} = p(\mathcal{A})y'$$
$$\Rightarrow \mathcal{A}x \in \mathcal{R}(p(\mathcal{A}))$$

p. 53

7.3 Eigenvalues and Eigenvectors

⋆ Matrix representation:

$$\mathcal{A}: \underset{\substack{\downarrow B \\ \text{basis set for } \mathcal{V}}}{\overset{n}{\mathcal{V}}} \to \underset{\substack{\downarrow B \\ \text{basis set for } \mathcal{V}}}{\overset{n}{\mathcal{V}}} \qquad A: \mathbb{C}^n \to \mathbb{C}^n \qquad \begin{array}{cc} y = \mathcal{A}x & \lambda e = \mathcal{A}e \\ \Downarrow & \Downarrow \\ [y]_B = A[x]_B & \lambda[e]_B = A[e]_B \end{array}$$

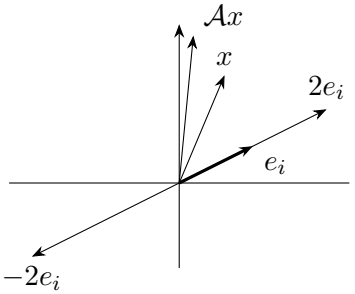
$$\det \mathcal{A} \triangleq \det A, \qquad \operatorname{tr} \mathcal{A} \triangleq \operatorname{tr} A$$

▷ *Well defined: another basis gives $\bar{A} = P^{-1}AP$, and $\det(P^{-1}AP) = \det A$, $\operatorname{tr}(P^{-1}AP) = \operatorname{tr} A$.*

Definition 7.6. Let A denote the matrix representation of a linear transformation $\mathcal{A}$ from $\mathcal{V}$ into $\mathcal{V}$ (A is a square matrix). The **eigenvalues** of $\mathcal{A}$ denoted by λ_i are defined as the n roots of the equation

$$\det(sI - A) = 0,$$

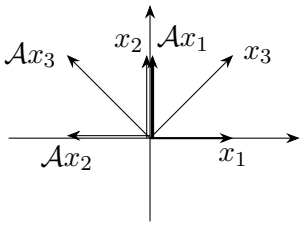
where $d(s) \triangleq \det(sI - A)$ is called the **characteristic polynomial** of A. The vectors $e_i \neq 0$ satisfying $Ae_i = \lambda_i e_i$ are called **eigenvectors** of A.



$$\mathcal{A}e_i = \lambda_i e_i$$
$$\lambda_i = 2$$
$$\lambda_i = -2$$

▷ *A general x is turned into a different direction ($\mathcal{A}x \not\parallel x$); an eigenvector e_i is only scaled, e.g. to $2e_i$ if $\lambda_i = 2$ or to $-2e_i$ if $\lambda_i = -2$, so the line $\operatorname{sp}\{e_i\}$ is invariant.*

$$A = \underbrace{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}_{\text{rotation matrix}} \qquad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$s^2 + 1 = 0 \qquad \lambda_{1,2} = \mp j$

$\underset{\text{real}}{A}\, e_i = \underset{\text{complex}}{\lambda_i}\, e_i \qquad R:$ rotation matrix

$Re_i = \lambda_i e_i$

$\|Re_i\| = \|\lambda_i e_i\|$

$\cancel{\|e_i\|} = |\lambda_i| \cancel{\|e_i\|} \qquad (e_i:$ nonzero vector)

$|\lambda_i| = 1$

$\therefore \quad \boxed{\lambda_i = e^{j\omega}}$

Explanation

No real vector keeps its direction under a 90° rotation, so there are no real eigenvectors; the eigenvectors are complex: $A \begin{bmatrix} 1 \\ -j \end{bmatrix} = \begin{bmatrix} j \\ 1 \end{bmatrix} = j \begin{bmatrix} 1 \\ -j \end{bmatrix}$ and $A \begin{bmatrix} 1 \\ j \end{bmatrix} = -j \begin{bmatrix} 1 \\ j \end{bmatrix}$. A rotation preserves length, $\|Re\| = \|e\|$ (also for complex e, since $R^H R = I$), which is what forces $|\lambda_i| = 1$; here $\lambda = \pm j = e^{\pm j\pi/2}$, i.e. $\omega = \pm\pi/2$ is the rotation angle.

p. 54

⋆ Existence of eigenvectors:

$$A \overset{\neq 0}{e_i} = \lambda e_i \qquad d(\lambda) = \det(\lambda_i I - A) = 0$$

$A = [c_1 \; c_2 \; \dots \; c_n]$

$\exists\ \alpha_1, \alpha_2, \dots, \alpha_n$ not all equal to zero satisfying

$\alpha_1 c_1 + \alpha_2 c_2 + \cdots + \alpha_n c_n = 0$

$$\underset{A}{[c_1 \; c_2 \; \dots \; c_n]} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = 0$$

▷ *A is not full column rank $\Rightarrow \exists \alpha \neq 0$ s.t. $A\alpha = 0$. Apply this to the singular matrix $\lambda_i I - A$ ($\det(\lambda_i I - A) = 0$ means its columns are dependent):*

$\exists \alpha$ s.t. $(\lambda_i I - A)\alpha = 0$

$A\alpha = \lambda_i \alpha \;\Rightarrow$ Eğer bir eigenvalue varsa en az 1 eigenvector vardır

▷ ***If there is an eigenvalue, then there is at least 1 eigenvector**: $\alpha \neq 0$ is an eigenvector for λ_i.*

Example 7.7.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad d(s) = \det(sI - A) = \det \begin{bmatrix} s-2 & -1 \\ -1 & s-2 \end{bmatrix}$$
$$= (s-2)^2 - 1 = s^2 - 4s + 3 = (s-3)(s-1) = 0$$
$$\Rightarrow \ \lambda_1 = 3, \quad \lambda_2 = 1$$

$$Ae_i = \lambda_i e_i$$
$$(A - \lambda_i I)e_i = 0$$
$$e_i \in \underbrace{\mathcal{N}(A - \lambda_i I)}_{\textbf{eigenspace}}, \quad \dim \geq 1 \qquad \begin{matrix} p(s) = s - \lambda_i \\ p(A) = A - \lambda_i I \end{matrix}$$

$$A - \lambda_1 I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$A - \lambda_2 I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathcal{N}(A - \lambda_1 I) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$
$$\mathcal{N}(A - \lambda_2 I) = \text{sp}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

$$\mathcal{N}(A - \lambda_1 I) \overset{\perp}{\oplus} \mathcal{N}(A - \lambda_2 I) = \mathbb{C}^2$$

▷ $\overset{\perp}{\oplus}$*: basis vectors are linearly indep. and orthogonal* ($[1\ 1]^T \perp [1\ -1]^T$)*; dimensions* $1+1 = 2 = \dim \mathbb{C}^2$.

$$\alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{matrix} \mathbb{R}^2 \\ \mathbb{C}^2 \end{matrix}$$

$$\alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{iff } \alpha = \beta = 0 \ \Rightarrow \ \text{subspaces are linearly indep.}$$

▷ *Orthogonality is special here because* $A = A^H$ *is Hermitian; in general eigenspaces are only linearly independent (see p. 57).*

p. 55

⋆ Change of basis:

$$y = Ax, \qquad B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \ \rightarrow \ \text{canonical basis set}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad [x]_B, \qquad [y]_B = A[x]_B \ \sim \ [y]_{\bar{B}} = \bar{A}[x]_{\bar{B}}, \qquad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\bar{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}, \qquad A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$x = \begin{bmatrix}1\\1\end{bmatrix}, \qquad [x]_{\bar{B}} = \begin{bmatrix}1\\0\end{bmatrix}, \qquad y = Ax = \begin{bmatrix}3\\3\end{bmatrix} = 3\begin{bmatrix}1\\1\end{bmatrix}, \qquad [y]_{\bar{B}} = \begin{bmatrix}3\\0\end{bmatrix}$$

$$\begin{bmatrix}3\\0\end{bmatrix} = \bar{A}\begin{bmatrix}1\\0\end{bmatrix} \qquad \Rightarrow \ \bar{A} = \begin{bmatrix}3 & \\ 0 & \end{bmatrix}$$

$$x = \begin{bmatrix}1\\-1\end{bmatrix}, \qquad [x]_{\bar{B}} = \begin{bmatrix}0\\1\end{bmatrix}, \qquad y = Ax = \begin{bmatrix}1\\-1\end{bmatrix} = 1\begin{bmatrix}1\\-1\end{bmatrix}, \qquad [y]_{\bar{B}} = \begin{bmatrix}0\\1\end{bmatrix}$$

$$\begin{bmatrix}0\\1\end{bmatrix} = \bar{A}\begin{bmatrix}0\\1\end{bmatrix} \qquad \Rightarrow \ \bar{A} = \begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}$$

$$P = [e_1 \mid e_2], \qquad \bar{A} = f(A) = ?$$

$$Ae_1 = \lambda_1 e_1$$
$$Ae_2 = \lambda_2 e_2$$

$$[\lambda_1 e_1, \lambda_2 e_2] = [Ae_1 \mid Ae_2] = A[e_1 \mid e_2]$$

$$\underbrace{[e_1 \ e_2]}_{P}\underbrace{\begin{bmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{bmatrix}}_{\bar{A}} = A \cdot P \ \Rightarrow \ P\bar{A} = AP$$

$$\Rightarrow \ \boxed{\bar{A} = P^{-1}AP}$$

▷ *P is invertible (full rank: its columns e_1, e_2 are linearly indep.).*

Explanation

The columns of P are the new basis vectors written in the old basis, so $[x]_B = P[x]_{\bar{B}}$. Then $[y]_B = A[x]_B$ becomes $P[y]_{\bar{B}} = AP[x]_{\bar{B}}$, i.e. $[y]_{\bar{B}} = P^{-1}AP\,[x]_{\bar{B}}$. Check: $P = \begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$, $P^{-1} = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$, $P^{-1}AP = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}3 & 1\\ 3 & -1\end{bmatrix} = \begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}$, as found above.

$$\mathbb{C}^2 = \underset{1}{\mathcal{N}}(A - \lambda_1 I) \overset{\perp}{\oplus} \underset{1}{\mathcal{N}}(A - \lambda_2 I_2), \qquad \left[\begin{array}{c|c} \overset{1\times 1}{\lambda_1} & 0\\ \hline 0 & \lambda_2\end{array}\right]$$

▷ *Each block is 1×1 as $\dim \mathcal{N}(A - \lambda_2 I_2) = 1$.*

7.4 Block Diagonal Representations

Theorem 7.8 (Block diagonal representation)**.** Consider the linear transformation

$$y = Ax \quad \text{where } A \in \mathbb{C}^{n\times n},\ y, x \in \mathbb{C}^n.$$

Suppose that

i) $\mathbb{C}^n = M_1 \oplus M_2 \oplus \cdots \oplus M_k$

ii) M_i is invariant under A $\forall i$.

Then, the transformation A has a **block diagonal representation**

$$\bar{A} = \begin{bmatrix} \bar{A}_1 & & & 0 \\ & \bar{A}_2 & & \\ & & \ddots & \\ 0 & & & \bar{A}_k \end{bmatrix}$$

where $\bar{A} = P^{-1}AP$, p. 56

$$P = [P_1 \mid P_2 \mid \cdots \mid P_k]$$
$$P_i = [b_i^1 \; b_i^2 \; b_i^3 \; \ldots \; b_i^{n_i}], \qquad n_i = \dim M_i$$

$b_i^1, \ldots, b_i^{n_i} \in \mathbb{C}^n$ are basis vectors of M_i. $\bar{A}_i$ is of size $n_i \times n_i$.

Sketch of a proof.

$$k = 2, \qquad \mathbb{C}^n = M_1 \oplus M_2, \qquad n = 5$$
$$M_1 = \operatorname{sp}\{b_1^1, b_1^2\}, \qquad \dim M_1 = 2$$
$$M_2 = \operatorname{sp}\{b_2^1, b_2^2, b_2^3\}, \qquad \dim M_2 = 3$$

$$P = [P_1 \mid P_2], \qquad \begin{aligned} P_1 &= [b_1^1 \mid b_1^2]_{5\times 2} \\ P_2 &= [b_2^1 \; b_2^2 \; b_2^3]_{5\times 3} \end{aligned} \qquad \bar{A} = P^{-1}AP \;\Rightarrow\; P\bar{A} = AP$$

$$\underbrace{[b_1^1 \, b_1^2 \mid b_2^1 \, b_2^2 \, b_2^3]}_{P} \underbrace{\left[\begin{array}{cc|ccc} \alpha_1^1 & \beta_1^1 & 0 & 0 & 0 \\ \alpha_1^2 & \beta_1^2 & 0 & 0 & 0 \\ \hline 0 & 0 & \alpha_2^1 & \beta_2^1 & \gamma_2^1 \\ 0 & 0 & \alpha_2^2 & \beta_2^2 & \gamma_2^2 \\ 0 & 0 & \alpha_2^3 & \beta_2^3 & \gamma_2^3 \end{array}\right]}_{\bar{A} \; \left(\begin{smallmatrix} 3\times 3 \\ \dim M_2 \times M_2 \end{smallmatrix}\right)} = A \overbrace{[b_1^1 \; b_1^2 \; b_2^1 \; b_2^2 \; b_2^3]}^{P}$$

$$= [\underbrace{Ab_1^1}_{\in M_1} \mid \underbrace{Ab_1^2}_{\in M_1} \mid \underbrace{Ab_2^1}_{\in M_2} \mid \underbrace{Ab_2^2}_{\in M_2} \mid \underbrace{Ab_2^3}_{\in M_2}]$$

▷ $\in M_1$: *lin. comb. of* M_1 *vectors;* $\in M_2$: *lin. comb. of* M_2 *vectors.*

Why the off-diagonal blocks are zero

Column j of $P\bar{A}$ is P times column j of $\bar{A}$, i.e. the combination of the basis vectors with the coefficients in that column. Since M_1 is invariant, $Ab_1^1 = \alpha_1^1 b_1^1 + \alpha_1^2 b_1^2$ needs *no* M_2 vectors, so the lower entries of columns 1–2 are zero; likewise $Ab_2^j \in M_2$ gives zeros in the upper part of columns 3–5. The coefficients are unique because the five vectors form a basis of $\mathbb{C}^5$ ($\mathbb{C}^n = M_1 \oplus M_2$), which also makes P invertible. Each block $\bar{A}_i$ is the matrix of A restricted to M_i.

Example 7.9.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 3 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M_1 \triangleq \operatorname{sp}\left\{ \overset{b_1^1}{\begin{bmatrix}1\\1\\0\end{bmatrix}}, \overset{b_1^2}{\begin{bmatrix}0\\1\\0\end{bmatrix}} \right\},\ \dim M_1 = 2, \qquad M_2 \triangleq \operatorname{sp}\left\{ \overset{b_2^1}{\begin{bmatrix}0\\1\\1\end{bmatrix}} \right\},\ \dim M_2 = 1$$

$$\begin{aligned} Ab_1^1 &= \begin{bmatrix}2\\2\\0\end{bmatrix} = 2b_1^1 + 0b_1^2 \in M_1 \\ Ab_1^2 &= \begin{bmatrix}1\\3\\0\end{bmatrix} = 1b_1^1 + 2b_1^2 \in M_1 \\ A(\alpha_1 b_1^1 + \alpha_2 b_1^2) &= \alpha_1 Ab_1^1 + \alpha_2 Ab_1^2 \\ &= \alpha_1(2b_1^1) + \alpha_2(1b_1^1 + 2b_1^2) \\ &= (2\alpha_1 + \alpha_2)b_1^1 + 2\alpha_2 b_1^2 \in M_1 \end{aligned}$$

$\therefore$ M_1 **is invariant under** A**.**

$$\left.\begin{array}{ll} Ab_1^1 = 2b_1^1 & -\ b_1^1 \text{ is an eigenvector} \\ Ab_1^2 = 1b_1^1 + 2b_1^2 & \Rightarrow \text{ not an eigenvector } (b_1^2) \end{array}\right\} \Rightarrow M_1 \text{ is not an eigenspace}$$

$$Ab_2^1 = \begin{bmatrix}0\\1\\1\end{bmatrix} = 1b_2^1 \in M_2 \ \Rightarrow \begin{array}{l} \therefore\ M_2 \text{ is invariant under } A \\ M_2 \text{ is eigenspace} \end{array}$$

p. 57

$$\mathbb{C}^3 = \underset{\checkmark}{M_1 \oplus M_2}, \qquad \alpha_1 b_1^1 + \alpha_2 b_1^2 + \beta_1 b_2^1 = 0$$

▷ *Linearly independent (only solution $\alpha_1 = \alpha_2 = \beta_1 = 0$: the 3rd entry gives $\beta_1 = 0$, the 1st $\alpha_1 = 0$, then $\alpha_2 = 0$). No orthogonality (no need for it actually).*

$$\bar{A} = \left[\begin{array}{c|c} & \\ \hline & \end{array}\right], \qquad \bar{A} = P^{-1}AP \Rightarrow P\bar{A} = AP, \qquad P = [b_1^1\ b_1^2 \mid b_2^1]$$

$$\underbrace{[b_1^1\ b_1^2\ b_2^1]}_{P} \underbrace{\left[\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 2 & 0 \\ \hline 0 & 0 & 1 \end{array}\right]}_{\bar{A}} = A[b_1^1\ b_1^2\ b_2^1]$$

$$= [Ab_1^1 \mid Ab_1^2 \mid Ab_2^1]$$

▷ *The columns of $\bar{A}$ are the coefficients found above: $Ab_1^1 = 2b_1^1$, $Ab_1^2 = 1b_1^1 + 2b_1^2$, $Ab_2^1 = 1b_2^1$.*

$$d(s) = \det(sI - A) = (s-2)^{②}(s-1), \qquad \lambda_1 = 2\ \&\ \lambda_2 = 1$$

▷ *Expanding along the 3rd row: $d(s) = (s-1)\big[(s-1)(s-3)+1\big] = (s-1)(s^2-4s+4) = (s-1)(s-2)^2$; the same follows from $\det(sI - \bar{A})$, since $\bar{A}$ is block triangular.*

$$\underbrace{\mathcal{N}(A - \lambda_1 I)}_{\dim=1} + \underbrace{\mathcal{N}(A - \lambda_2 I)}_{\dim=1} \neq \mathbb{C}^3$$

▷ ⇒ *only two linearly indep. eigenvectors (*$[1\ 1\ 0]^T$ *and* $[0\ 1\ 1]^T$*)* ⇒ ***Not diagonalizable*** *as dim's does not hold (*$1+1<3$*).*

$$\underset{3\ \checkmark}{\mathbb{C}^3} = \underbrace{\mathcal{N}\big((A-\lambda_1 I)^{\textcircled{2}}\big)}_{2} \oplus \underbrace{\mathcal{N}(A-\lambda_2 I)}_{1}$$

Explanation

The circled exponent is the same number in both places: $\lambda_1 = 2$ is a double root of $d(s)$, and the right invariant subspace for it is $\mathcal{N}((A-2I)^2)$ (a generalized eigenspace), not the eigenspace. Indeed

$$(A-2I)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \mathcal{N}((A-2I)^2) = \{x:\ x_3 = 0\} = \mathrm{sp}\{b_1^1, b_1^2\} = M_1.$$

By the result on p. 52 ($\mathcal{N}(p(A))$ is invariant), this subspace is automatically invariant under A. So A cannot be diagonalized, but it is block diagonal in this basis, with an upper-triangular 2×2 block for λ_1 (a Jordan block).

7.5 Distinct Eigenvalues and Diagonalization

Let $A \in \mathbb{C}^{n\times n}$, $\lambda_1 \neq \lambda_2$, $Ae_1 = \lambda_1 e_1$, $Ae_2 = \lambda_2 e_2$.

$$\begin{aligned} & \alpha_1 e_1 + \alpha_2 e_2 = 0 \\ \Rightarrow\ & (A-\lambda_1 I)(\alpha_1 e_1 + \alpha_2 e_2) = 0 \\ \Rightarrow\ & \alpha_1 \underbrace{\cancel{(A-\lambda_1 I)e_1}}_{0} + \alpha_2 \underbrace{(A-\lambda_1 I)e_2}_{\neq 0 \text{ (see below)}} = 0 \ \Rightarrow\ \boxed{\alpha_2 = 0} \end{aligned}$$

where

$$\begin{aligned} (A-\lambda_1 I)e_1 &= Ae_1 - \lambda_1 e_1 = \lambda_1 e_1 - \lambda_1 e_1 = 0 \\ (A-\lambda_1 I)e_2 &= Ae_2 - \lambda_1 e_2 = \lambda_2 e_2 - \lambda_1 e_2 = (\lambda_2 - \lambda_1)e_2 \neq 0 \end{aligned}$$

Then $\alpha_1 e_1 = 0$ with $e_1 \neq 0$ gives $\alpha_1 = 0$ as well, so e_1, e_2 **are linearly independent**.

Eigenvectors of distinct eigenvalues are independent

The same trick works for k distinct eigenvalues: if $\sum_{i=1}^k \alpha_i e_i = 0$, apply $\prod_{i\neq j}(A-\lambda_i I)$. Every term except the j-th is annihilated, leaving $\alpha_j \prod_{i\neq j}(\lambda_j - \lambda_i)\, e_j = 0$, so $\alpha_j = 0$ for every j. Consequently, **if** $A \in \mathbb{C}^{n\times n}$ **has** n **distinct eigenvalues, it is diagonalizable**: its eigenvectors form a basis, $\mathbb{C}^n = \mathcal{N}(A-\lambda_1 I)\oplus\cdots\oplus\mathcal{N}(A-\lambda_n I)$, and $P^{-1}AP = \mathrm{diag}(\lambda_1,\ldots,\lambda_n)$ with $P = [e_1\ \ldots\ e_n]$. The example above shows this can fail when an eigenvalue is repeated.

Theorem 7.10. Let A be an $n\times n$ matrix with n distinct eigenvalues, $\lambda_1, \lambda_2, \ldots, \lambda_n$ where $\lambda_i \neq \lambda_j$ when $i \neq j$. Then p. 58

$$\begin{array}{ccccccccc} \mathbb{C}^n = & \underset{\mathrm{sp}\{e_1\}}{\mathcal{N}(A-\lambda_1 I)} & \oplus & \underset{\mathrm{sp}\{e_2\}}{\mathcal{N}(A-\lambda_2 I)} & \oplus\cdots\oplus & \underset{\mathrm{sp}\{e_n\}}{\mathcal{N}(A-\lambda_n I)} & & \\ & 1 & + & 1 & +\cdots+ & 1 & = & n \end{array}$$

Since $\mathcal{N}(A-\lambda_i I)$ is invariant under A, A **has a diagonal representation**:

$$\bar{A}=\begin{bmatrix}\lambda_1 & & & 0\\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n\end{bmatrix}, \qquad \bar{A}=P^{-1}AP \quad \text{where } P=\big[\,e_1 \mid e_2 \mid \cdots \mid e_n\,\big].$$

Linear independence of $e_1,\ e_2,\ \ldots,\ e_n$. Suppose

$$\alpha_1 e_1+\alpha_2 e_2+\cdots+\alpha_n e_n=0$$

and use

$$(A-\lambda_i I)e_j=Ae_j-\lambda_i e_j=\lambda_j e_j-\lambda_i e_j=(\lambda_j-\lambda_i)e_j.$$

Multiply with $(A-\lambda_1 I)$, then with $(A-\lambda_2 I)$, and so on:

$$\begin{aligned}
\times(A-\lambda_1 I):&\quad 0+(\lambda_2-\lambda_1)\alpha_2 e_2+\alpha_3(\lambda_3-\lambda_1)e_3+\cdots+\alpha_n(\lambda_n-\lambda_1)e_n=0\\
\times(A-\lambda_2 I):&\quad 0+0+\alpha_3(\lambda_3-\lambda_1)(\lambda_3-\lambda_2)e_3+\cdots+\alpha_n(\lambda_n-\lambda_1)(\lambda_n-\lambda_2)e_n=0\\
&\quad\ \vdots\\
&\quad \alpha_n\underbrace{(\lambda_n-\lambda_1)(\lambda_n-\lambda_2)\cdots(\lambda_n-\lambda_{n-1})}_{\neq 0}\underbrace{e_n}_{\neq 0}=0 \quad\Longrightarrow\quad \alpha_n=0 \Longrightarrow \forall\alpha_i=0
\end{aligned}$$

▷ *The product is nonzero for distinct eigenvalues (this case), and e_n is nonzero.*

Completing the argument

After the $n-1$ multiplications only the e_n-term survives, so $\alpha_n=0$. The original relation then has only the terms $e_1,\ldots,e_{n-1}$; repeating the procedure with the factors $(A-\lambda_1 I),\ldots,(A-\lambda_{n-2}I)$ gives $\alpha_{n-1}=0$, and so on; hence all $\alpha_i=0$. Thus $\{e_1,\ldots,e_n\}$ is a set of n linearly independent vectors in $\mathbb{C}^n$, i.e. a basis, and each $\mathcal{N}(A-\lambda_i I)$ is exactly one-dimensional (it is at least 1-dimensional, and the dimensions cannot add up to more than n). This proves the direct sum in the theorem.

$$P=\big[\,e_1 \mid e_2 \mid \cdots \mid e_n\,\big], \qquad \bar{A}=P^{-1}AP \;\Rightarrow\; P\bar{A}=AP$$

$$\begin{aligned}
\big[\,e_1 \mid e_2 \mid \cdots \mid e_n\,\big]\begin{bmatrix}\lambda_1 & 0 & \cdots & 0\\ 0 & \lambda_2 & & \vdots\\ \vdots & & \ddots & \vdots\\ 0 & \cdots & \cdots & \lambda_n\end{bmatrix} &= A\big[\,e_1 \mid e_2 \mid \cdots \mid e_n\,\big]\\
&= \big[\,Ae_1 \mid Ae_2 \mid \cdots \mid Ae_n\,\big]\\
&= \big[\,\lambda_1 e_1 \;\; \lambda_2 e_2 \;\; \ldots \;\; \lambda_n e_n\,\big]
\end{aligned}$$

$$\begin{aligned}
\therefore\ \text{Distinct eigenvalues} &\Rightarrow \text{Diagonalizable matrix}\\
&\Rightarrow \text{we can find } P \text{ matrix \& } \bar{A}\\
\text{Distinct eigenvalues} &\not\Leftarrow \text{Diagonalizable}
\end{aligned}$$

Contrapositive ($p\Rightarrow q \;\equiv\; \bar{q}\Rightarrow\bar{p}$): Non-diagonalizable $\Rightarrow$ repeated eigenvalues.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad d(s) = (s-1)^3 \qquad \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

▷ *Diagonal ⇒ but repeated eigenvalues.*

Both matrices are already diagonal (so trivially diagonalizable with $P = I$), yet their eigenvalues are repeated: this is why the converse implication fails.

8 Characteristic and Minimal Polynomials

Syllabus: Week 10 — Cayley–Hamilton theorem, characteristic and minimal polynomials

8.1 Similar Matrices and the Characteristic Polynomial

Similar matrices have the same characteristic polynomial: p. 59

$$\begin{aligned} \bar{d}(s) &= \det(sI - \bar{A}) \\ &= \det(sI - P^{-1}AP) \\ &= \det(sP^{-1}P - P^{-1}AP) = \det\left(P^{-1}(sI - A)P\right) \\ &= \det P^{-1}\, \det(sI - A)\, \det P \\ &= \frac{1}{\cancel{\det P}} \cdot \det(sI - A) \cdot \cancel{\det P} \\ &= \det(sI - A) = d(s) \end{aligned}$$

▷ *Uses* $\det(XY) = \det X \det Y$ *and* $\det P^{-1} = 1/\det P$. *Hence* ***similar matrices have the same eigenvalues*** *(with the same algebraic multiplicities); in particular the eigenvalues of A are the diagonal entries of $\bar{A}$.*

Example 8.1.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}, \qquad sI - A = \begin{bmatrix} s-1 & 0 & 0 \\ 1 & s-2 & -1 \\ 0 & 0 & s-3 \end{bmatrix}$$

$$\det(sI - A) = (s-1)\begin{vmatrix} s-2 & -1 \\ 0 & s-3 \end{vmatrix} = (s-1)(s-2)(s-3) = 0 \quad \Rightarrow \quad \lambda = 1, \quad \lambda = 2, \quad \lambda = 3$$

$$A - \lambda_1 I = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \mathcal{N}(A - \lambda_1 I) = \operatorname{sp}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$A - \lambda_2 I = \begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathcal{N}(A - \lambda_2 I) = \operatorname{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$A - \lambda_3 I = \begin{bmatrix} -2 & 0 & 0 \\ -1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \mathcal{N}(A - \lambda_3 I) = \operatorname{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \bar{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = P^{-1}AP$$

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \bar{A} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Check for the second choice ($P\bar{A} = AP$):

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \overbrace{\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}}^{\bar{A}} = A \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 2 & 1 \\ 3 & 0 & 0 \end{bmatrix}$$

▷ *Both sides agree, so $P^{-1}AP = \bar{A}$. Reordering the eigenvectors in P only reorders the eigenvalues on the diagonal of $\bar{A}$.*

Characteristic polynomial and equation.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad d(s) = \det(sI - A) = \left| \begin{bmatrix} s-2 & -1 \\ -1 & s-2 \end{bmatrix} \right|$$

$$\begin{aligned} d(s) &= s^2 - 4s + 3 && \Rightarrow \textbf{ charac. poly.} \\ d(s) &= 0 && \Rightarrow \textbf{ charac. eqn.} \end{aligned}$$

$$\begin{aligned} d(A) = A^2 - 4A + 3I &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 4\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} + 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$\underline{\underline{d(A) = 0_{n\times n}}}$$

8.2 The Cayley–Hamilton Theorem

Theorem 8.2 (Cayley–Hamilton, CHT)**. Every $n \times n$ matrix satisfies its characteristic equation.** p. 60

$$\therefore\ d(A) = 0_{n\times n}$$

$$\begin{aligned} d(s) &= s^n + d_1 s^{n-1} + d_2 s^{n-2} + \cdots + d_{n-1}s + d_n \\ d(A) &= A^n + d_1 A^{n-1} + d_2 A^{n-2} + \cdots + d_{n-1}A + d_n I = 0 \\ A^n &= -d_1 A^{n-1} - \cdots - d_{n-1}A - d_n \underbrace{I}_{A^0} \end{aligned}$$

▷ $\Rightarrow$ *among the powers $I, A, A^2, \ldots$ we can have at most n linearly independent matrices.*

Remark. CHT says that n^{th} **and higher order powers of an** $n \times n$ **matrix can be written as a linear combination of its lower powers.**

$$A^2 - 4A + 3I = 0 \;\Rightarrow\; A^2 = 4A - 3I$$

$$\begin{aligned} A^3 = AA^2 = A(4A - 3I) = 4A^2 - 3A = 4(4A - 3I) - 3A \\ = 13A - 12I \end{aligned}$$

For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:

$$\begin{aligned} A^2 - 4A + 3I &= 0 \\ A - 4I + 3A^{-1} &= 0 \qquad \text{(multiply by } A^{-1}\text{; } A \text{ is invertible since } d(0) = 3 \neq 0) \\ A^{-1} &= -\frac{A}{3} + \frac{4I}{3} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \\ \therefore\; A^i &= \alpha_1 A^{n-1} + \alpha_2 A^{n-2} + \cdots + \alpha_{n-1} A + \alpha_n I \end{aligned}$$

$A = 2 \times 2$**:** By CHT, for a 2×2 matrix every power is a combination of A and I. To find the coefficients we use eigenvectors. For an eigenpair $Ae_1 = \lambda_1 e_1$:

$$\begin{aligned} Ae_1 &= \lambda_1 e_1 \\ A^2 e_1 &= A(Ae_1) = \lambda_1^2 e_1 \\ A^i e_1 &= \lambda_1^i e_1 \\ p(A)e_1 &= p(\lambda_1)e_1 \end{aligned}$$

▷ *For any polynomial* p*:* $p(A)$ *acts on an eigenvector as multiplication by* $p(\lambda_1)$*.*

$$\boxed{A^i = \alpha_i A + \beta_i I} \qquad \lambda_1 = 3\ (e_1), \qquad \lambda_2 = 1\ (e_2)$$

$$\underbrace{A^i e_1}_{\lambda_1^i e_1} = \alpha_i \underbrace{Ae_1}_{\lambda_1 e_1} + \beta_i \underbrace{Ie_1}_{e_1} \qquad\qquad A^i e_2 = \alpha_i A e_2 + \beta_i I e_2$$

$$3^i e_1 = 3\alpha_i e_1 + \beta_i e_1 \qquad\qquad \lambda_2^i e_2 = \alpha_i \lambda_2 e_2 + \beta_i e_2$$

$$3^i e_1 = (3\alpha_i + \beta_i) e_1 \qquad\qquad \lambda_2^i = \alpha_i \lambda_2 + \beta_i$$

Since $e_1, e_2 \neq 0$ we may compare coefficients; with $\lambda_2 = 1$ this gives

$$\begin{aligned} 3\alpha_i + \beta_i &= 3^i \\ \alpha_i + \beta_i &= 1 \qquad (= 1^i) \end{aligned}$$

▷ *Subtracting:* $2\alpha_i = 3^i - 1$*.*

$$\alpha_i = \frac{3^i - 1}{2}, \qquad \beta_i = 1 - \alpha_i = 1 - \frac{3^i - 1}{2} = \frac{3 - 3^i}{2}$$

$$\boxed{A^i = \frac{3^i - 1}{2} A + \frac{3 - 3^i}{2} I}$$

Check

$i = 0$: $0 \cdot A + 1 \cdot I = I$; $i = 1$: $A + 0 \cdot I = A$; $i = 2$: $4A - 3I$ (as found from CHT); $i = 3$: $13A - 12I$ (as above). The formula holds for every $i \geq 0$, and in fact also for $i = -1$: $\frac{3^{-1}-1}{2}A + \frac{3-3^{-1}}{2}I = -\frac{A}{3} + \frac{4I}{3} = A^{-1}$.

8.3 The Minimal Polynomial

p. 61

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad d(s) = (s-2)^2(s-1)$$

$$\begin{aligned} d(A) &= (A-2I)^2(A-I) \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$m(s) = (s-2)(s-1)$$

$$m(A) = (A-2I)(A-I) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0_{n \times n}$$

▷ *Lowest order ⇒ minimal polynomial. (A degree-1 monic $A - cI$ cannot vanish, since A is not a multiple of I.)*

Definition 8.3. A **monic polynomial** is a polynomial whose highest degree of polynomial coefficient is unity ($= 1$).

- $s^2 + 2s + 3$ (Monic)
- $2s^2 + 2s + 3$ (Not monic)

Definition 8.4. For an $n \times n$ matrix A, the **minimal polynomial** is the monic polynomial $m(s)$ with the smallest degree/order s.t.

$$m(A) = 0_{n \times n}$$

Theorem 8.5. Given $A \in \mathbb{C}^{n \times n}$ let $m(s)$ be its minimal polynomial then

i) $m(s)$ is **unique**

ii) $m(s)$ **divides** $d(s)$ without any remainder: $\exists q(s)$ s.t. $d(s) = m(s)q(s)$

iii) **Every root of** $d(s)$ **is also a root of** $m(s)$

Candidates for $d(s) = (s-1)(s-2)^2$:

- ~~$m(s) = (s-1)$~~, ~~$m(s) = (s-2)$~~, ~~$m(s) = (s-2)^2$~~
  ▷ not possible, property III (a root of $d(s)$ is missing)
- ~~$m(s) = (s-1)(s-3)$~~, ~~$m(s) = (s-2)^3$~~
  ▷ not possible, property II (they do not divide $d(s)$)
- $m(s) = (s-1)(s-2)$ ✓ $\qquad m(s) = (s-1)(s-2)^2$ ✓
  ▷ the only possibilities (i.e. $d(s) = \underbrace{m(s)}_{(s-1)(s-2)} \underbrace{q(s)}_{(s-2)}$ or $q(s) = 1$)

In general, for $A \in \mathbb{C}^{n\times n}$:

$$d(s) = (s-\lambda_1)^{r_1}(s-\lambda_2)^{r_2}\cdots(s-\lambda_k)^{r_k}, \qquad r_1 + r_2 + \cdots + r_k = n$$
$$m(s) = (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}, \qquad \boxed{1 \le m_i \le r_i}$$

▷ *Lower bound ($1 \le m_i$) by property III; upper bound by property II.*

Proof. **I)** $p \to q \;\equiv\; p \wedge \bar{q} = \text{False} \;\triangleq$ proof by contradiction. p. 62

Suppose $m(s)$ is not unique and minimal polynomial ($p \wedge \bar{q}$):

$$m_1(s) \longrightarrow m_1(A) = 0_{n\times n}$$
$$m_2(s) \longrightarrow m_2(A) = 0_{n\times n}$$
$$m_1(s) \neq m_2(s)$$

$$m_1(s) = s^l + a_1 s^{l-1} + \cdots + a_l$$
$$m_2(s) = s^l + b_1 s^{l-1} + \cdots + b_l, \qquad \exists i \text{ s.t. } a_i \neq b_i$$
$$r(s) = m_1(s) - m_2(s)$$
$$r(s) = (a_1 - b_1)s^{l-1} + \cdots + (a_l - b_l)$$
$$\bar{r}(s) = \frac{r(s)}{\alpha}$$

▷ $r(s) \neq 0$ *since* $m_1 \neq m_2$, *and* $\deg r < l$ *since the* s^l *terms cancel;* α*: coefficient of the highest order term* $\Rightarrow \bar{r}(s)$ *is monic.*

$$\deg \bar{r}(s) < \big(\deg m_1(s) = \deg m_2(s)\big)$$
$$\bar{r}(A) = \frac{r(A)}{\alpha} = \frac{m_1(A) - m_2(A)}{\alpha} = \frac{0_{n\times n} - 0_{n\times n}}{\alpha} = 0_{n\times n}$$

$\bar{r}(A) = 0_{n\times n}$: we assumed m_1, m_2 are minimal, but found a monic $\bar{r}(s)$ of lower degree that also annihilates $A \Rightarrow$ contradiction with minimality. Hence $m(s)$ is unique.

II) Divide $d(s)$ by $m(s)$ (polynomial division):

$$d(s) = m(s)q(s) + r(s), \qquad \deg r(s) < \deg m(s).$$

Suppose $r(s) \neq 0(s)$ (proof by contradiction). Then

$$\exists\, \bar{r}(s) = \frac{r(s)}{\alpha} \text{ and } \bar{r}(s) \text{ is monic} \quad (\alpha: \text{ leading coefficient of } r(s)). \qquad \bar{r}(A) = ?$$

$$\underbrace{d(A)}_{0_{n\times n}\ (\text{CHT})} = \underbrace{m(A)}_{0_{n\times n}} q(A) + r(A) \qquad \Rightarrow \qquad r(A) = 0_{n\times n} \ \Rightarrow\ \bar{r}(A) = 0_{n\times n}$$

▷ *$\bar{r}$ is monic, annihilates A and has lower degree than m: contradiction. So $r(s) = 0$, i.e. $m(s)$ divides $d(s)$.*

The same argument in general

The proof of II only used $d(A) = 0$. Hence: $m(s)$ **divides every polynomial** $p(s)$ **with** $p(A) = 0$**.**

III) Roots of $d(s)$ are eigenvalues of A.

$$\begin{aligned}
&d(\lambda_i) = 0\\
&Ae_i = \lambda_i e_i, \qquad e_i \neq 0\\
&m(\lambda_i) \stackrel{?}{=} 0\\
&m(s) = s^l + a_1 s^{l-1} + \cdots + a_l\\
&m(A) = A^l + a_1 A^{l-1} + \cdots + a_l I = 0_{n\times n}\\
&(A^l + a_1 A^{l-1} + \cdots + a_l I)e_i = 0_{n\times 1}\\
&A^l e_i + a_1 A^{l-1} e_i + \cdots + a_{l-1} A e_i + a_l e_i = 0_{n\times 1}\\
&(\lambda_i^l + a_1 \lambda_i^{l-1} + \cdots + a_{l-1}\lambda_i + a_l)\ \underbrace{e_i}_{\neq 0} = 0_{n\times 1}\\
&\lambda_i^l + a_1 \lambda_i^{l-1} + \cdots + a_{l-1}\lambda_i + a_l = 0 \ \Rightarrow\ m(\lambda_i) = 0
\end{aligned}$$

▷ *$A^j e_i = \lambda_i^j e_i$; a scalar times a nonzero vector is zero only if the scalar is zero.*

$$\therefore\ \lambda_i \text{ is a root of } m(s). \qquad \square$$

8.4 Minimal Polynomial and Null-Space Decomposition

p. 63

$$\begin{aligned}
\therefore\ d(s) &= (s-\lambda_1)^{r_1}(s-\lambda_2)^{r_2}\cdots(s-\lambda_k)^{r_k}\\
m(s) &= (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}\\
1 \le m_i &\le r_i, \qquad r_1 + r_2 + \cdots + r_k = n
\end{aligned}$$

Distinct eigenvalue case.

$$\begin{aligned}
d(s) &= (s-\lambda_1)^1(s-\lambda_2)^1\cdots(s-\lambda_n)^1\\
r_i = 1 &\Rightarrow m_i = 1\\
m(s) &= d(s)
\end{aligned}$$

$$A = \begin{bmatrix} 1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3 \end{bmatrix}, \qquad d(s) = (s-1)(s-2)(s-3) = m(s)$$

$$\mathbb{C}^3 = \underbrace{\mathcal{N}(A-\lambda_1 I)}_{\text{dim=1}} \oplus \underbrace{\mathcal{N}(A-\lambda_2 I)}_{\text{dim=1}} \oplus \underbrace{\mathcal{N}(A-\lambda_3 I)}_{\text{dim=1}}$$

$$\mathbb{C}^n = \underbrace{\mathcal{N}\big((A-\lambda_1 I)^1\big)}_{\dim=1} \oplus \cdots \oplus \underbrace{\mathcal{N}\big((A-\lambda_n I)^1\big)}_{\dim 1}$$

Example 8.6.

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Matrix A_1.

$$\begin{aligned} d_1(s) &= (s-1)(s-2)^2 \\ m_1(s) &= \boxed{(s-1)(s-2)} \ \checkmark \\ &= \cancel{(s-1)(s-2)^2} \end{aligned}$$

$$A_1 - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \text{rank} = 2,\ \dim = 3,\ \text{null} = 1$$

$$A_1 - 2I = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \text{rank} = 1,\ \dim = 3,\ \text{null} = 2$$

▷ *Columns 2, 3 of $A_1 - I$: lin. indep vectors.*

$$(A_1 - I)(A_1 - 2I) = 0_{3\times 3}$$

$$\mathcal{N}(A_1 - I) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\mathcal{N}(A_1 - 2I) = \text{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

▷ *Linearly indep $\Rightarrow$ direct sum.*

$$\underbrace{\mathcal{N}\big((A_1 - I)^1\big)}_{\dim=1} \oplus \underbrace{\mathcal{N}\big((A_1 - 2I)^1\big)}_{\dim=2} = \mathbb{C}^3$$

Matrix A_2.

$$\begin{aligned} d_2(s) &= (s-1)(s-2)^2 \\ m_2(s) &= \cancel{(s-1)(s-2)} \\ &= \boxed{(s-1)(s-2)^2} \ \checkmark \end{aligned}$$

$$A_2 - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad r = 2,\ n = 1,\ d = 3$$

$$A_2 - 2I = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \neq 0_{3\times 3}, \qquad r = 2,\ n = 1,\ d = 3$$

$$(A_2 - 2I)^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(A_2 - I)(A_2 - 2I) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \neq 0_{3\times 3}$$

▷ *Hence* $(s-1)(s-2)$ *is not the minimal polynomial; only* $(A_2 - I)(A_2 - 2I)^2 = 0_{3\times 3}$.

$$\mathcal{N}((A_2 - I)) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\mathcal{N}((A_2 - 2I)) = \text{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$\mathbb{C}^3 \neq \underbrace{\mathcal{N}((A_2 - I))}_{1} \oplus \underbrace{\mathcal{N}((A_2 - 2I))}_{1}$$

$$\mathcal{N}((A_2 - 2I)^2) = \text{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\mathbb{C}^3 = \underbrace{\mathcal{N}((A_2 - I)^1)}_{\dim=1} \oplus \underbrace{\mathcal{N}((A_2 - 2I)^2)}_{\dim=2}$$

What the example shows

In both cases $\mathbb{C}^3$ splits into null spaces of powers of $A - \lambda_i I$, and the power needed for λ_i is exactly the exponent m_i in the minimal polynomial. For A_1 ($m_i = 1$) the eigenvectors alone span $\mathbb{C}^3$, so A_1 is diagonalizable. For A_2 there are only two independent eigenvectors; the missing direction $\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T$ is not an eigenvector but satisfies $(A_2 - 2I)^2 x = 0$ (a *generalized eigenvector*). This is made precise in the next theorem.

9 Jordan Canonical Form

Syllabus: Weeks 11–12 — Jordan canonical forms

9.1 Generalized Eigenspaces and the Jordan Form

Theorem 9.1 (Null-space decomposition)**.**

$$d(s) = (s-\lambda_1)^{r_1}(s-\lambda_2)^{r_2}\cdots(s-\lambda_k)^{r_k}$$

$$\mathbb{C}^n = \overbrace{\mathcal{N}\big((A-\lambda_1 I)^{m_1}\big)}^{\dim=r_1} \oplus \overbrace{\mathcal{N}\big((A-\lambda_2 I)^{m_2}\big)}^{\dim=r_2} \oplus \cdots \oplus \overbrace{\mathcal{N}\big((A-\lambda_k I)^{m_k}\big)}^{\dim=r_k}$$

$$m(s) = (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}$$

$*$ If we form a P matrix out of the columns which are the basis vectors for the null spaces $\mathcal{N}\big((A-\lambda_i I)^{m_i}\big)$ then we can have

$$\begin{aligned} \bar{A} &= P^{-1}AP, \\ P &= \big[\, P_1 \mid P_2 \mid \cdots \mid P_k \,\big]_{n\times n}, \\ P_i &= \big[\, \underbrace{b_1^i \;\; b_2^i \;\; \ldots \;\; b_{r_i}^i}_{\substack{\text{basis vectors for} \\ \mathcal{N}((A-\lambda_i I)^{m_i})}} \,\big]_{n\times r_i} \end{aligned}$$

▷ *P has $r_1 + r_2 + \cdots + r_k = n$ columns.*

$$\bar{A} = \begin{bmatrix} \bar{A}_1 & & & 0 \\ & \bar{A}_2 & & \\ & & \ddots & \\ 0 & & & \bar{A}_k \end{bmatrix}_{n\times n}$$

where the size of $\bar{A}_i$ is $r_i \times r_i$.

$*$ If P_i is selected in a specific way, we can have

$$\bar{A}_i = \begin{bmatrix} \lambda_i & * & & 0 \\ & \lambda_i & * & \\ & & \ddots & * \\ 0 & & & \lambda_i \end{bmatrix}_{r_i\times r_i}, \qquad * \in \{0,1\}$$

In this case $\bar{A}$ is called the **Jordan Canonical Form** of A.

Why $\bar{A}$ is block diagonal

Each subspace $\mathcal{N}\big((A-\lambda_i I)^{m_i}\big)$ is invariant under A: if $(A-\lambda_i I)^{m_i}x = 0$ then, since A commutes with $(A-\lambda_i I)^{m_i}$, also $(A-\lambda_i I)^{m_i}(Ax) = A(A-\lambda_i I)^{m_i}x = 0$. So A maps the columns of P_i into the span of the columns of P_i, and in the basis given by the columns of P the matrix $\bar{A} = P^{-1}AP$ has no entries coupling different blocks. Each $\bar{A}_i$ has the single eigenvalue λ_i (with multiplicity r_i). The "specific way" of choosing P_i is via Jordan chains x, $(A-\lambda_i I)x$, $(A-\lambda_i I)^2 x, \ldots$; every chain produces one Jordan block (a λ_i-diagonal with 1's above it).

Example 9.2.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \left.\begin{aligned} d(s) &= (s-1)(s-2)^2 \\ m(s) &= (s-1)(s-2)^2 \end{aligned}\right\} \text{ found before}$$

$$\mathbb{C}^3 = \underbrace{\mathcal{N}((A-I))}_{\dim 1} \oplus \underbrace{\mathcal{N}((A-2I)^2)}_{\dim 2}$$

Powers of $A-I$.

$$(A-I) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad r=2, \qquad \dim \mathcal{N}(A-I) = 1, \qquad \mathcal{N}(A-I) = \operatorname{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$(A-I)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \quad r=2,\ \text{Null}=1, \quad \dim \mathcal{N}((A-I)^2) = 1, \quad \mathcal{N}((A-I)^2) = \operatorname{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\vdots$$

$$(A-I)^l = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & l \\ 0 & 0 & 1 \end{bmatrix}, \qquad \dim \mathcal{N}((A-I)^l) = 1, \quad \mathcal{N}((A-I)^l) = \operatorname{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\mathcal{N}(A-I) = \mathcal{N}((A-I)^2) = \cdots = \mathcal{N}((A-I)^l)$$

p. 65

Powers of $A-2I$.

$$(A-2I) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \dim \mathcal{N}(A-2I) = 1, \qquad \mathcal{N}(A-2I) = \operatorname{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$(A-2I)^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \dim \mathcal{N}((A-2I)^2) = 2, \quad \mathcal{N}((A-2I)^2) = \operatorname{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\therefore\ \mathcal{N}(A-2I) \subsetneq \mathcal{N}((A-2I)^2) \text{ (proper subset) } = \mathcal{N}((A-2I)^3) \cdots$$

▷ $\subsetneq$: $\subset$ & $\neq$, *i.e. subset and not equal.*

$$(A-2I)^3 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \dim \mathcal{N}((A-2I)^3) = 2, \quad \mathcal{N}((A-2I)^3) = \operatorname{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

In general:

$$\begin{array}{ccccccccc} \dim & < & \dim & < \cdots < & \dim & = & \dim & = \cdots \\ \mathcal{N}(A-\lambda_i I) & \subsetneq & \mathcal{N}((A-\lambda_i I)^2) & \subsetneq \cdots \subsetneq & \underbrace{\mathcal{N}((A-\lambda_i I)^{m_i})}_{\dim r_i} & = & \mathcal{N}((A-\lambda_i I)^{m_i+1}) & = \cdots \end{array}$$

▷ $\dim r_i$: *largest dimension* $\hookrightarrow$ *exponent* r_i *in the characteristic polynomial.*

Example 9.3.

$$A = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 2 \end{array}\right], \qquad \begin{array}{l} d(s) = (s-1)^3(s-2) \\ m(s) = ? \quad 3 \text{ possibilities:} \end{array}$$

1) $(s-1)(s-2)$

2) $(s-1)^2(s-2)$ $\Leftarrow$ This is the minimal polynomial

3) $(s-1)^3(s-2)$

$$(A-I) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \dim\left(\mathcal{N}(A-I)\right) = 2$$

$$(A-I)^2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \dim\left(\mathcal{N}((A-I)^2)\right) = 3 \checkmark$$

▷ $\dim\mathcal{N}\big((A-I)^2\big) = 3 = r_1$ *is reached at power* 2, *so* $m_1 = 2$; *for* $\lambda_2 = 2$, $r_2 = 1$ *forces* $m_2 = 1$. *Hence* $m(s) = (s-1)^2(s-2)$; *indeed* $(A-I)^2(A-2I) = 0$ *but* $(A-I)(A-2I) \neq 0$.

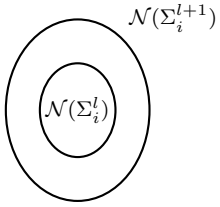
9.2 Chains of Null Spaces

∗ Let $\Sigma_i \triangleq A - \lambda_i I$. Then p. 66

$$\begin{array}{ccccccccccc} \mathcal{N}(\Sigma_i) & \subsetneq & \mathcal{N}(\Sigma_i^2) & \subsetneq & \mathcal{N}(\Sigma_i^3) & \cdots & \subsetneq & \mathcal{N}(\Sigma_i^{m_i}) & = & \mathcal{N}(\Sigma_i^{m_i+1}) & \cdots \\ 1 \leq \dim & < & \dim & < & \dim & \cdots & < & \dim r_i & = & r_i & \end{array}$$

Proof.

Claim. $\mathcal{N}(\Sigma_i^l) \subset \mathcal{N}(\Sigma_i^{l+1})$



$$\begin{aligned} x \in \mathcal{N}(\Sigma_i^l) &\Rightarrow \Sigma_i^l x = 0 \\ &\Rightarrow \Sigma_i \cdot \Sigma_i^l x = \Sigma_i \cdot 0 = 0 \\ &\Rightarrow \Sigma_i^{l+1} x = 0 \Rightarrow x \in \mathcal{N}(\Sigma_i^{l+1}) \end{aligned}$$

Claim. If $\mathcal{N}(\Sigma_i^l) = \mathcal{N}(\Sigma_i^{l+1})$ then $\mathcal{N}(\Sigma_i^{l+1}) = \mathcal{N}(\Sigma_i^{l+2})$

Assume $\mathcal{N}(\Sigma_i^l) = \mathcal{N}(\Sigma_i^{l+1})$. Need to show $\mathcal{N}(\Sigma_i^{l+1}) = \mathcal{N}(\Sigma_i^{l+2})$, i.e.

✓ $\mathcal{N}(\Sigma_i^{l+1}) \subset \mathcal{N}(\Sigma_i^{l+2})$
$\mathcal{N}(\Sigma_i^{l+2}) \subset \mathcal{N}(\Sigma_i^{l+1})$

$$\begin{aligned} x \in \mathcal{N}(\Sigma_i^{l+2}) &\Rightarrow \Sigma_i^{l+2} x = 0 \\ &\Rightarrow \Sigma_i^{l+1}\,\Sigma_i x = 0 \Rightarrow \Sigma_i x \in \mathcal{N}(\Sigma_i^{l+1}) \\ &\Rightarrow \Sigma_i x \in \mathcal{N}(\Sigma_i^l) \\ &\Rightarrow \Sigma_i^l\,\Sigma_i x = 0 \Rightarrow \Sigma_i^{l+1} x = 0 \Rightarrow x \in \mathcal{N}(\Sigma_i^{l+1}) \end{aligned}$$ □

Consequence

By the two claims the chain of null spaces grows strictly as long as it grows at all; once two consecutive null spaces coincide it is constant forever. Since the dimensions are bounded by n, the chain must stop growing, and it **stops exactly at the power** m_i (the exponent in $m(s)$),

where the dimension equals the algebraic multiplicity r_i. Because each step adds at least one dimension, $m_i \leq r_i$ again.

Example 9.4.

$$A = \begin{bmatrix} \boxed{\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{smallmatrix}} & & & & & \\ & \boxed{\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}} & & & & \\ & & \boxed{2} & & & \\ & & & \boxed{2} & & \\ & & & & \boxed{\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}} & \\ & & & & & \boxed{3} \end{bmatrix}_{10\times 10}, \qquad \begin{aligned} &\dim \mathcal{N}\big((A-\lambda_i I)^l\big) = ? \\ &d(s) = ? \\ &m(s) = ? \end{aligned}$$

▷ *Block diagonal; all entries outside the boxed blocks are zero.*

First block.

$$A_1 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad d_1(s) = (s-2)^3, \qquad m_1(s) = (s-2)^3$$

$$A_1 - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \text{rank} = 2,\ \mathcal{N} = 1, \qquad \dim\big(\mathcal{N}(A_1 - 2I)\big) = 1$$

$$(A_1 - 2I)^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \dim\big(\mathcal{N}((A_1 - 2I)^2)\big) = 2$$

$$(A_1 - 2I)^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \dim\big(\mathcal{N}((A_1 - 2I)^3)\big) = 3$$

$$(A_1 - 2I)^4 = 0_{3\times 3}, \qquad \dim\big(\mathcal{N}((A_1 - 2I)^4)\big) = 3 \ \ldots$$

▷ *For $(A_1 - 2I)^3$: no need to calculate as it can be max 3 and is not equal to 1 or 2.*
▷ *Each Jordan block of size p behaves the same way:* $\dim \mathcal{N}\big((J-\lambda I)^l\big) = \min(l, p)$. *Each block of A contributes to the dimensions of A independently.*

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For A:

$$d(s) = (s-2)^7(s-3)^3$$

▷ *$\lambda = 2$ appears in blocks of sizes $3, 2, 1, 1$ $(3+2+1+1=7)$; $\lambda = 3$ in blocks of sizes $2, 1$ $(2+1=3)$.*

$$\begin{aligned} &\Sigma_1 = A - 2I \\ &\dim\big(\mathcal{N}(\Sigma_1)\big) = 4 \\ &\dim\big(\mathcal{N}(\Sigma_1^2)\big) = 6 \quad (+2) \\ &\dim\big(\mathcal{N}(\Sigma_1^3)\big) = 7 \quad (+1) \\ &\Rightarrow (s-2)^3 \quad (+0) \end{aligned} \qquad \begin{aligned} &\Sigma_2 = A - 3I \\ &\dim\big(\mathcal{N}(\Sigma_2)\big) = 2 \\ &\dim\big(\mathcal{N}(\Sigma_2^2)\big) = 3 \quad (+1) \\ &\Rightarrow (s-3)^2 \quad (+0) \end{aligned}$$

$$\boxed{\therefore\ m(s) = (s-2)^3(s-3)^2}$$

▷ *Exponent 3: largest JB for $\lambda_i = 2$; exponent 2: largest JB for $\lambda_i = 3$.*

9.3 Counting Jordan Blocks

(JB: Jordan block.)

$$\begin{aligned}\dim \mathcal{N}(\Sigma_i) &= \#\text{ of JB's for } \lambda_i \\ &= \#\text{ of JB's with size} \geq 1 \text{ for } \lambda_i \\ &= \max \#\text{ of linearly ind. eigenvectors for } \lambda_i\end{aligned}$$

$$\begin{aligned}\dim \mathcal{N}(\Sigma_i^2) - \dim \mathcal{N}(\Sigma_i^1) &= \#\text{ of JB's with size} \geq 2 \text{ for } \lambda_i \\ \dim \mathcal{N}(\Sigma_i^3) - \dim \mathcal{N}(\Sigma_i^2) &= \#\text{ of JB's with size} \geq 3 \text{ for } \lambda_i \\ &\vdots \\ \dim \mathcal{N}(\Sigma_i^l) - \dim \mathcal{N}(\Sigma_i^{l-1}) &= \#\text{ of JB's with size} \geq l \text{ for } \lambda_i.\end{aligned}$$

For $l = 1$:

$$\dim \mathcal{N}(\Sigma_i) - \underbrace{\dim \mathcal{N}(\overbrace{\Sigma_i^0}^{I})}_{0} = \#\text{ of JB's with size} \geq 1 \text{ for } \lambda_i$$

▷ *Works for $l = 1$ also.*

m_i is the size of the largest JB.

Reading off the block sizes

Write $\delta_l = \dim \mathcal{N}(\Sigma_i^l) - \dim \mathcal{N}(\Sigma_i^{l-1})$ (the increments, $\delta_l = \#$ JB's of size $\geq l$). Then

$$\#\text{ of JB's of size exactly } l \;=\; \delta_l - \delta_{l+1}.$$

Reason: a block of size p contributes 1 to $\dim \mathcal{N}(\Sigma_i^l)$ for each $l \leq p$, so it raises the dimension by one at each of the steps $l = 1, \ldots, p$ and not after. The increments therefore stop being positive at $l = m_i$, the size of the largest block.

$\Rightarrow$ For diagonalizable matrix, minimal polynomial should have the power at most 1:

JCF is diagonal iff $m_i = 1\ \forall i$

$$\begin{aligned}\mathbb{C}^n &= \mathcal{N}(\Sigma_1^{m_1}) \oplus \cdots \oplus \mathcal{N}(\Sigma_k^{m_k}) \\ \mathbb{C}^n &= \underbrace{\mathcal{N}(\Sigma_1)}_{r_1} \oplus \cdots \oplus \underbrace{\mathcal{N}(\Sigma_k)}_{r_k}\end{aligned}$$

▷ *$\mathcal{N}(\Sigma_1)$: at most r_1 linearly indep eigenvectors.*

The second decomposition holds exactly when $m_i = 1$ for all i, i.e. when $\dim \mathcal{N}(\Sigma_i) = r_i$ for every eigenvalue.

$\Rightarrow$ **A is diagonalizable iff there exist n linearly indep eigenvectors for A.**

Example 9.5. p. 68

$$A = \begin{bmatrix} 1 & -1 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & -2 & -2 & 0 \end{bmatrix} \quad \Rightarrow \quad \begin{array}{l} d(s) = s^4 \\ m(s) = ? \\ J = ? \end{array}$$

▷ $d(s) = s^4$*: Learn to calculate this! (Shortcut:* $A^3 = 0$ *below, so every eigenvalue satisfies* $\lambda^3 = 0$*, i.e.* $\lambda = 0$ *with multiplicity 4.)*

$$\Sigma_1 = A - \cancel{\lambda_1}^{0} I = A, \qquad \dim \mathcal{N}(\Sigma_1) = 2, \quad \dim \mathcal{N}(\Sigma_1^2) = 3, \quad \dim \mathcal{N}(\Sigma_1^3) = 4$$

$$A^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{bmatrix}, \qquad A^3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

▷ $\operatorname{rank} A = 2$ *(row 4 =* $-2\cdot$*row 1+*$4\cdot$*row 2, row 3 =* 0*)* $\Rightarrow \dim \mathcal{N}(A) = 2$*;* $\operatorname{rank} A^2 = 1 \Rightarrow \dim \mathcal{N}(A^2) = 3$*;* $A^3 = 0 \Rightarrow \dim \mathcal{N}(A^3) = 4$*.*

$$\begin{array}{ll} \therefore \ m(s) = s^3 & \\ \#\text{ of JB's} = 2 & (2-0) \\ \#\text{ of JB's with size} \geq 2 = 1 & (3-2) \\ \#\text{ of JB's with size} \geq 3 = 1 & (4-3) \\ \#\text{ of JB's with size} \geq 4 = 0 & (4-4) \end{array}$$

$\Rightarrow$ one JB with size $= 3$, zero JB with size $= 2$, one JB with size $= 1$:

$$\text{JCF} = \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right]$$

$\Rightarrow$ **Not diagonalizable!** (only $\dim \mathcal{N}(A) = 2 < 4$ independent eigenvectors; equivalently $m_1 = 3 \neq 1$)

Example 9.6. Let $A \in \mathbb{R}^{8\times 8}$ have the following subspace dimensions

$$\begin{array}{rl} \dim\left(\mathcal{N}(A - 3I)\right) = 5 & (+5) \\ \dim\left(\mathcal{N}((A-3I)^2)\right) = 7 & (+2) \\ \dim\left(\mathcal{N}((A-3I)^3)\right) = 8 & (+1) \\ & (+0) \end{array}$$

$d(s) = ?$ $\quad m(s) = ?$ $\quad \text{JCF} = ?$

$$\begin{aligned} m(s) &= (s-3)^3 \\ d(s) &= (s-3)^8 \end{aligned}$$

▷ $\Rightarrow$ *max linearly indep eigenvectors is 5.*
▷ $\dim \mathcal{N}\big((A-3I)^3\big) = 8 = n$*, so* 3 *is the only eigenvalue (*$r_1 = 8$*) and the chain stops at power* 3 *(*$m_1 = 3$*).*

Block sizes from the increments $5, 2, 1, 0$: 5 JB's in total; $5 - 2 = 3$ of size 1, $2 - 1 = 1$ of size 2, $1 - 0 = 1$ of size 3 (check: $3 \cdot 1 + 1 \cdot 2 + 1 \cdot 3 = 8$).

$$\text{JCF} = \left[\begin{array}{ccc|ccc|c|c} 3 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 3 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \end{array}\right]$$

Example 9.7. Let $A \in \mathbb{R}^{4\times 4}$. Single eigenvalue at $\lambda = 7$, $m_1 = 2$. Find all possible JCF the matrix A can have.

$$d(s) = (s-7)^4$$
$$m(s) = (s-7)^2$$

$$\begin{array}{ccc} \mathcal{N}(A-7I) & \subsetneq & \mathcal{N}((A-7I)^2) \\ 1 \leq\ ? & & 4 \\ \text{dim=2 or 3 or 1} & & \end{array}$$

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Case I: $\dim \mathcal{N}(A-7I) = 3$**.**

$$\begin{array}{rccccccc} & \{0\} & \subsetneq & \mathcal{N}(A-7I) & \subsetneq & \mathcal{N}(A-7I)^2 & \subsetneq & \cdots \\ \dim: & 0 & & 3 & & 4 & & 4 \\ \text{increment:} & & +3 & & +1 & & +0 & \\ & & \downarrow & & \downarrow & & & \\ & & \text{2 JB size 1} & & \text{1 JB size 2} & & & \end{array}$$

$$A = \left[\begin{array}{cc|c|c} 7 & 1 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ \hline 0 & 0 & 7 & 0 \\ \hline 0 & 0 & 0 & 7 \end{array}\right] \qquad \text{① 1st possibility}$$

Case II: $\dim \mathcal{N}(A-7I) = 2$**.**

$$\begin{array}{rccccccc} & \{0\} & \subsetneq & \mathcal{N}(A-7I) & \subsetneq & \mathcal{N}(A-7I)^2 & \subsetneq & \cdots \\ \dim: & 0 & & 2 & & 4 & & 4 \\ \text{increment:} & & +2 & & +2 & & +0 & \end{array}$$

$$A = \left[\begin{array}{cc|cc} 7 & 1 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ \hline 0 & 0 & 7 & 1 \\ 0 & 0 & 0 & 7 \end{array}\right] \qquad \text{② 2nd possibility}$$

Case III: $\dim \mathcal{N}(A-7I) = 1$**.**

$$\begin{array}{rccccccc} & \{0\} & \subsetneq & \mathcal{N}(A-7I) & \subsetneq & \mathcal{N}(A-7I)^2 & \subsetneq & \cdots \\ \dim: & 0 & & 1 & & 4 & & 4 \\ \text{increment:} & & +1 & & +3 & & +0 & \end{array}$$

▷ *These differences either should decrease or stay constant ⇒ No A matrix exists!*
∴ Two possibilities.

Why the increments cannot grow

The increment $\dim\mathcal{N}(A-7I)^k - \dim\mathcal{N}(A-7I)^{k-1}$ equals the number of Jordan blocks of size $\geq k$. Every block of size $\geq k+1$ also has size $\geq k$, so the increments form a non-increasing sequence. In Case III they would be $1, 3, 0$: there would be one block in total but three blocks of size ≥ 2, which is impossible. Equivalently: with $\dim\mathcal{N}(A-7I) = 1$ there is a single Jordan block, so it must be 4×4; then $m(s) = (s-7)^4$ and $\dim\mathcal{N}(A-7I)^2 = 2$, not 4.

9.4 Generalized Eigenvectors and Jordan Chains

Let $x \in \mathcal{N}(\Sigma_i^{m_i})$, $x \notin \mathcal{N}(\Sigma_i^{m_i-1})$:

$$\Sigma_i^{m_i} x = 0, \qquad \Sigma_i^{m_i-1} x \neq 0 \quad \Rightarrow \quad x\text{: } \textbf{generalized eigenvector}$$

$$\{\underbrace{\Sigma_i^{m_i-1}x}_{\text{not zero}},\ \Sigma_i^{m_i-2}x,\ \ldots,\ \Sigma_i x,\ x\} \quad \Rightarrow \quad \text{none of the vectors are equal to zero.} \quad \textit{(Jordan chain)}$$

▷ *If $\Sigma_i^{m_i-2}x$ were zero, then $\Sigma_i(\Sigma_i^{m_i-2}x) = \Sigma_i^{m_i-1}x$ would be zero ⇒ it cannot be.*

Linear independence: suppose

$$\alpha_1\Sigma_i^{m_i-1}x + \alpha_2\Sigma_i^{m_i-2}x + \cdots + \alpha_{m_i-1}\Sigma_i x + \alpha_{m_i}x = 0.$$

Multiplying by $\Sigma_i^{m_i-1}$:

$$\alpha_1\underbrace{\cancel{\Sigma_i^{2m_i-2}}}_{0}x + \alpha_2\underbrace{\cancel{\Sigma_i^{2m_i-3}}}_{0}x + \cdots + \alpha_{m_i-1}\underbrace{\cancel{\Sigma_i^{m_i}}}_{0}x + \alpha_{m_i}\Sigma_i^{m_i-1}x = 0$$
$$\Rightarrow\ \alpha_{m_i}\Sigma_i^{m_i-1}x = 0\ \Rightarrow\ \alpha_{m_i} = 0.$$

Multiplying the remaining equation

$$\alpha_1\Sigma_i^{m_i-1}x + \alpha_2\Sigma_i^{m_i-2}x + \cdots + \alpha_{m_i-1}\Sigma_i x = 0$$

by $\Sigma_i^{m_i-2}$, every term except the last again contains a power $\geq m_i$ of Σ_i:

$$\alpha_{m_i-1}\Sigma_i^{m_i-1}x = 0\ \Rightarrow\ \alpha_{m_i-1} = 0.$$

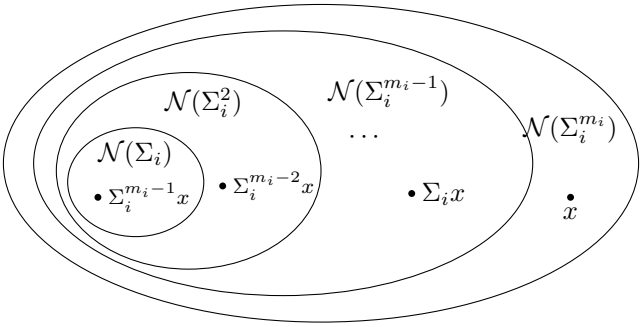
Continuing in this way (multiplying by $\Sigma_i^{m_i-3}, \ldots, \Sigma_i, I$) gives $\alpha_{m_i-2} = \cdots = \alpha_1 = 0$.

∴ **All vectors of the chain are linearly independent.**

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$$\begin{aligned}
\Sigma_i(\Sigma_i^{m_i-1}x) = 0 \quad &\Rightarrow\ \Sigma_i^{m_i-1}x \in \mathcal{N}(\Sigma_i)\\
\Sigma_i^{m_i-1}(\Sigma_i x) = 0 \quad &\Rightarrow\ \Sigma_i x \in \mathcal{N}(\Sigma_i^{m_i-1})\\
\Sigma_i^{m_i-2}\Sigma_i x = \Sigma_i^{m_i-1}x \neq 0 \quad &\Rightarrow\ \Sigma_i x \notin \mathcal{N}(\Sigma_i^{m_i-2})
\end{aligned}$$

▷ *In general $\Sigma_i^k x \in \mathcal{N}(\Sigma_i^{m_i-k})$ but $\Sigma_i^k x \notin \mathcal{N}(\Sigma_i^{m_i-k-1})$: the k-th vector of the chain sits exactly in the "layer" $\mathcal{N}(\Sigma_i^{m_i-k}) \setminus \mathcal{N}(\Sigma_i^{m_i-k-1})$.*



9.5 Computing J and P: Worked Examples

Example 9.8.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix} \qquad d(s) =? \quad m(s) =? \quad \text{JCF} =? \quad P =?$$

$$sI - A = \begin{bmatrix} s-1 & -1 & 1 \\ 1 & s-3 & -2 \\ 0 & 0 & s-1 \end{bmatrix}$$

Expanding along the last row (signs $+\ -\ +$; only the $(3,3)$ entry is nonzero):

$$\begin{aligned} d(s) = \det(sI - A) &= (s-1)\begin{vmatrix} s-1 & -1 \\ 1 & s-3 \end{vmatrix} \\ &= (s-1)\big[(s-1)(s-3)+1\big] \\ &= (s-1)(s^2 - 4s + 4) \\ &\boxed{d(s) = (s-1)(s-2)^2} \qquad \lambda_1 = 2, \quad \lambda_2 = 1 \end{aligned}$$

$$\Sigma_1 = A - 2I = \begin{bmatrix} -1 & 1 & -1 \\ -1 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix} \qquad \text{rank} = 2, \quad N = 1$$

$m(s)$ possibilities:

1- $(s-1)(s-2)$ $\quad\times$

2- $(s-1)(s-2)^2$ $\quad\checkmark$

$$\dim \mathcal{N}(\Sigma_1) = 1 \ \Rightarrow\ \text{not } 2 \quad \Rightarrow \quad \boxed{m(s) = (s-1)(s-2)^2}$$

▷ *If $m_1 = 1$, then $\mathcal{N}(\Sigma_1) = \mathcal{N}(\Sigma_1^{m_1})$ would have dimension $r_1 = 2$; but $\dim \mathcal{N}(\Sigma_1) = 3 - \operatorname{rank}\Sigma_1 = 1$, so $m_1 = 2$.*

$$\Sigma_1^2 = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Sigma_2 = A - I = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \quad \text{rank} = 2, \quad \dim \mathcal{N}(\Sigma_2) = 1$$

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	$\mathcal{N}(\Sigma_1)$	$\subsetneq$	$\mathcal{N}(\Sigma_1^2)$	$=$	$\mathcal{N}(\Sigma_1^3)$
dim :	1		2		2
increment:	$+1$		$+1$		$+0$

	$\mathcal{N}(\Sigma_2)$	$=$	$\mathcal{N}(\Sigma_2^2)$
dim :	1		1

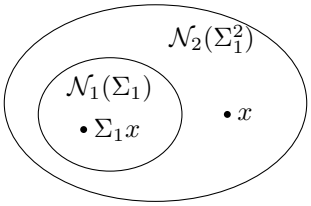
$\Rightarrow$ 1 JB with size 1 for λ_2.

$$(\#\text{ of JB for } \lambda_1) = \dim\mathcal{N}(\Sigma_1) = 1$$
$$(\#\text{ of JB with size} \geq 2 \text{ for } \lambda_1) = \dim\mathcal{N}(\Sigma_1^2) - \dim\mathcal{N}(\Sigma_1) = 1$$
$$(\#\text{ of JB with size} \geq 3 \text{ for } \lambda_1) = \dim\mathcal{N}(\Sigma_1^3) - \dim\mathcal{N}(\Sigma_1^2) = 0$$

▷ *Differences of consecutive counts:* $1-1=0$ *JB of size exactly* 1, $1-0=1$ *JB of size exactly* 2.
$\Rightarrow$ 1 JB with size equal to 2 for λ_1:

$$J = \left[\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 2 & 0 \\ \hline 0 & 0 & 1 \end{array}\right]$$

$$x \in \mathcal{N}(\Sigma_1^2), \quad x \notin \mathcal{N}(\Sigma_1):$$



$$x = \begin{bmatrix}0\\1\\0\end{bmatrix} \quad \text{or} \quad x = \begin{bmatrix}1\\0\\0\end{bmatrix} \quad \cdots$$

P **matrix is not unique**, e.g.:

$$J = P^{-1}AP, \qquad P_1 = 2P$$
$$P_1^{-1}AP_1 = \tfrac{1}{2}P^{-1}AP \cdot 2 = P^{-1}AP = J$$

Choosing the first one:

$$x = \begin{bmatrix}0\\1\\0\end{bmatrix}, \qquad \Sigma_1 x = \begin{bmatrix}1\\1\\0\end{bmatrix}, \qquad \{\Sigma_1 x, x\}$$

- x: generalized eigenvector; $\Sigma_1 x$**: should be eigenvector.**
- $\{\Sigma_1 x, x\}$: linearly indep., form a basis for $\mathcal{N}(\Sigma_1^2)$.

$$y \in \mathcal{N}(\Sigma_2), \quad y \notin \mathcal{N}(\Sigma_2^0) = \mathcal{N}(I) = \{0\} \quad (y \text{ should be nonzero})$$



$$y = \begin{bmatrix}4\\1\\1\end{bmatrix}$$

$$\bar{A} = P^{-1}AP, \qquad P\bar{A} = AP, \qquad P = \left[\begin{array}{cc|c} \Sigma_1 x & x & y \end{array}\right]$$

$$\underbrace{\begin{bmatrix}1&0&4\\1&1&1\\0&0&1\end{bmatrix}}_{P} \underbrace{\left[\begin{array}{cc|c} 2&1&0\\0&2&0\\ \hline 0&0&1\end{array}\right]}_{\bar{A}=J} = A\begin{bmatrix}1&0&4\\1&1&1\\0&0&1\end{bmatrix} = \begin{bmatrix}2&1&4\\2&3&1\\0&0&1\end{bmatrix}$$

▷ *Check: column by column,* $PJ = [\,2\Sigma_1 x \mid \Sigma_1 x + 2x \mid y\,]$ *and* $AP = [\,A\Sigma_1 x \mid Ax \mid Ay\,]$ *both equal the matrix on the right, and* $\det P = 1 \neq 0$*, so* $P^{-1}AP = J$.
For $P = \left[\begin{array}{cc|c} x & \Sigma_1 x & y \end{array}\right]$:

$$J = \left[\begin{array}{cc|c} 2&0& \\ 1&2& \\ \hline & & 1\end{array}\right] \qquad \times \text{ not accepted}$$

▷ *This is also a Jordan CF but not responsible in this course.*

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$$P = \left[\begin{array}{c|cc} y & \Sigma_1 x & x \end{array}\right]: \quad \bar{A} = \left[\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right] \checkmark \qquad P = \left[\begin{array}{c|cc} y & x & \Sigma_1 x \end{array}\right]: \quad \bar{A} = \left[\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & 2 & 0 \\ 0 & 1 & 2 \end{array}\right] \times$$

▷ *The second one (crossed out in the notes): this will not be expected etc. in this course.*

Recipe used in this example

1. Factor $d(s)$; for each λ_i form $\Sigma_i = A - \lambda_i I$ and compute $\dim \mathcal{N}(\Sigma_i^k)$, $k = 1, 2, \ldots$ until it reaches r_i; the first such k is m_i.
2. Increments of these dimensions give the number of JBs of size $\geq k$; their differences give the number of JBs of each size.
3. For each block of size p pick a vector $x \in \mathcal{N}(\Sigma_i^p) \setminus \mathcal{N}(\Sigma_i^{p-1})$ and put the chain $\Sigma_i^{p-1}x, \ldots, \Sigma_i x, x$ (eigenvector first) into consecutive columns of P, choosing the chains so that all columns are linearly independent.

The ordering *eigenvector first* is what produces the 1's above the diagonal; reversing a chain puts them below the diagonal.

9.6 Special Cases

Case I: $A \in \mathbb{C}^{n\times n}$, single eigenvalue λ_i. We have $m_i = r_i$.

$$d(s) = (s-\lambda_i)^{r_i}, \qquad m(s) = (s-\lambda_i)^{m_i = r_i} = d(s)$$

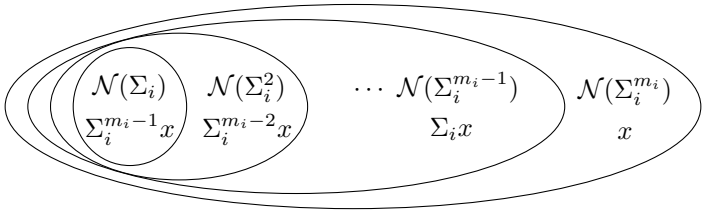
$$\underbrace{\mathbb{C}^n}_{\dim n} = \mathcal{N}(\Sigma_i^{m_i}), \qquad \dim \mathcal{N}(\Sigma_i^{m_i}) = r_i = n, \qquad n = r_i = m_i$$

$$\begin{array}{ccccccccc} \mathcal{N}(\Sigma_i) & \subsetneq & \mathcal{N}(\Sigma_i^2) & \subsetneq & \cdots & \subsetneq & \mathcal{N}(\Sigma_i^{m_i-1}) & \subsetneq & \mathcal{N}(\Sigma_i^{m_i}) \\ 1 & \underbrace{<}_{+1} & 2 & < & \cdots & < & m_i - 1 & \underbrace{<}_{+1} & r_i = m_i = n \end{array}$$

▷ *n dimensions have to be gained in $m_i = n$ strict steps, each of at least $+1$ $\Rightarrow$ every increment is exactly $+1$, in particular* $\dim \mathcal{N}(\Sigma_i) = 1$.

$$\begin{aligned} \#\text{ of JB's} &= 1 \\ \#\text{ of JB's with size} \geq 2 &= 1 \\ \#\text{ of JB's with size} \geq 3 &= 1 \\ &\vdots \\ \#\text{ of JB's with size} \geq m_i &= 1 \\ \#\text{ of JB's with size} \geq m_i + 1 &= 0 \end{aligned} \quad \Rightarrow \textbf{ single JB with size } m_i = r_i = n$$

$$\text{JB} = \begin{bmatrix} \lambda_i & 1 & & & \\ & \lambda_i & 1 & & 0 \\ & & \lambda_i & \ddots & \\ & & & \ddots & 1 \\ 0 & & & & \lambda_i \end{bmatrix}_{n\times n} \qquad J = P^{-1}AP \;\Rightarrow\; P = ?$$



$\underbrace{\{\Sigma_i^{m_i-1}x,\ \Sigma_i^{m_i-2}x,\ \ldots,\ \Sigma_i x,\ x\}}_{m_i=r_i=n \text{ l. indep. vectors}}$ 1 chain with n elements

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$$P = \left[\ \Sigma_i^{m_i-1}x \ \middle|\ \cdots \ \middle|\ \Sigma_i x \ \middle|\ x\ \right], \qquad P\bar{A} = AP$$

$$\left[\ \Sigma_i^{m_i-1}x \ \middle|\ \cdots \ \middle|\ \Sigma_i x \ \middle|\ x\ \right] \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 \\ 0 & \lambda_i & 1 & \cdots & \vdots \\ \vdots & 0 & \lambda_i & \ddots & \vdots \\ \vdots & \vdots & 0 & \ddots & 1 \\ 0 & 0 & 0 & \cdots & \lambda_i \end{bmatrix} = A\left[\ \Sigma_i^{m_i-1}x \ \middle|\ \cdots \ \middle|\ \Sigma_i x \ \middle|\ x\ \right]$$
$$= \left[\ A\Sigma_i^{m_i-1}x \ \middle|\ \cdots \ \middle|\ A\Sigma_i x \ \middle|\ Ax\ \right]$$

$\Rightarrow x \in \mathcal{N}(\Sigma_i^{m_i})$ & $x \notin \mathcal{N}(\Sigma_i^{m_i-1})$:

$$\begin{aligned} &\Sigma_i^{m_i}x = 0, \qquad \Sigma_i^{m_i-1}x \neq 0 \\ &\Sigma_i(\Sigma_i^{m_i-1}x) = 0 \\ &(A-\lambda_i I)(\Sigma_i^{m_i-1}x) = A\Sigma_i^{m_i-1}x - \lambda_i \underbrace{I}_{1}\Sigma_i^{m_i-1}x = 0 \\ &A\Sigma_i^{m_i-1}x = \lambda_i\Sigma_i^{m_i-1}x \end{aligned}$$

$\Rightarrow$ Next vector of the chain:

$$\begin{aligned} &\Sigma_i(\Sigma_i^{m_i-2}x) = \Sigma_i^{m_i-1}x \\ &(A-\lambda_i I)(\Sigma_i^{m_i-2}x) = \Sigma_i^{m_i-1}x \\ &A\Sigma_i^{m_i-2}x = \lambda_i\Sigma_i^{m_i-2}x + \Sigma_i^{m_i-1}x \end{aligned}$$

$\Rightarrow$ and similarly:

$$\begin{aligned} &\Sigma_i(\Sigma_i^{m_i-3}x) = \Sigma_i^{m_i-2}x \\ &(A-\lambda_i I)(\Sigma_i^{m_i-3}x) = \Sigma_i^{m_i-2}x \ \Rightarrow\ A\Sigma_i^{m_i-3}x = \lambda_i\Sigma_i^{m_i-3}x + \Sigma_i^{m_i-2}x \end{aligned}$$

▷ *Hence column k of AP equals $\lambda_i \cdot$ (column k of P) + (column $k-1$ of P), which is exactly column k of PJ: a λ_i on the diagonal and a 1 just above it.*

$\{\Sigma_i^{m_i-1}x,\ \Sigma_i^{m_i-2}x,\ \ldots,\ \Sigma_i x,\ x\}$ Single chain with size m_i.

Case II: $A \in \mathbb{C}^{n\times n}$, single eigenvalue λ_i, $m_i = 1$.

$$d(s) = (s-\lambda_i)^{r_i}, \qquad m(s) = (s-\lambda_i)$$

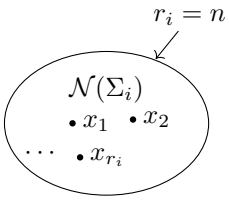
$$\mathbb{C}^n = \underset{r_i=n}{\mathcal{N}(\Sigma_i^1)} \qquad \begin{array}{ccccc} \mathcal{N}(\Sigma_i) & = & \mathcal{N}(\Sigma_i^2) & = & \cdots \\ r_i & \underbrace{=}_{+0} & r_i & & \end{array}$$

$$\#\text{ of JB} = r_i = n$$
$$\#\text{ of JB size} \geq 2 = 0 \quad \Rightarrow \; n \text{ JB size } = 1$$

▷ m_i ***is the size of the largest JB for*** λ_i**.**

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$$J = \begin{bmatrix} \boxed{\lambda_i} & & & \\ & \boxed{\lambda_i} & & \\ & & \boxed{\lambda_i} & \\ & & & \ddots \; \boxed{\lambda_i} \end{bmatrix}$$



$x_1 \in \mathcal{N}(\Sigma_i), \quad x_1 \notin \underbrace{\mathcal{N}(\Sigma_i^0)}_{I} \Rightarrow x_1 \neq 0$: chain $\{x_1\}$

$x_2 \in \mathcal{N}(\Sigma_i), \quad x_2 \notin \mathcal{N}(\Sigma_i^0)$: chain $\{x_2\}$

$\Rightarrow x_1$ & x_2 are linearly indep.

$\vdots$

$x_{r_i} \in \mathcal{N}(\Sigma_i), \quad x_{r_i} \notin \mathcal{N}(\Sigma_i^0)$: chain $\{x_{r_i}\}$

$\Rightarrow r_i = n$ chains with size 1.

Each one of the vectors are eigenvectors.

$$P = \left[\begin{array}{c|c|c|c} x_1 & x_2 & \cdots & x_{r_i} \end{array}\right]$$
$$P\bar{A} = AP$$

$$\left[\begin{array}{c|c|c|c} x_1 & x_2 & \cdots & x_{r_i} \end{array}\right] \underbrace{\begin{bmatrix} \lambda_i & & 0 \\ & \ddots & \\ 0 & & \lambda_i \end{bmatrix}}_{\bar{A}} = \begin{bmatrix} Ax_1 & Ax_2 & \cdots & Ax_{r_i} \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_i x_1 & \lambda_i x_2 & \cdots & \lambda_i x_{r_i} \end{bmatrix} \quad \text{(as they are eigenvectors)}$$

$$J = \lambda_i I$$
$$J = P^{-1}AP \;\Rightarrow\; A = PJP^{-1} = P(\lambda_i I)P^{-1} = \lambda_i PP^{-1} = \lambda_i I$$

Equivalently, from the minimal polynomial:

$$m(s) = s - \lambda_i$$
$$m(A) = 0_{n\times n} \;\Rightarrow\; A - \lambda_i I = 0_{n\times n} \;\Rightarrow\; A = \lambda_i I$$

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9.7 Further Examples

Example 9.9. A is a 6×6 matrix, λ_i is the only eigenvalue.

$$d(s) = (s-\lambda_i)^6, \qquad \dim \mathcal{N}(\Sigma_i) = 2,$$
$$m(s) = (s-\lambda_i)^3, \qquad \dim \mathcal{N}(\Sigma_i^2) = 4.$$

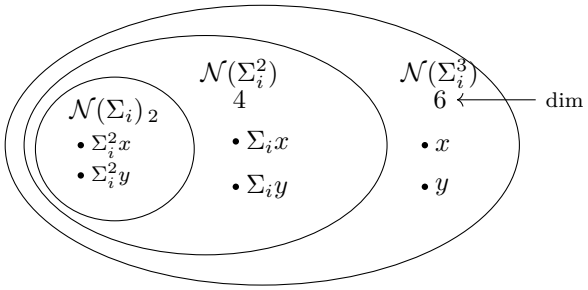
$$\underset{2}{\mathcal{N}(\Sigma_i)} \subsetneq \underset{4}{\mathcal{N}(\Sigma_i^2)} \subsetneq \underset{6}{\mathcal{N}(\Sigma_i^3)} = \underset{6}{\mathcal{N}(\Sigma_i^4)} \qquad \mathbb{C}^6 = \mathcal{N}(\Sigma_i^3)$$

- $m(s) = (s-\lambda_i)$ should be happen for diagonalizability.
- At most 2 linearly indep. eigenvectors.
- $\Rightarrow$ **not diagonalizable** (it should be 6).

Jordan blocks:

$$\left.\begin{array}{l} \#\text{ JB} = 2 \\ \#\text{ JB size} \geq 2 = 2 \\ \#\text{ JB size} \geq 3 = 2 \\ \#\text{ JB size} \geq 4 = 0 \end{array}\right\} \text{ 2 JB w.s. 3} \qquad J = \left[\begin{array}{ccc|ccc} \lambda_i & 1 & 0 & & & \\ 0 & \lambda_i & 1 & & 0 & \\ & & \lambda_i & & & \\ \hline & & & \lambda_i & 1 & 0 \\ & 0 & & & \lambda_i & 1 \\ & & & 0 & & \lambda_i \end{array}\right]$$

▷ *# JB of size* $\geq k = \dim \mathcal{N}(\Sigma_i^k) - \dim \mathcal{N}(\Sigma_i^{k-1})$*: here* $2-0$, $4-2$, $6-4$, $6-6$.



Chains:

$$x \in \mathcal{N}(\Sigma_i^3),\ x \notin \mathcal{N}(\Sigma_i^2): \qquad \{\Sigma_i^2 x, \Sigma_i x, x\} \Rightarrow \text{ alone not sufficient to form a basis}$$
$$y \in \mathcal{N}(\Sigma_i^3),\ y \notin \mathcal{N}(\Sigma_i^2): \qquad \{\Sigma_i^2 y, \Sigma_i y, y\}$$

▷ *The two chains together should be linearly indep. to form a basis.*

$\Rightarrow$ Condition: $\Sigma_i^2 x$ **&** $\Sigma_i^2 y$ **should be linearly indep.**

Why this condition suffices

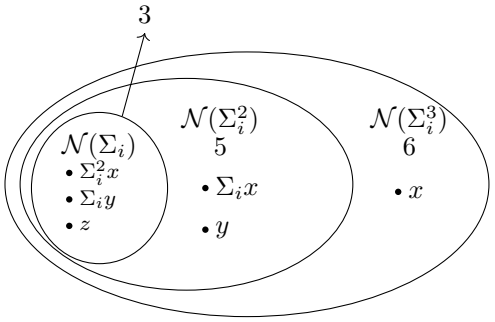
Suppose a combination of the six vectors vanishes. Applying Σ_i^2 kills everything except the coefficients of x and y, leaving $a\Sigma_i^2 x + b\Sigma_i^2 y = 0$, so $a = b = 0$ by the condition. Applying Σ_i to what remains gives the same relation for the coefficients of $\Sigma_i x, \Sigma_i y$, and finally for $\Sigma_i^2 x, \Sigma_i^2 y$. So independence of the eigenvectors at the bottom of the chains implies independence of the whole set (the same argument as for a single chain).

$$P = \left[\begin{array}{ccc|ccc} \Sigma_i^2 x & \Sigma_i x & x & \Sigma_i^2 y & \Sigma_i y & y \end{array}\right]$$

$\Rightarrow$ Continue same example but $\dim \mathcal{N}(\Sigma_i) = 3$, $\dim \mathcal{N}(\Sigma_i^2) = 5$.

$$\begin{array}{rcl} \#\text{ JB} = 3 & & m_i = \text{size of largest JB} = 3 \\ \#\text{ JB size} \geq 2 = 2 & & \text{1 JB size 3} \\ \geq 3 = 1 & \Longrightarrow & \text{1 JB size 2} \\ \geq 4 = 0 & & \text{1 JB size 1} \end{array}$$

$$J = \left[\begin{array}{ccc|cc|c} \lambda_i & 1 & 0 & & & \\ 0 & \lambda_i & 1 & & & \\ 0 & 0 & \lambda_i & & & \\ \hline & & & \lambda_i & 1 & \\ & & & 0 & \lambda_i & \\ \hline & & & & & \lambda_i \end{array}\right]$$



Chains:

$$\begin{aligned} &x \in \mathcal{N}(\Sigma_i^3),\ x \notin \mathcal{N}(\Sigma_i^2): && \{\Sigma_i^2 x, \Sigma_i x, x\} \\ &y \in \mathcal{N}(\Sigma_i^2),\ y \notin \mathcal{N}(\Sigma_i): && \{\Sigma_i y, y\} \\ &z \in \mathcal{N}(\Sigma_i),\ z \notin \mathcal{N}(\Sigma_i^0) \Rightarrow z \neq 0: && \{z\} \end{aligned}$$

$\Rightarrow \Sigma_i^2 x,\ \Sigma_i y,\ z$ **should be linearly indep.**

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$$P = \left[\begin{array}{ccc|c|c} \Sigma_i^2 x & \Sigma_i x & x & \Sigma_i y & y & z \end{array}\right]$$

Example 9.10.

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 2 \end{bmatrix}$$

$d(s) = (s-1)^3(s-2)$ (know to calculate), $\lambda_1 = 1,\ \lambda_2 = 2$

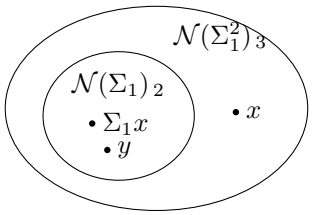
$m(s)\,?\quad J\,?\quad P\,?$

$$\Sigma_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}, \quad \Sigma_1^2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

$\dim \mathcal{N}(\Sigma_1) = 2, \quad \text{rank} = 2$
it should increase (to $r_1 = 3$) $\quad \Rightarrow \quad (s-1)^2(s-2) = m(s)$

$\underset{2}{\mathcal{N}(\Sigma_1)} \subsetneq \underset{3}{\mathcal{N}(\Sigma_1^2)} \Rightarrow$ 1 JB w.s. 2, 1 JB w.s. 1 $\qquad \underset{1}{\mathcal{N}(\Sigma_2)} = \underset{1}{\mathcal{N}(\Sigma_2^2)} \Rightarrow$ 1 JB w.s. 1

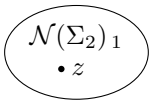
$$\begin{aligned} \#\text{ of JB} &= 2 \\ \text{size JB} \geq 2 &= 1 \\ \geq 3 &= 0 \end{aligned} \qquad \mathbb{C}^4 = \underset{3}{\mathcal{N}(\Sigma_1^2)} \oplus \underset{1}{\mathcal{N}(\Sigma_2)} \qquad J = \left[\begin{array}{cc|c|c} 1 & 1 & & \\ 0 & 1 & & \\ \hline & & 1 & \\ \hline & & & 2 \end{array}\right]$$



$x \in \mathcal{N}(\Sigma_1^2) \notin \mathcal{N}(\Sigma_1)$
$y \in \mathcal{N}(\Sigma_1)$ & $y \neq 0$; $\quad y$ & $\Sigma_1 x$ lin. indep.

$$x = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \Sigma_1 x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{(eigenvector)}$$

Chains: $\{\Sigma_1 x,\ x\}$ and $\{y\}$.
$z \in \mathcal{N}(\Sigma_2)$, $z \neq 0$:



$$z = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \qquad P = \left[\begin{array}{c|c|c} \Sigma_1 x & x & y \mid z \end{array}\right]$$

Check

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \qquad AP = PJ = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 2 \end{bmatrix}, \qquad \det P = 1 \neq 0,$$

so indeed $P^{-1}AP = J$. (Σ_1^2 has rank 1, so $\dim \mathcal{N}(\Sigma_1^2) = 3 = r_1$, confirming $m_1 = 2$.)

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10 Hermitian and Positive Definite Matrices

Syllabus: Week 13 — Hermitian, positive definite and positive semi-definite matrices

10.1 Hermitian Matrices and Their Eigenstructure

* An $n \times n$, in general, complex matrix $A \in \mathbb{C}^{n\times n}$ is called **Hermitian** iff

$$\bar{A}^T = A, \qquad \text{i.e.} \qquad A^H = A,$$

where $^{-T} \triangleq {}^{H}$: the complex conjugate transpose is denoted as Hermitian.

* If $A \in \mathbb{R}^{n\times n}$:

$$A^T = A: \text{ Symmetric matrices are Hermitian.}$$

* Example:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \lambda_1 = 3,\ \lambda_2 = 1, \qquad e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

* In addition to being linearly independent, eigenvectors of the Hermitian matrices are **orthogonal**.

* **Eigenvalues are always real** in Hermitian matrices.

Theorem 10.1. Let $A \in \mathbb{C}^{n\times n}$ be Hermitian. Then, $\langle x, Ax\rangle$ **is real** $\forall x \in \mathbb{C}^n$.

$*$ $\langle x, y\rangle = y^H x$: ℓ_2-inner product. Properties:

$$\begin{aligned} &\langle \alpha x, y\rangle = \alpha \langle x, y\rangle, && \langle x, x\rangle \text{ is real,} \\ &\langle x, \alpha y\rangle = \bar{\alpha} \langle x, y\rangle, && \langle x, x\rangle \geq 0 \ \ \forall x \in \mathbb{C}^n, \\ &\langle x, y\rangle = \overline{\langle y, x\rangle}, && \langle x, x\rangle = 0 \iff x = 0, \end{aligned}$$

$$\langle x, Ay\rangle = (Ay)^H x = y^H (A^H x) = \left\langle A^H x, y \right\rangle.$$

Proof.

$$\langle x, Ax\rangle = (Ax)^H x = x^H A^H x \overset{A^H = A}{=} x^H A x : \textbf{ Quadratic Form}$$

$$\begin{aligned} \overline{\langle x, Ax\rangle} &= \langle Ax, x\rangle \\ &= \left\langle x, A^H x \right\rangle \qquad \text{(identity above for } A^H \text{, with } y = x) \\ &= \langle x, Ax\rangle \qquad \text{if } A \text{ is Hermitian} \end{aligned}$$

The complex conjugate is equal to itself $\Rightarrow \langle x, Ax\rangle$ is real. □

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Theorem 10.2. If A is Hermitian, then, **its eigenvalues are all real**.

Proof. λ_i, e_i: one eigenvalue of A and corresponding eigenvector.

$$\begin{aligned} \langle x, Ax\rangle \Big|_{x=e_i} &= \langle e_i, Ae_i\rangle \\ &= \left\langle A^H e_i, e_i \right\rangle \\ &= \langle Ae_i, e_i\rangle \\ &= \langle \lambda_i e_i, e_i\rangle \\ &= \lambda_i \underbrace{\langle e_i, e_i\rangle}_{\text{always real}} \ \Rightarrow \text{ Should be real} \end{aligned}$$

$$\Rightarrow \ \underline{\underline{\lambda_i \text{ is real}}}$$

▷ *By the previous theorem $\langle e_i, Ae_i\rangle$ is real, and $\langle e_i, e_i\rangle = \|e_i\|_2^2 > 0$ since $e_i \neq 0$; hence $\lambda_i = \langle e_i, Ae_i\rangle \,/\, \langle e_i, e_i\rangle$ is real.*

□

Theorem 10.3. Let A be Hermitian, & λ_i, λ_j are two distinct eigenvalues of A, with eigenvectors e_i & e_j. e_i **&** e_j **are orthogonal.**

$$\lambda_i \neq \lambda_j \ \Rightarrow \ \langle e_i, e_j\rangle = 0$$

Proof.

$$\begin{aligned} \langle e_i, Ae_j\rangle &= \langle e_i, \lambda_j e_j\rangle = \lambda_j \langle e_i, e_j\rangle \\ \| \\ \langle Ae_i, e_j\rangle &= \langle \lambda_i e_i, e_j\rangle = \lambda_i \langle e_i, e_j\rangle \end{aligned}$$

$$\begin{aligned} \lambda_j \langle e_i, e_j\rangle &= \lambda_i \langle e_i, e_j\rangle \\ \underbrace{(\lambda_i - \lambda_j)}_{\neq 0} \underbrace{\langle e_i, e_j\rangle}_{=0} &= 0 \end{aligned}$$

▷ *Top row:* $\langle e_i, \lambda_j e_j\rangle = \bar{\lambda}_j \langle e_i, e_j\rangle = \lambda_j \langle e_i, e_j\rangle$ *since* λ_j *is real. Left equality:* $\langle e_i, Ae_j\rangle = \langle A^H e_i, e_j\rangle = \langle Ae_i, e_j\rangle$.

$$\mathcal{N}(A-\lambda_i I) \perp \mathcal{N}(A-\lambda_j I) \ : \text{ Eigenspaces are also orthogonal.}$$

□

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Theorem 10.4. Let A be Hermitian. Then, its minimal polynomial is

$$m(s) = (s-\lambda_1)(s-\lambda_2)\cdots(s-\lambda_k)$$

when A has k distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$, i.e. $m_i = 1\ \forall i$.

In general:

$$\begin{aligned} d(s) &= (s-\lambda_1)^{r_1}(s-\lambda_2)^{r_2}\cdots(s-\lambda_k)^{r_k} \\ m(s) &= (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}, \qquad 1 \le m_i \le r_i \text{ in general case.} \end{aligned}$$

Proof. Let $\Sigma_i = A - \lambda_i I$. In general

$$\mathcal{N}(\Sigma_i) \subsetneq \mathcal{N}(\Sigma_i^2) \subsetneq \cdots \subsetneq \mathcal{N}(\Sigma_i^{m_i}) = \mathcal{N}(\Sigma_i^{m_i+1}).$$

$$\mathcal{N}(\Sigma_i) = \mathcal{N}(\Sigma_i^2) \ : \text{ Theorem's Claim.}$$

Show that

$$\begin{aligned} &\mathcal{N}(\Sigma_i) \subset \mathcal{N}(\Sigma_i^2): \quad \Sigma_i x = 0 \text{ implies } \Sigma_i(\Sigma_i x) = 0 \qquad \checkmark \\ &\mathcal{N}(\Sigma_i^2) \subset \mathcal{N}(\Sigma_i): \end{aligned}$$

$$\begin{aligned} x \in \mathcal{N}(\Sigma_i^2) \ &\Rightarrow\ \Sigma_i^2 x = 0 \\ &\Rightarrow\ \left\langle x, \Sigma_i^2 x\right\rangle = 0 \\ &\Rightarrow\ \langle x, \Sigma_i \Sigma_i x\rangle = 0 \\ &\Rightarrow\ \left\langle \Sigma_i^H x, \Sigma_i x\right\rangle = 0 \\ &\Rightarrow\ \langle \Sigma_i x, \Sigma_i x\rangle = 0 \\ &\Rightarrow\ \|\Sigma_i x\|_2^2 = 0 \ \Rightarrow\ \Sigma_i x = 0 \ \Rightarrow\ x \in \mathcal{N}(\Sigma_i) \end{aligned}$$

▷ *Used:* $\Sigma_i^H = (A-\lambda_i I)^H = A^H - \bar{\lambda}_i I^H = A - \lambda_i I = \Sigma_i$ *(since* λ_i *is real).*

Hence $\mathcal{N}(\Sigma_i) = \mathcal{N}(\Sigma_i^2)$, the chain of null spaces stops at the first step, and $m_i = 1$ for every i. □

- Direct sum decompositions:

$$\begin{aligned} \mathbb{C}^n &= \mathcal{N}(\Sigma_1^{m_1}) \oplus \mathcal{N}(\Sigma_2^{m_2}) \oplus \cdots \oplus \mathcal{N}(\Sigma_k^{m_k}) \\ \mathbb{C}^n &= \mathcal{N}(\Sigma_1) \overset{\perp}{\oplus} \mathcal{N}(\Sigma_2) \overset{\perp}{\oplus} \cdots \overset{\perp}{\oplus} \mathcal{N}(\Sigma_k) \end{aligned}$$

  ▷ *Orthogonal as every subspace defines a unique eigenspace: since* $m_i = 1$, $\mathcal{N}(\Sigma_i^{m_i}) = \mathcal{N}(\Sigma_i)$ *is the eigenspace of* λ_i*, and eigenspaces of distinct eigenvalues are orthogonal.*

- m_i is the size of the largest Jordan block for λ_i.

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∗ For a Hermitian matrix:

$$\underset{\dim r_i}{\mathcal{N}(\Sigma_i)} = \underset{\dim r_i}{\mathcal{N}(\Sigma_i^2)} \quad \Rightarrow \quad \begin{array}{l}(\#\text{ of JB's for } \lambda_i) = r_i \\ (\#\text{ of JB's with size} \geq 2) = 0 \\ r_i \text{ JB's w.s.} = 1 \text{ for } \lambda_i\end{array}$$

$$J = \left[\begin{array}{ccc|ccc|c|ccc} \lambda_1 & 0 & 0 & & & & & & & \\ 0 & \lambda_1 & 0 & & & & & & & \\ & & \lambda_1 & & & & & & & \\ \hline & & & \lambda_2 & 0 & 0 & & & & \\ & & & & \lambda_2 & 0 & & & & \\ & & & 0 & & \lambda_2 & & & & \\ \hline & & & & & & \ddots & & & \\ \hline & & & & & & & \lambda_k & 0 & 0 \\ & & & & & & & & \lambda_k & 0 \\ & & & & & & & 0 & & \lambda_k \end{array}\right] : \text{Diagonal Matrix.}$$

▷ *The λ_1 block is $r_1 \times r_1$, the λ_2 block $r_2 \times r_2$, ..., the λ_k block $r_k \times r_k$.*

∗ J is diagonal for a Hermitian matrix. **Any Hermitian matrix is diagonalizable**:

$$J = \begin{bmatrix} \lambda_1 I_{r_1 \times r_1} & & & \\ & \lambda_2 I_{r_2 \times r_2} & & \\ & & \ddots & \\ & & & \lambda_k I_{r_k \times r_k} \end{bmatrix}_{n \times n}$$

∗ Any Hermitian matrix has n linearly independent eigenvectors.

∗ For diagonalization with orthonormal eigenvectors (unitary diagonalization), we need to have:

$N^H N = N N^H$: **Normal matrix** (Outside of the scope)
$N^H N = N^2 = N N^H$ for Hermitian matrices (Inherently satisfy normal matrix condition).

Summary.

$$\text{Hermitian} \implies \left\{\begin{array}{l}\lambda_i\text{'s are real} \\ e_i \perp e_j \text{ when } \lambda_i \neq \lambda_j \\ m_i = 1 \ \ \forall i \\ J \text{ is diagonal} \\ n \text{ linearly independent eigenvectors}\end{array}\right\} \impliedby \begin{array}{l}A \text{ is normal} \\ (\text{for these 3 statements})\end{array}$$

▷ *The three bracketed statements are equal and they imply diagonalizability.*

▷ *$\not\Longleftarrow$: the reverse implication (to Hermitian) is not true!*

Hermitian, normal, diagonalizable

- Hermitian ⇒ normal ⇒ diagonalizable; neither arrow can be reversed.
- A is normal $\iff$ A has n *orthonormal* eigenvectors $\iff$ $A = U\Lambda U^H$ with U unitary. Hermitian matrices are exactly the normal matrices with real eigenvalues.

- Counterexamples: $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ is normal (and diagonalizable over $\mathbb{C}$, eigenvalues $\pm j$) but not Hermitian; $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ is diagonalizable (two distinct eigenvalues, so $m_i = 1$) but not normal, its eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are not orthogonal.

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10.2 Unitary Diagonalization

Theorem 10.5 (Unitary diagonalization)**.** Let A be an $n \times n$ Hermitian matrix. Then, **there exists a unitary matrix** P s.t.

$$J = P^H A P.$$

What J looks like here

For a Hermitian matrix the Jordan form J is *diagonal* with *real* entries: the eigenvalues of $A = A^H$ are real, and A is always diagonalizable (every eigenvalue has $m_j = 1$ in the minimal polynomial). The theorem adds that the diagonalizing matrix can be chosen *unitary*, i.e. $P^{-1} = P^H$, so that $A = PJP^H$ (unitary diagonalization).

⋆ **Unitary matrix**: $P^{-1} = P^H$.

→ **Orthogonal matrix** (real-valued unitary matrix): $P^{-1} = P^T$.

⋆ $P^{-1}P = PP^{-1} = I$, i.e. $P^H P = PP^H = I$.

Columns of a unitary matrix.

$$P = \begin{bmatrix} p_1 \vdots p_2 \vdots \cdots \vdots p_n \end{bmatrix}, \qquad P^H = \begin{bmatrix} p_1^H \\ p_2^H \\ \vdots \\ p_n^H \end{bmatrix}$$

$$P^H P = \begin{bmatrix} p_1^H \\ p_2^H \\ \vdots \\ p_n^H \end{bmatrix} \begin{bmatrix} p_1 \vdots p_2 \vdots \cdots \vdots p_n \end{bmatrix} = \begin{bmatrix} p_1^H p_1 & p_1^H p_2 & \cdots & p_1^H p_n \\ p_2^H p_1 & p_2^H p_2 & \cdots & p_2^H p_n \\ \vdots & & & \vdots \\ p_n^H p_1 & & & p_n^H p_n \end{bmatrix}$$

$$= \begin{bmatrix} \langle p_1, p_1 \rangle & \langle p_2, p_1 \rangle & \cdots & \langle p_n, p_1 \rangle \\ \langle p_1, p_2 \rangle & & & \vdots \\ \vdots & & & \\ \langle p_1, p_n \rangle & & & \langle p_n, p_n \rangle \end{bmatrix}$$

▷ *All inner products, similar to Gram matrix.*

$\Rightarrow$ **Columns of P are orthonormal!**

$$\langle p_i, p_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

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Rows of a unitary matrix.

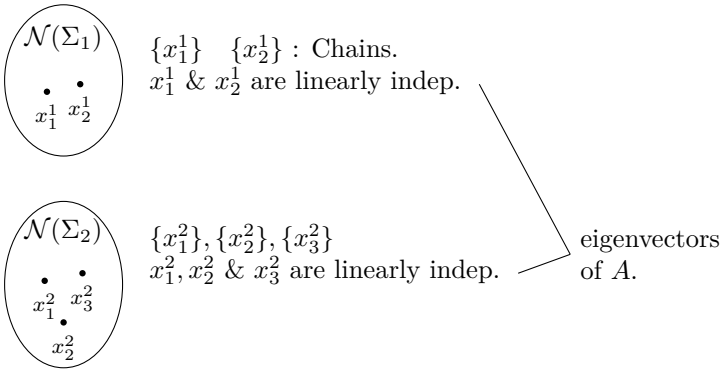
$$P = \begin{bmatrix} \overline{r_1} \\ \overline{r_2} \\ \overline{\vdots} \\ \overline{r_n} \end{bmatrix}, \qquad \underbrace{\begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}}_{P} \underbrace{\begin{bmatrix} r_1^H \vdots r_2^H \vdots \cdots \vdots r_n^H \end{bmatrix}}_{P^H} = I$$

$\Rightarrow$ **Rows are also orthonormal!!**

Proof (kind of proof, not an actual one). A is Hermitian, $A \in \mathbb{C}^{5\times 5}$; λ_1, λ_2 are eigenvalues of A, $\lambda_1 \neq \lambda_2$, and $\Sigma_j = A - \lambda_j I$:

$$d(s) = (s-\lambda_1)^2(s-\lambda_2)^3$$
$$m(s) = (s-\lambda_1)(s-\lambda_2)$$

$$\mathbb{C}^5 = \underbrace{\mathcal{N}(\Sigma_1)}_{\dim 2} \overset{\perp}{\oplus} \underbrace{\mathcal{N}(\Sigma_2)}_{\dim 3}$$



$$J = \begin{bmatrix} \lambda_1 & & & & \\ & \lambda_1 & & 0 & \\ & & \lambda_2 & & \\ & 0 & & \lambda_2 & \\ & & & & \lambda_2 \end{bmatrix}$$

$$P = \begin{bmatrix} x_1^1 \vdots x_2^1 \vdots x_1^2 \vdots x_2^2 \vdots x_3^2 \end{bmatrix}_{5\times 5}$$

P is not directly unitary. The theorem states that there exists a unitary matrix P, which can be found. How can we convert P to a unitary one?

$$\underbrace{x_1^1 \quad x_2^1}_{} \qquad \underbrace{x_1^2 \quad x_2^2 \quad x_3^2}_{}$$

▷ *Vectors are orthogonal between the two sets. Vectors inside the sets are not necessarily orthogonal.*

Why eigenvectors of different eigenvalues are orthogonal

If $Ax = \lambda_1 x$, $Ay = \lambda_2 y$ with $A = A^H$ (so $\lambda_1, \lambda_2 \in \mathbb{R}$), then

$$\lambda_1\, y^H x = y^H A x = (Ay)^H x = \lambda_2\, y^H x \;\Rightarrow\; (\lambda_1 - \lambda_2)\, y^H x = 0 \;\Rightarrow\; y^H x = 0.$$

Gram–Schmidt inside each eigenspace keeps the vectors in that eigenspace (they remain eigenvectors), so the orthonormalized set still diagonalizes A.

⋆ We can apply **Gram–Schmidt orthonormalization to the two basis sets** and obtain the unitary matrix

$$P = [\, e_1^1 \,\vdots\, e_2^1 \,\vdots\, e_1^2 \,\vdots\, e_2^2 \,\vdots\, e_3^2 \,].$$

□

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10.3 Spectral Decomposition

With $J = P^H A P$, we obtain the **spectral decomposition** of A:

$$\begin{aligned}
A &= PJP^H \\
&= [\, e_1 \,\vdots\, e_2 \,\vdots\, \cdots \,\vdots\, e_n \,] \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} e_1^H \\ \hline e_2^H \\ \hline \vdots \\ \hline e_n^H \end{bmatrix} \\
&= [\, \lambda_1 e_1 \,\vdots\, \lambda_2 e_2 \,\vdots\, \cdots \,\vdots\, \lambda_n e_n \,] \begin{bmatrix} e_1^H \\ \hline e_2^H \\ \hline \vdots \\ \hline e_n^H \end{bmatrix} \\
&= \lambda_1 e_1 e_1^H + \lambda_2 e_2 e_2^H + \cdots + \lambda_n e_n e_n^H
\end{aligned}$$

$$\boxed{A = \sum_{i=1}^{n} \lambda_i e_i e_i^H}$$

Explanation

Each $e_i e_i^H$ is the orthogonal projection onto $\mathrm{sp}\{e_i\}$ (rank one, $(e_i e_i^H)^2 = e_i e_i^H$). So a Hermitian matrix is a weighted sum of orthogonal projections onto its eigen-directions, weighted by the (real) eigenvalues. Also $\sum_i e_i e_i^H = PP^H = I$.

10.4 Positive Definite and Positive Semi-Definite Matrices

Definition 10.6. A Hermitian matrix is said to be **positive definite** iff

$$\langle x, Ax \rangle > 0 \quad \forall x \neq 0 \qquad (x^H A x > 0).$$

A Hermitian matrix is said to be **positive semi-definite** iff

$$\langle x, Ax \rangle \geq 0 \quad \forall x \qquad (x^H A x \geq 0).$$

Theorem 10.7. If a Hermitian matrix is positive definite, or positive semi-definite, then

$$\underset{\text{(positive definite)}}{\lambda_i > 0} \quad \text{or} \quad \underset{\text{(positive semi-definite)}}{\lambda_i \geq 0} \quad \text{for } \forall i.$$

Proof (only for positive-definite case). Let λ_i, e_i be an eigenpair, $e_i \neq 0$.

$$\begin{aligned} \langle e_i, Ae_i \rangle &> 0 \\ \langle e_i, \lambda_i e_i \rangle &> 0 \\ \lambda_i \langle e_i, e_i \rangle &> 0 \\ \lambda_i \|e_i\|^2 &> 0 \end{aligned}$$

▷ $\|e_i\|^2$ *is positive (as* $e_i \neq 0$*)* $\Rightarrow \lambda_i$ *should be positive.*

The semi-definite case is identical with "$>$" replaced by "$\geq$". □

Converse

The converse also holds: with the spectral decomposition, $x^H Ax = \sum_i \lambda_i x^H e_i e_i^H x = \sum_i \lambda_i \left| e_i^H x \right|^2$. If all $\lambda_i > 0$ this is > 0 for $x \neq 0$ (not all $e_i^H x$ vanish since $\{e_i\}$ is a basis). Hence **a Hermitian A is positive definite $\iff$ all $\lambda_i > 0$** (semi-definite $\iff$ all $\lambda_i \geq 0$).

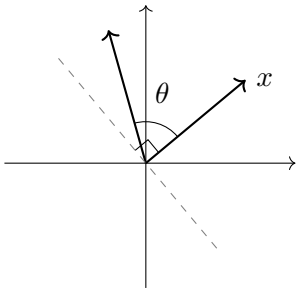
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Geometric interpretation. $\star$ $\langle x, y \rangle = \|x\| \, \|y\| \cdot \cos\theta$, so

$$\langle x, Ax \rangle = \underbrace{\|x\|}_{>0} \underbrace{\|Ax\|}_{>0} \cos\theta > 0 \quad \forall x \neq 0.$$

▷ $\|Ax\| > 0$*, otherwise the inner product would be 0.*

▷ $\cos\theta > 0 \hookrightarrow -90° < \theta < 90°$.



11 Functions of a Matrix

Syllabus: Week 14 — functions of a matrix

11.1 Definition via Power Series

Functions of a matrix are *not* applied entrywise:

$$\exp\left(\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}\right) \neq \begin{bmatrix}e^2 & e\\ e & e^2\end{bmatrix} \qquad \text{MATLAB: } \cancel{\texttt{exp(A)}},\ \texttt{expm(A)}\ \checkmark$$

$$\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}^{1/2} \neq \begin{bmatrix}\sqrt{2} & 1\\ 1 & \sqrt{2}\end{bmatrix} \qquad \text{MATLAB: } \cancel{\texttt{sqrt(A)}},\ \texttt{sqrtm(A)}\ \checkmark$$

Motivation:

$$\begin{aligned}&\dot{x} = Ax, && \dot{x} = ax,\\ &x(t) = e^{At}x_0, && x(t) = e^{at}x_0,\end{aligned} \qquad e^{At} = \sum_{i=0}^{\infty} \frac{(At)^i}{i!}.$$

Taylor series.

$$f(s) = \sum_{i=0}^{\infty} \frac{f^{(i)}(s_0)}{i!}(s - s_0)^i, \qquad f^{(i)}(s_0) = \frac{d^i f}{ds^i}(s_0)$$

If $s_0 = 0$:

$$f(s) = \sum_{i=0}^{\infty} \underbrace{\frac{f^{(i)}(0)}{i!}}_{\alpha_i} s^i \quad \Longrightarrow \quad f(s) = \sum_{i=0}^{\infty} \alpha_i s^i$$

For a matrix:

$$(*) \qquad f(A) = \sum_{i=0}^{\infty} \frac{f^{(i)}(s_0)}{i!}(A - s_0 I)^i$$

$$f(A) = \sum_{i=0}^{\infty} \alpha_i A^i$$

Reduction by Cayley–Hamilton.

$$\begin{aligned}d(s) &= (s - \lambda_1)^{r_1} \cdots (s - \lambda_k)^{r_k}\\ d(s) &= s^n + d_1 s^{n-1} + \cdots + d_n\\ d(A) &= 0_{n\times n}\\ A^n &= -d_1 A^{n-1} - d_2 A^{n-2} - \cdots - d_n I\end{aligned}$$

Hence **all powers A^i, $i \geq n$, can be written using $I, A, \ldots, A^{n-1}$**:

$$f(A) = \sum_{i=0}^{\infty} \alpha_i A^i = c_0 I + c_1 A + c_2 A^2 + \cdots + c_{n-1} A^{n-1}$$

▷ *$c_0, c_1, \ldots, c_{n-1}$: n unknown coefficients.*

Using the minimal polynomial instead: p. 85

$$\begin{aligned}m(s) &= (s - \lambda_1)^{m_1} \cdots (s - \lambda_k)^{m_k}, && \ell = m_1 + m_2 + \cdots + m_k\\ m(s) &= s^\ell + g_1 s^{\ell-1} + \cdots + g_\ell, && m(A) = 0_{n\times n}\\ A^\ell &= -g_1 A^{\ell-1} - g_2 A^{\ell-2} - \cdots - g_\ell I\end{aligned}$$

$$f(A) = c_0 I + c_1 A + \cdots + c_{\ell-1} A^{\ell-1}$$

▷ $c_0, c_1, \ldots, c_{\ell-1}$: *$\ell$ unknowns, $\ell \leq n$.*

$$\left.\begin{aligned} f(s) &= \sum_{i=0}^{\infty} \alpha_i s^i, \quad & f(A) &= \sum_{i=0}^{\infty} \alpha_i A^i \\ p(s) &= \sum_{i=0}^{\ell-1} c_i s^i, \quad & P(A) &= \sum_{i=0}^{\ell-1} c_i A^i \end{aligned}\right\rangle \boxed{f(A) = P(A)}.$$

Why a polynomial of degree $\ell - 1$ suffices

Repeatedly substituting $A^\ell = -g_1 A^{\ell-1} - \cdots - g_\ell I$ reduces every power A^i $(i \geq \ell)$ to a combination of $I, A, \ldots, A^{\ell-1}$; summing the series gives $f(A) = \sum_{i=0}^{\ell-1} c_i A^i$. The coefficients c_i depend on f *and* on A (through $m(s)$), and the scalar polynomial $p(s)$ is in general *not* equal to $f(s)$ — only the matrix values agree. The next pages show that the c_i are fixed by matching f and p (and derivatives) at the eigenvalues.

Two methods to find $f(A)$:

- **Method I:** Interpolating polynomial
- **Method II:** Jordan canonical form

11.2 Method I: Interpolating Polynomial (Diagonalizable Case)

If A is diagonalizable $\Rightarrow m(s) = (s - \lambda_1)(s - \lambda_2)\cdots(s - \lambda_k)$, $\ell = k$:

$$p(s) = \sum_{i=0}^{\ell-1} c_i s^i, \qquad \ell = k \text{ is the \# of unknowns.}$$

Let λ_j, e_j be the eigenpairs, $j = 1, 2, \ldots, k$. First note

$$\begin{aligned} A^0 e_j &= \lambda_j^0 e_j = I e_j \\ A e_j &= \lambda_j e_j \\ A^2 e_j &= A(A e_j) = A\lambda_j e_j = \lambda_j A e_j = \lambda_j^2 e_j \\ A^i e_j &= \lambda_j^i e_j \end{aligned}$$

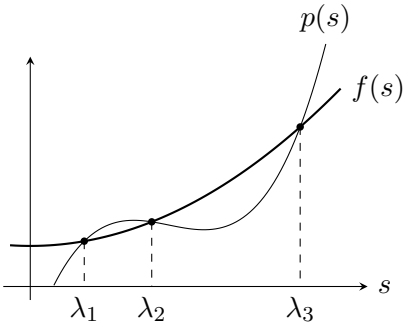
Then

$$\begin{aligned} f(A)e_j &= P(A)e_j \\ \sum_{i=0}^{\infty} \alpha_i A^i e_j &= \sum_{i=0}^{\ell-1} c_i A^i e_j \\ \Big(\sum_{i=0}^{\infty} \alpha_i \lambda_j^i\Big) e_j &= \Big(\sum_{i=0}^{\ell-1} c_i \lambda_j^i\Big) e_j \\ f(\lambda_j)e_j &= p(\lambda_j)e_j \\ f(\lambda_j)e_j - p(\lambda_j)e_j &= 0 \end{aligned}$$

▷ *e_j is a nonzero vector.*

$$\boxed{f(\lambda_j) = p(\lambda_j)}, \qquad j = 1, 2, \ldots, k$$

$\Rightarrow$ k **equations and** k **unknowns**; we can solve these.



Although $p(s)$ & $f(s)$ are different, they have the same values at the eigenvalues. Thus $p(s)$ is called the **interpolating polynomial**.

$$\left.\begin{array}{l} f(s) \neq p(s) \\ f(A) = P(A) \end{array}\right\rangle \text{ satisfied only at eigenvalues}$$

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$$\underbrace{f(A)\cdot e_j}_{\text{matrix}\times\text{vector}} = \underbrace{f(\lambda_j)\, e_j}_{\text{scalar}\times\text{vector}} \quad \Leftarrow \quad A\cdot e_j = \lambda_j e_j$$

▷ *The eigenvalue is modified by the function f.*

$$\begin{array}{lll} e^A & e^{\lambda_j} & e_j \\ \log A & \log \lambda_j & e_j \end{array}$$

▷ ***Eigenvectors are invariant under functions**: $f(A)$ has the same eigenvectors e_j as A, with eigenvalues $f(\lambda_j)$.*

Example 11.1.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow \lambda_1 = 3, \quad \lambda_2 = 1, \qquad e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

▷ *Bunları nasıl buluyorduk? (How did we find these?)* $Ae_1 = \lambda_1 e_1$, $[A - \lambda_1 I]e_1 = 0$.
▷ $d(s) = (s-2)^2 - 1 = (s-3)(s-1)$; $A - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \Rightarrow e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $A - I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow e_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

$$\begin{aligned} m(s) &= (s-3)(s-1) \\ \ell &= 2 \\ p(s) &= c_0 + c_1 s \end{aligned}$$

Find e^A & $\log A$.

e^A. $\rightarrow f(A) = p(A)$, $f(s) = e^s$, $f(\lambda_j) = p(\lambda_j)$, $j = 1, 2$:

$$\begin{aligned} f(3) = e^3 = p(3) &= c_0 + 3c_1 \\ f(1) = e = p(1) &= c_0 + c_1 \end{aligned}$$

$$\begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} e^3 \\ e \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} e^3 \\ e \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} e^3 - 3e \\ -e^3 + e \end{bmatrix}$$

$$\Rightarrow \; c_0 = \frac{3e - e^3}{2}, \qquad c_1 = \frac{e^3 - e}{2}$$

▷ $\det \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix} = -2$, *so the inverse is* $-\frac{1}{2}\begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix}$.

$$\boxed{e^A = c_0 I + c_1 A = \frac{3e - e^3}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{e^3 - e}{2}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}} = \frac{1}{2}\begin{bmatrix} e^3 + e & e^3 - e \\ e^3 - e & e^3 + e \end{bmatrix}$$

Check

$e^A e_1 = \frac{1}{2}\begin{bmatrix} 2e^3 \\ 2e^3 \end{bmatrix} = e^3 e_1$ and $e^A e_2 = \frac{1}{2}\begin{bmatrix} 2e \\ -2e \end{bmatrix} = e\, e_2$, as required by $f(A)e_j = f(\lambda_j)e_j$. Numerically $e^A \approx \begin{bmatrix} 11.402 & 8.684 \\ 8.684 & 11.402 \end{bmatrix}$, which agrees with summing the series $\sum_i A^i/i!$ (MATLAB `expm(A)`), while the entrywise $\begin{bmatrix} e^2 & e \\ e & e^2 \end{bmatrix}$ is different.

$\log A$**.** $f(s) = \log s$:

$$\begin{aligned} \log 3 &= c_0 + 3c_1 \\ \log 1 &= c_0 + c_1 = 0 \;\Rightarrow\; c_0 = -c_1 \\ \log 3 &= 2c_1 \;\Rightarrow\; c_1 = \frac{\log 3}{2}, \qquad c_0 = -\frac{\log 3}{2} \end{aligned}$$

▷ *Subtracting the second equation from the first gives* $\log 3 = 2c_1$.

$$\boxed{\log A = -\frac{\log 3}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{\log 3}{2}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}} = \frac{\log 3}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

▷ *Check:* $\log A\, e_1 = \log 3\, e_1$, $\log A\, e_2 = 0 = \log 1\, e_2$; *and* $e^{\log A} = A$.

$e^{At} = ?$ $f(s) = e^{st}$, $f(\lambda_j) = p(\lambda_j)$:

$$\begin{aligned} e^{3t} &= c_0 + 3c_1 = c_0(t) + 3c_1(t) \\ e^{t} &= c_0 + c_1 = c_0(t) + c_1(t) \end{aligned}$$

▷ *Same linear system as for* e^A *with* $e^3 \to e^{3t}$, $e \to e^t$; *the coefficients now depend on* t.

$$c_0(t) = \frac{3e^t - e^{3t}}{2}, \qquad c_1(t) = \frac{e^{3t} - e^t}{2}$$

$$\boxed{e^{At} = \frac{3e^t - e^{3t}}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{e^{3t} - e^t}{2}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}} = \frac{1}{2}\begin{bmatrix} e^{3t} + e^t & e^{3t} - e^t \\ e^{3t} - e^t & e^{3t} + e^t \end{bmatrix}$$

This solves

$$\dot{x} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} x, \quad x(0) = x_0 \qquad \Rightarrow \qquad x(t) = e^{At} x_0.$$

▷ *Check: at* $t = 0$, $e^{A\cdot 0} = I$; *and* $\frac{d}{dt}e^{At} = Ae^{At}$ *(e.g.* $(1,1)$ *entry:* $\frac{3e^{3t}+e^t}{2} = 2\cdot\frac{e^{3t}+e^t}{2} + \frac{e^{3t}-e^t}{2}$*).*

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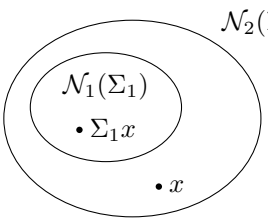
11.3 Method I: Non-Diagonalizable and General Case

Case II: A is not diagonalizable.

Example 11.2. Let us have $A \in \mathbb{R}^{3\times 3}$ and

$$m(s) = (s-\lambda_1)^2(s-\lambda_2)$$
$$d(s) = (s-\lambda_1)^2(s-\lambda_2) \qquad \text{as } A \in \mathbb{R}^{3\times 3}$$

$$J = \left[\begin{array}{cc|c} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ \hline 0 & 0 & \lambda_2 \end{array}\right], \qquad \underbrace{\mathcal{N}(A-\lambda_1 I)}_{1} \subsetneq \underbrace{\mathcal{N}(A-\lambda_1 I)^2}_{2} = \mathcal{N}(A-\lambda_1 I)^3$$



$x \in \mathcal{N}(\Sigma_1^2), \quad x \notin \mathcal{N}(\Sigma_1)$
$\{\Sigma_1 x,\ x\}$



$\mathcal{N}(A-\lambda_2 I), \quad 0 \neq y \in \mathcal{N}(\Sigma_2)$
$\{y\}$

$$? \begin{cases} P = [\,\Sigma_1 x \,\vdots\, x \,\vdots\, y\,] \\ P = [\,e_1 \,\vdots\, f_1 \,\vdots\, e_2\,] \end{cases} \qquad f(A) = p(A)$$

$$f(A)e_1 = p(A)e_1$$
$$f(\lambda_1) = p(\lambda_1)$$
$$f(\lambda_2) = p(\lambda_2)$$

$$\ell = 3, \qquad \boxed{p(s) = c_0 + c_1 s + c_2 s^2}$$

▷ *2 equations, but c_0, c_1, c_2: 3 unknowns ⇒ we need one more.*

Use the generalized eigenvector f_1:

$$f(A)\cdot f_1 = p(A)\cdot f_1, \qquad \sum_{i=0}^{\infty} \alpha_i A^i f_1 = \sum_{i=0}^{\ell-1} c_i A^i f_1$$

$$A^0 f_1 = \lambda_1^0 f_1$$
$$A f_1 = ?$$

$$\Sigma_1 f_1 = e_1, \qquad (A-\lambda_1 I) f_1 = e_1$$
$$\boxed{A f_1 = \lambda_1 f_1 + e_1}$$

▷ *As f_1 is not an eigenvector, Af_1 should include e_1.*

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$$A^2 f_1 = A\cdot(A f_1) = A(\lambda_1 f_1 + e_1)$$

$$
\begin{aligned}
&= \lambda_1 A f_1 + A e_1 \\
&= \lambda_1(\lambda_1 f_1 + e_1) + \lambda_1 e_1 \\
&= \lambda_1^2 f_1 + 2\lambda_1 e_1 \\
A^3 f_1 = A(A^2 f_1) &= A(\lambda_1^2 f_1 + 2\lambda_1 e_1) \\
&= \lambda_1^2(\lambda_1 f_1 + e_1) + 2\lambda_1^2 e_1 \\
&= \lambda_1^3 f_1 + 3\lambda_1^2 e_1
\end{aligned}
$$

$$\boxed{A^2 f_1 = \lambda_1^2 f_1 + 2\lambda_1 e_1} \qquad \boxed{A^3 f_1 = \lambda_1^3 f_1 + 3\lambda_1^2 e_1} \qquad \boxed{A^i f_1 = \lambda_1^i f_1 + i\lambda_1^{i-1} e_1}$$

Recall

$$p(s) = \sum_{i=0}^{\ell-1} c_i s^i, \qquad f(s) = \sum_{i=0}^{\infty} \alpha_i s^i, \qquad f(\lambda_1) = p(\lambda_1), \qquad f(\lambda_2) = p(\lambda_2).$$

Then

$$\Big(\sum_{i=0}^{\infty} \alpha_i \lambda_1^i\Big) f_1 + \Big(\sum_{i=0}^{\infty} \alpha_i\, i\lambda_1^{i-1}\Big) e_1 = \Big(\sum_{i=0}^{\ell-1} c_i \lambda_1^i\Big) f_1 + \Big(\sum_{i=0}^{\ell-1} c_i\, i\lambda_1^{i-1}\Big) e_1$$

$$\cancel{f(\lambda_1) f_1} + f'(\lambda_1) e_1 = \cancel{p(\lambda_1) f_1} + p'(\lambda_1) e_1$$

$$\boxed{f'(\lambda_1) = p'(\lambda_1)}$$

3 equations:

$$
\begin{aligned}
p(\lambda_1) &= f(\lambda_1) \\
p(\lambda_2) &= f(\lambda_2) \\
p'(\lambda_1) &= f'(\lambda_1)
\end{aligned}
$$

If $m(s) = (s-\lambda_1)^3(s-\lambda_2)$:

$$
\begin{aligned}
p(\lambda_1) &= f(\lambda_1), \qquad p(\lambda_2) = f(\lambda_2) \\
p'(\lambda_1) &= f'(\lambda_1) \\
p''(\lambda_1) &= f''(\lambda_1)
\end{aligned}
$$

11.3.1 General Case

$$
\begin{aligned}
&m(s) = (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k} \\
&\underbrace{c_0, c_1, \ldots, c_{\ell-1}}_{\ell \text{ unknowns}}, \qquad \ell = m_1 + m_2 + \cdots + m_k
\end{aligned}
$$

$$f^{(t)}(\lambda_j) = p^{(t)}(\lambda_j), \qquad t = 0, \ldots, m_j - 1 \ \ (m_j \text{ equations}), \quad j = 1, \ldots, k$$

Example 11.3. Find $\sin(\pi A)$ in terms of A, where

$$A = \left[\begin{array}{ccc|cc|cc} 0 & 1 & 0 & & & & \\ 0 & 0 & 1 & & & & \\ 0 & 0 & 0 & & & & \\ \hline & & & 0 & 1 & & \\ & & & 0 & 0 & & \\ \hline & & & & & 1 & 1 \\ & & & & & 0 & 1 \end{array}\right].$$

$$
\begin{aligned}
d(s) &= (s-1)^2 s^5 \\
m(s) &= (s-1)^2 s^3, \qquad \ell = 5
\end{aligned}
$$

$$p(s) = c_0 + c_1 s + c_2 s^2 + c_3 s^3 + c_4 s^4 \qquad (4^{\text{th}} \text{ order})$$

Conditions and derivatives:

- $f(0) = p(0)$
- $f'(0) = p'(0)$
- $f''(0) = p''(0)$
- $f(1) = p(1)$
- $f'(1) = p'(1)$

$$f(s) = \sin(\pi s)$$
$$f'(s) = \pi \cos(\pi s)$$
$$f''(s) = -\pi^2 \sin(\pi s)$$

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$$p'(s) = c_1 + 2c_2 s + 3c_3 s^2 + 4c_4 s^3, \qquad p''(s) = 2c_2 + 6c_3 s + 12c_4 s^2$$

$$\begin{aligned}
0 &= f(0) = p(0) = c_0 \\
\pi &= f'(0) = p'(0) = c_1 \\
0 &= f''(0) = p''(0) = 2c_2 \\
0 &= f(1) = p(1) = c_1 + \cancel{c_2} + c_3 + c_4 + \cancel{c_0} = \pi + c_3 + c_4 \\
-\pi &= f'(1) = p'(1) = c_1 + \cancel{2c_2} + 3c_3 + 4c_4 = \pi + 3c_3 + 4c_4
\end{aligned}$$

▷ $f(0) = \sin 0 = 0$, $f'(0) = \pi \cos 0 = \pi$, $f''(0) = -\pi^2 \sin 0 = 0$, $f(1) = \sin \pi = 0$, $f'(1) = \pi \cos \pi = -\pi$.

$$\boxed{\begin{aligned} c_0 &= 0 \\ c_1 &= \pi \\ c_2 &= 0 \end{aligned}} \quad \Rightarrow \quad \begin{aligned} c_3 + c_4 &= -\pi \\ 3c_3 + 4c_4 &= -2\pi \end{aligned} \quad \Rightarrow \quad \boxed{\begin{aligned} c_3 &= -2\pi \\ c_4 &= \pi \end{aligned}}$$

$$\boxed{\sin(\pi A) = \pi A - 2\pi A^3 + \pi A^4}$$

We still need to take 3rd (and 4th) powers of A. If A is in JCF, Method II will be much more efficient.

Check

From $c_3 + c_4 = -\pi$: $c_4 = -\pi - c_3$, so $3c_3 - 4\pi - 4c_3 = -2\pi \Rightarrow c_3 = -2\pi$, $c_4 = \pi$. Evaluating $\pi A - 2\pi A^3 + \pi A^4$ blockwise: for the nilpotent blocks N (3×3 and 2×2) $N^3 = 0$, so the result is πN; for the block $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ it gives $\begin{bmatrix} 0 & -\pi \\ 0 & 0 \end{bmatrix}$. This matches Method II below: $\sin(\pi N) = \pi N - \frac{\pi^3}{6} N^3 + \cdots = \pi N$, and $\sin\left(\pi \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} \sin \pi & \pi \cos \pi \\ 0 & \sin \pi \end{bmatrix} = \begin{bmatrix} 0 & -\pi \\ 0 & 0 \end{bmatrix}$.

$$f^{(t)}(\lambda_j), \qquad t = 0, \ldots, m_j - 1, \quad j = 1, \ldots, k$$

If any derivative does not exist (∞ or div 0), then $f(A)$ **does not exist**.

Remark. $f(A)$ does not exist when one of the $f^{(t)}(\lambda_j)$, $j = 1, 2, \ldots, k$; $t = 0, \ldots, m_j - 1$ does not exist.

Example 11.4.

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \lambda_1 = 0, \quad \lambda_2 = 1$$

- $\cancel{\log A = ?}$, $\cancel{\log s}$: gives infinity (at $\lambda_1 = 0$).
- $\cancel{(I - A)^{-1} = ?}$, $\cancel{(1 - s)^{-1} = \frac{1}{1-s}} \Rightarrow$ for $\lambda = 1$: $\frac{1}{0} = \infty$.

- $A^{-1} \Longrightarrow$ function of a matrix. $s^{-1} = \dfrac{1}{s} \Rightarrow$ if one of the eigenvalues is zero, the inverse does not exist.

Example 11.5. $f(s) = s^{1/2} = \sqrt{s}, \qquad A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad m(s) = s^2.$

$$f(0) = p(0) \quad \Rightarrow \text{ no problem}$$
$$\cancel{f'(0) = p'(0)} \quad \Rightarrow \text{ problem: } f'(s) = \frac{1}{2}\frac{1}{\sqrt{s}} \text{ @ } s = 0 \ \rightarrow \ \infty$$

▷ *The square root function does not apply here: its derivative is not defined at the eigenvalue $s = 0$.*

$\Rightarrow \sqrt{A}$ does not exist for $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

Direct confirmation

Suppose $X^2 = A$ with $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then $X^4 = A^2 = 0$, so X is nilpotent; a nilpotent 2×2 matrix satisfies $X^2 = 0$ (its minimal polynomial divides s^2). Hence $A = X^2 = 0$, a contradiction. So *no* matrix squares to this A. In contrast, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (with $m(s) = s$, only $f(0)$ needed) has the square root 0: whether $f(A)$ exists depends on which derivatives the minimal polynomial requires, not only on the eigenvalues.

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11.4 Method II: Jordan Canonical Form

$$J = \underbrace{P^{-1}AP}_{\substack{\text{similarity} \\ \text{transformation}}}, \qquad J \text{ is similar to } A.$$

Similar matrices have the same characteristic polynomial:

$$B = P^{-1}AP \qquad \text{(scalar analogue: } b = \cancel{p^{-1}}a\cancel{p} = a)$$
$$d_B(s) = \det(sI - B) = \det(sI - P^{-1}AP)$$
$$= \det(P^{-1}(sI - A)P) = \underbrace{\cancel{\det(P^{-1})}}_{\frac{1}{\det P}} \cdot \det(sI - A) \cdot \cancel{\det(P)}$$
$$d_B(s) = d_A(s)$$

Powers of $A = PJP^{-1}$:

$$(PJP^{-1})^0 = I = PP^{-1} = P \cdot J^0 P^{-1}$$
$$(PJP^{-1})^1 = PJP^{-1}$$
$$(PJP^{-1})^2 = PJ\cancel{P^{-1}}\cancel{P}JP^{-1} = PJ^2P^{-1}$$
$$(PJP^{-1})^n = PJ^nP^{-1}$$

Hence

$$f(A) = \sum_{i=0}^{\infty} \alpha_i (PJP^{-1})^i = \sum_{i=0}^{\infty} \alpha_i P(J^i)P^{-1}$$
$$= P \cdot \underbrace{\left(\sum_{i=0}^{\infty} \alpha_i J^i\right)}_{f(J)} P^{-1}$$

$$\boxed{f(A) = Pf(J)P^{-1}}$$

Block structure.

$$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_\sigma \end{bmatrix}, \qquad J_t = \begin{bmatrix} \lambda_j & 1 & & 0 \\ & \lambda_j & 1 & \\ & & \lambda_j & \ddots \\ & & & \ddots & 1 \\ 0 & & & & \lambda_j \end{bmatrix}_{d_t \times d_t}$$

▷ *J_t ($t = 1, \dots, \sigma$) denotes a generic Jordan block of size d_t belonging to the eigenvalue λ_j.*

$$J^i = \begin{bmatrix} J_1^i & & & \\ & J_2^i & & \\ & & \ddots & \\ & & & J_\sigma^i \end{bmatrix}, \qquad f(J) = \sum_{i=0}^{\infty} \alpha_i J^i = \begin{bmatrix} \sum_{i=0}^{\infty} \alpha_i J_1^i & & & 0 \\ & \sum_{i=0}^{\infty} \alpha_i J_2^i & & \\ & & \ddots & \\ 0 & & & \sum_{i=0}^{\infty} \alpha_i J_\sigma^i \end{bmatrix}$$

$$f(J) = \begin{bmatrix} f(J_1) & & & 0 \\ & f(J_2) & & \\ & & \ddots & \\ 0 & & & f(J_\sigma) \end{bmatrix}$$

One Jordan block. Expanding around $s_0 = \lambda_j$ (advantageous):

$$f(J_t) = \sum_{i=0}^{\infty} \alpha_i J_t^i \underset{s_0=\lambda_j}{=} \sum_{i=0}^{\infty} \frac{f^{(i)}(\lambda_j)}{i!} (J_t - \lambda_j I)^i$$

$$J_t - \lambda_j I = \begin{bmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix}_{d_t \times d_t}, \qquad (J_t - \lambda_j I)^2 = \begin{bmatrix} 0 & 0 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ & & & & 0 \end{bmatrix}_{d_t \times d_t}$$

▷ *Each power shifts the line of 1's one diagonal up (↗).*

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$$(*) \qquad (J_t - \lambda_j I)^d = 0, \qquad d \geq d_t$$

$$(J_t-\lambda_j I)^{d_t-1}=\begin{bmatrix} & 0 & 1\\ 0 & & 0\\ & & \end{bmatrix},\qquad (J_t-\lambda_j I)^0=\begin{bmatrix}1 & & 0\\ & \ddots & \\ 0 & & 1\end{bmatrix}$$

$$f(J_t)=\sum_{i=0}^{d_t-1}\frac{f^{(i)}(\lambda_j)}{i!}(J_t-\lambda_j I)^i$$

▷ *Upper limit d_t-1 due to $(*)$.*

$$f(J_t)=\begin{bmatrix} f(\lambda_j) & f'(\lambda_j) & \frac{f''(\lambda_j)}{2} & \cdots & \frac{f^{(d_t-1)}(\lambda_j)}{(d_t-1)!}\\ & f(\lambda_j) & f'(\lambda_j) & \frac{f''(\lambda_j)}{2!} & \vdots\\ & & \ddots & \ddots & \frac{f''(\lambda_j)}{2}\\ & 0 & & \ddots & f'(\lambda_j)\\ & & & & f(\lambda_j)\end{bmatrix}$$

- 0th power contributes to the diagonal.
- 1st super diagonal is formed by the first power.

  ⋮

Link with Method I

$f(J_t)$ only uses $f(\lambda_j), f'(\lambda_j), \ldots, f^{(d_t-1)}(\lambda_j)$. The largest block of λ_j has size m_j (the exponent in $m(s)$), so Methods I and II use exactly the same data $f^{(t)}(\lambda_j)$, $t=0,\ldots,m_j-1$; in particular $f(A)$ exists iff these derivatives exist. Method II avoids solving for $c_0,\ldots,c_{\ell-1}$ and computing powers of A, but needs P and P^{-1}.

Example 11.6.

$$A=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix},\qquad \lambda_1=3,\ \lambda_2=1,\qquad e_1=\begin{bmatrix}1\\1\end{bmatrix},\quad e_2=\begin{bmatrix}1\\-1\end{bmatrix}$$

$$P=[\,e_1 \vdots e_2\,],\qquad A=P\underbrace{\left[\begin{array}{c|c}3 & 0\\ \hline 0 & 1\end{array}\right]}_{J}P^{-1}\qquad\Rightarrow\qquad f(A)=P\left[\begin{array}{c|c}f(3) & 0\\ \hline 0 & f(1)\end{array}\right]P^{-1}$$

With $P=\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$, $P^{-1}=\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$:

$$f(A)=\frac{1}{2}\begin{bmatrix}f(3)+f(1) & f(3)-f(1)\\ f(3)-f(1) & f(3)+f(1)\end{bmatrix}.$$

▷ *For $f(s)=e^s$, $\log s$, e^{st} this reproduces the Method I results e^A, $\log A$, e^{At} above.*

Example 11.7. $\sin \pi A = ?$ for

$$A = \left[\begin{array}{ccc|cc|c} \frac{1}{2} & 1 & 0 & & & \\ 0 & \frac{1}{2} & 1 & & & \\ 0 & 0 & \frac{1}{2} & & & \\ \hline & & & \frac{3}{2} & 1 & \\ & & & 0 & \frac{3}{2} & \\ \hline & & & & & \frac{5}{2} \end{array}\right] = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & J_3 \end{bmatrix}.$$

$$A = PJP^{-1}, \quad P = I \text{ chosen} \Rightarrow A = J$$
$$d(s) = \left(s - \tfrac{1}{2}\right)^3 \left(s - \tfrac{3}{2}\right)^2 \left(s - \tfrac{5}{2}\right)$$
$$m(s) = \left(s - \tfrac{1}{2}\right)^3 \left(s - \tfrac{3}{2}\right)^2 \left(s - \tfrac{5}{2}\right)^1$$
$$p(s) = c_0 + c_1 s + c_2 s^2 + c_3 s^3 + c_4 s^4 + c_5 s^5$$

▷ *Method I ↪ too complicated; use Method II blockwise.*

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$$f(s) = \sin \pi s, \qquad f'(s) = \cos \pi s \cdot \pi, \qquad f''(s) = -\pi^2 \sin \pi s$$

$$\sin \pi J_1 = \begin{bmatrix} 1 & 0 & -\frac{\pi^2}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \sin \pi J_2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad \sin \pi J_3 = \begin{bmatrix} 1 \end{bmatrix}$$

▷ J_1 *(*$\lambda = \frac{1}{2}$*, size 3):* $f(\frac{1}{2}) = 1$, $f'(\frac{1}{2}) = \pi \cos \frac{\pi}{2} = 0$, $\frac{f''(\frac{1}{2})}{2} = -\frac{\pi^2}{2}$. J_2 *(*$\lambda = \frac{3}{2}$*):* $f = -1$, $f' = \pi \cos \frac{3\pi}{2} = 0$. J_3*:* $\sin \frac{5\pi}{2} = 1$.

$$\sin \pi A = \left[\begin{array}{ccc|cc|c} 1 & 0 & -\frac{\pi^2}{2} & & & \\ 0 & 1 & 0 & & & \\ 0 & 0 & 1 & & & \\ \hline & & & -1 & 0 & \\ & & & 0 & -1 & \\ \hline & & & & & 1 \end{array}\right]$$

Example 11.8 (Square root of a matrix)**.**

$$f(s) = \sqrt{s} = s^{1/2}, \qquad A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad A^{1/2} = P \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{1} \end{bmatrix} P^{-1} \;\Rightarrow\; \text{unique}$$

▷ ↪ *unique p.def. sqrt of A: MATLAB* ~~`sqrt(A)`~~*,* `sqrtm(A)` ✓.
But if we only require $A^{1/2} \cdot A^{1/2} = A \Rightarrow$ not unique anymore (compare: $x = \sqrt{3}$ vs. $y \cdot y = 3$):

$$A^{1/2} \cdot A^{1/2} = P \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} P^{-1}$$
$$A^{1/2} = P \begin{bmatrix} -\sqrt{3} & 0 \\ 0 & \sqrt{1} \end{bmatrix} P^{-1} \;\Rightarrow\; A^{1/2} A^{1/2} = P \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} P^{-1}$$
$$A^{1/2} = P \begin{bmatrix} \pm\sqrt{3} & 0 \\ 0 & \pm\sqrt{1} \end{bmatrix} P^{-1}$$

▷ *These are all valid for the definition* $A^{1/2} A^{1/2} = A$ *(signs chosen independently: 4 matrices).*

The numbers

With $P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $P\,\mathrm{diag}(a,b)P^{-1} = \frac{1}{2}\begin{bmatrix} a+b & a-b \\ a-b & a+b \end{bmatrix}$ and its square is $\frac{1}{2}\begin{bmatrix} a^2+b^2 & a^2-b^2 \\ a^2-b^2 & a^2+b^2 \end{bmatrix}$, which equals A whenever $a^2 = 3$, $b^2 = 1$. The principal (function) square root is

$$A^{1/2} = \texttt{sqrtm(A)} = \frac{1}{2}\begin{bmatrix} \sqrt{3}+1 & \sqrt{3}-1 \\ \sqrt{3}-1 & \sqrt{3}+1 \end{bmatrix} \approx \begin{bmatrix} 1.366 & 0.366 \\ 0.366 & 1.366 \end{bmatrix},$$

whose eigenvalues $\sqrt{3}, 1$ are positive: it is **the unique positive-definite square root**. The other three choices $(a,b) = (-\sqrt{3},1), (\sqrt{3},-1), (-\sqrt{3},-1)$ also square to A but are not positive definite.

If A is positive definite $\Rightarrow$ the square root is wanted to be positive definite, which is given by the function.